symembed: simple cases written out in full

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Each section shows one case. The slots of w are listed with horizontal rules between G-orbits. Each group element's action is shown as the row $g \cdot w$, which holds the coefficients of $q_g(x) = \sum_j (g \cdot w)_j \, x^j$. Then every Bursarial output $E_{c,j}$, where $\prod_g \big(y + q_g(x)\big) = \sum_c y^{|G|-c} \sum_j E_{c,j} \, x^j$, is shown expanded in the d's and e's. The status column shows whether the safe reductions keep the output, and otherwise the exact identity used to drop it. No determinants are omitted here, because no case contains parity. Expansions with more than 100 terms show only their term count. Every expansion appears in full in `simple_cases_full.txt`. This file is generated by `symembed/explore/simple_cases.py`; do not edit it by hand.

1 $S(2)$ acting on (a, b), $k = 2$

$n = 2$, $k = 2$, $|G| = 2$, $m = 4$ slots. Bursarial outputs: 11; after safe reductions: 8 (exact linear dependencies checked for $c \le 2$). Quotient dimension (the minimum number of outputs, for generic inputs): 3.

slot j	w_j	degree in L
0	d_{aa}	2
1	d_{bb}	2
2	d_{ab}	2
3	e_{ab}	2

g	$g \cdot w$
e	$(d_{aa},\ d_{bb},\ d_{ab},\ e_{ab})$
(ab)	$(d_{bb},\ d_{aa},\ d_{ab},\ -e_{ab})$

$c = 1$

output	expanded	safe reductions
$E_{1,0}$	$= d_{aa} + d_{bb}$	kept
$E_{1,1}$	$= d_{aa} + d_{bb}$	dropped: $= E_{1,0}$
$E_{1,2}$	$= 2d_{ab}$	kept

$E_{1,3}$	$= 0$	dropped: $= 0$

$c = 2$

output	expanded	safe reductions
$E_{2,0}$	$= d_{aa}d_{bb}$	kept
$E_{2,1}$	$= d_{aa}^2 + d_{bb}^2$	kept
$E_{2,2}$	$= d_{aa}d_{ab} + d_{aa}d_{bb} + d_{ab}d_{bb}$	kept
$E_{2,3}$	$= d_{aa}d_{ab} - d_{aa}e_{ab} + d_{ab}d_{bb} + d_{bb}e_{ab}$	kept
$E_{2,4}$	$= d_{aa}e_{ab} + d_{ab}^2 - d_{bb}e_{ab}$	kept
$E_{2,5}$	$= 0$	dropped: $= 0$
$E_{2,6}$	$= -e_{ab}^2$	kept

2 $S(3)$ acting on (a, b, c), $k = 2$

$n = 3$, $k = 2$, $|G| = 6$, $m = 9$ slots. Bursarial outputs: 174; after safe reductions: 155 (exact linear dependencies checked for $c \leq 6$). Quotient dimension (the minimum number of outputs, for generic inputs): 5.

slot j	w_j	degree in L
0	d_{aa}	2
1	d_{bb}	2
2	d_{cc}	2
3	d_{ab}	2
4	d_{ac}	2
5	d_{bc}	2
6	e_{ab}	2
7	e_{ac}	2
8	e_{bc}	2

g	$g \cdot w$
e	$(d_{aa},\ d_{bb},\ d_{cc},\ d_{ab},\ d_{ac},\ d_{bc},\ e_{ab},\ e_{ac},\ e_{bc})$
(bc)	$(d_{aa},\ d_{cc},\ d_{bb},\ d_{ac},\ d_{ab},\ d_{bc},\ e_{ac},\ e_{ab},\ -e_{bc})$
(ab)	$(d_{bb},\ d_{aa},\ d_{cc},\ d_{ab},\ d_{bc},\ d_{ac},\ -e_{ab},\ e_{bc},\ e_{ac})$
(abc)	$(d_{bb},\ d_{cc},\ d_{aa},\ d_{bc},\ d_{ab},\ d_{ac},\ e_{bc},\ -e_{ab},\ -e_{ac})$
(acb)	$(d_{cc},\ d_{aa},\ d_{bb},\ d_{ac},\ d_{bc},\ d_{ab},\ -e_{ac},\ -e_{bc},\ e_{ab})$
(ac)	$(d_{cc},\ d_{bb},\ d_{aa},\ d_{bc},\ d_{ac},\ d_{ab},\ -e_{bc},\ -e_{ac},\ -e_{ab})$

$c = 1$

output	expanded	safe reductions
$E_{1,0}$	$= 2d_{aa} + 2d_{bb} + 2d_{cc}$	kept
$E_{1,1}$	$= 2d_{aa} + 2d_{bb} + 2d_{cc}$	dropped: $= E_{1,0}$
$E_{1,2}$	$= 2d_{aa} + 2d_{bb} + 2d_{cc}$	dropped: $= E_{1,0}$
$E_{1,3}$	$= 2d_{ab} + 2d_{ac} + 2d_{bc}$	kept
$E_{1,4}$	$= 2d_{ab} + 2d_{ac} + 2d_{bc}$	dropped: $= E_{1,3}$
$E_{1,5}$	$= 2d_{ab} + 2d_{ac} + 2d_{bc}$	dropped: $= E_{1,3}$
$E_{1,6}$	$= 0$	dropped: $= 0$
$E_{1,7}$	$= 0$	dropped: $= 0$
$E_{1,8}$	$= 0$	dropped: $= 0$

$c = 2$

output	expanded	safe reductions
$E_{2,0}$	$= d_{aa}^2 + 4d_{aa}d_{bb} + 4d_{aa}d_{cc} + d_{bb}^2 + 4d_{bb}d_{cc} + d_{cc}^2$	kept
$E_{2,1}$	$= 4d_{aa}^2 + 6d_{aa}d_{bb} + 6d_{aa}d_{cc} + 4d_{bb}^2 + 6d_{bb}d_{cc} + 4d_{cc}^2$	kept
$E_{2,2}$	$= 5d_{aa}^2 + 10d_{aa}d_{bb} + 10d_{aa}d_{cc} + 5d_{bb}^2 + 10d_{bb}d_{cc} + 5d_{cc}^2$	dropped: $= E_{2,0} + E_{2,1}$
$E_{2,3}$	$= 4d_{aa}^2 + 3d_{aa}d_{ab} + 3d_{aa}d_{ac} + 6d_{aa}d_{bb} + 4d_{aa}d_{bc} + 6d_{aa}d_{cc} + 3d_{ab}d_{bb} + 4d_{ab}d_{cc} + 4d_{ac}d_{bb} + 3d_{ac}d_{cc} + 4d_{bb}^2 + 3d_{bb}d_{bc} + 6d_{bb}d_{cc} + 3d_{bc}d_{cc} + 4d_{cc}^2$	kept
$E_{2,4}$	$= d_{aa}^2 + 6d_{aa}d_{ab} + 6d_{aa}d_{ac} + 4d_{aa}d_{bb} + 8d_{aa}d_{bc} + 4d_{aa}d_{cc} + 6d_{ab}d_{bb} + 8d_{ab}d_{cc} + 8d_{ac}d_{bb} + 6d_{ac}d_{cc} + d_{bb}^2 + 6d_{bb}d_{bc} + 4d_{bb}d_{cc} + 6d_{bc}d_{cc} + d_{cc}^2$	dropped: $= E_{2,0} - 2E_{2,1} + 2E_{2,3}$
$E_{2,5}$	$= 12d_{aa}d_{ab} + 12d_{aa}d_{ac} + 6d_{aa}d_{bc} + 12d_{ab}d_{bb} + 6d_{ab}d_{cc} + 6d_{ac}d_{bb} + 12d_{ac}d_{cc} + 12d_{bb}d_{bc} + 12d_{bc}d_{cc}$	kept
$E_{2,6}$	$= 6d_{aa}d_{ab} + 6d_{aa}d_{ac} + 8d_{aa}d_{bc} - d_{aa}e_{ab} - d_{aa}e_{ac} + d_{ab}^2 + 4d_{ab}d_{ac} + 6d_{ab}d_{bb} + 4d_{ab}d_{bc} + 8d_{ab}d_{cc} + d_{ac}^2 + 8d_{ac}d_{bb} + 4d_{ac}d_{bc} + 6d_{ac}d_{cc} + 6d_{bb}d_{bc} + d_{bb}e_{ab} - d_{bb}e_{bc} + d_{bc}^2 + 6d_{bc}d_{cc} + d_{cc}e_{ac} + d_{cc}e_{bc}$	kept
$E_{2,7}$	$= 3d_{aa}d_{ab} + 3d_{aa}d_{ac} + 4d_{aa}d_{bc} + 4d_{ab}^2 + 6d_{ab}d_{ac} + 3d_{ab}d_{bb} + 6d_{ab}d_{bc} + 4d_{ab}d_{cc} + 4d_{ac}^2 + 4d_{ac}d_{bb} + 6d_{ac}d_{bc} + 3d_{ac}d_{cc} + 3d_{bb}d_{bc} + 4d_{bc}^2 + 3d_{bc}d_{cc}$	kept
$E_{2,8}$	$= 5d_{ab}^2 + 10d_{ab}d_{ac} + 10d_{ab}d_{bc} + 5d_{ac}^2 + 10d_{ac}d_{bc} + 5d_{bc}^2$	kept
$E_{2,9}$	$= 4d_{ab}^2 + 6d_{ab}d_{ac} + 6d_{ab}d_{bc} + 4d_{ac}^2 + 6d_{ac}d_{bc} + 4d_{bc}^2$	dropped: $= E_{2,1} - E_{2,3} + E_{2,7}$
$E_{2,10}$	$= d_{aa}e_{ab} + d_{aa}e_{ac} + d_{ab}^2 + 4d_{ab}d_{ac} + 4d_{ab}d_{bc} - 2d_{ab}e_{ac} - 2d_{ab}e_{bc} + d_{ac}^2 + 4d_{ac}d_{bc} - 2d_{ac}e_{ab} + 2d_{ac}e_{bc} - d_{bb}e_{ab} + d_{bb}e_{bc} + d_{bc}^2 + 2d_{bc}e_{ab} + 2d_{bc}e_{ac} - d_{cc}e_{ac} - d_{cc}e_{bc}$	kept
$E_{2,11}$	$= 0$	dropped: $= 0$
$E_{2,12}$	$= 2d_{ab}e_{ac} + 2d_{ab}e_{bc} + 2d_{ac}e_{ab} - 2d_{ac}e_{bc} - 2d_{bc}e_{ab} - 2d_{bc}e_{ac} - e_{ab}^2 - e_{ac}^2 - e_{bc}^2$	kept
$E_{2,13}$	$= -2e_{ab}e_{ac} + 2e_{ab}e_{bc} - 2e_{ac}e_{bc}$	kept
$E_{2,14}$	$= -e_{ab}^2 + 2e_{ab}e_{ac} - 2e_{ab}e_{bc} - e_{ac}^2 + 2e_{ac}e_{bc} - e_{bc}^2$	dropped: $= 4E_{2,1} - 4E_{2,3} + E_{2,6} + 2E_{2,7} - 2E_{2,8} + E_{2,10} + E_{2,12} - E_{2,13}$
$E_{2,15}$	$= -2e_{ab}e_{ac} + 2e_{ab}e_{bc} - 2e_{ac}e_{bc}$	dropped: $= E_{2,13}$
$E_{2,16}$	$= -e_{ab}^2 - e_{ac}^2 - e_{bc}^2$	dropped: $= 4E_{2,1} - 4E_{2,3} + E_{2,6} + 2E_{2,7} - 2E_{2,8} + E_{2,10} + E_{2,12}$

$c = 3$

output	expanded	safe reductions
$E_{3,0}$	$= 2d_{aa}^2d_{bb} + 2d_{aa}^2d_{cc} + 2d_{aa}d_{bb}^2 + 8d_{aa}d_{bb}d_{cc} + 2d_{aa}d_{cc}^2 + 2d_{bb}^2d_{cc} + 2d_{bb}d_{cc}^2$	kept
$E_{3,1}$	$= 2d_{aa}^3 + 7d_{aa}^2d_{bb} + 7d_{aa}^2d_{cc} + 7d_{aa}d_{bb}^2 + 12d_{aa}d_{bb}d_{cc} + 7d_{aa}d_{cc}^2 + 2d_{bb}^3 + 7d_{bb}^2d_{cc} + 7d_{bb}d_{cc}^2 + 2d_{cc}^3$	kept
$E_{3,2}$	$= 4d_{aa}^3 + 14d_{aa}^2d_{bb} + 14d_{aa}^2d_{cc} + 14d_{aa}d_{bb}^2 + 24d_{aa}d_{bb}d_{cc} + 14d_{aa}d_{cc}^2 + 4d_{bb}^3 + 14d_{bb}^2d_{cc} + 14d_{bb}d_{cc}^2 + 4d_{cc}^3$	dropped: $= 2E_{3,1}$
$E_{3,3}$	$= 8d_{aa}^3 + d_{aa}^2d_{ab} + d_{aa}^2d_{ac} + 14d_{aa}^2d_{bb} + 2d_{aa}^2d_{bc} + 14d_{aa}^2d_{cc} + 4d_{aa}d_{ab}d_{bb} + 6d_{aa}d_{ab}d_{cc} + 6d_{aa}d_{ac}d_{bb} + 4d_{aa}d_{ac}d_{cc} + 14d_{aa}d_{bb}^2 + 6d_{aa}d_{bb}d_{bc} + 32d_{aa}d_{bb}d_{cc} + 6d_{aa}d_{bc}d_{cc} + 14d_{aa}d_{cc}^2 + d_{ab}d_{bb}^2 + 6d_{ab}d_{bb}d_{cc} + 2d_{ab}d_{cc}^2 + 2d_{ac}d_{bb}^2 + 6d_{ac}d_{bb}d_{cc} + d_{ac}d_{cc}^2 + 8d_{bb}^3 + d_{bb}^2d_{bc} + 14d_{bb}^2d_{cc} + 4d_{bb}d_{bc}d_{cc} + 14d_{bb}d_{cc}^2 + d_{bc}d_{cc}^2 + 8d_{cc}^3$	kept

$E_{3,4}$ $= 4d_{aa}^3 + 5d_{aa}^2 d_{ab} + 5d_{aa}^2 d_{ac} + 14d_{aa}^2 d_{bb} + 10d_{aa}^2 d_{bc} + 14d_{aa}^2 d_{cc} + 12d_{aa} d_{ab} d_{bb} + 14d_{aa} d_{ab} d_{cc} + 14d_{aa} d_{ac} d_{bb} +$ kept
$12d_{aa} d_{ac} d_{cc} + 14d_{aa} d_{bb}^2 + 14d_{aa} d_{bb} d_{bc} + 24d_{aa} d_{bb} d_{cc} + 14d_{aa} d_{bc} d_{cc} + 14d_{aa} d_{cc}^2 + 5d_{ab} d_{bb}^2 + 14d_{ab} d_{bb} d_{cc} +$
$10d_{ab} d_{cc}^2 + 10d_{ac} d_{bb}^2 + 14d_{ac} d_{bb} d_{cc} + 5d_{ac} d_{cc}^2 + 4d_{bb}^3 + 5d_{bb}^2 d_{bc} + 14d_{bb}^2 d_{cc} + 12d_{bb} d_{bc} d_{cc} + 14d_{bb} d_{cc}^2 + 5d_{bc} d_{cc}^2 + 4d_{cc}^3$

$E_{3,5}$ $= 2d_{aa}^3 + 15d_{aa}^2 d_{ab} + 15d_{aa}^2 d_{ac} + 7d_{aa}^2 d_{bb} + 10d_{aa}^2 d_{bc} + 7d_{aa}^2 d_{cc} + 28d_{aa} d_{ab} d_{bb} + 26d_{aa} d_{ab} d_{cc} + 26d_{aa} d_{ac} d_{bb} +$ kept
$28d_{aa} d_{ac} d_{cc} + 7d_{aa} d_{bb}^2 + 26d_{aa} d_{bb} d_{bc} + 12d_{aa} d_{bb} d_{cc} + 26d_{aa} d_{bc} d_{cc} + 7d_{aa} d_{cc}^2 + 15d_{ab} d_{bb}^2 + 26d_{ab} d_{bb} d_{cc} +$
$10d_{ab} d_{cc}^2 + 10d_{ac} d_{bb}^2 + 26d_{ac} d_{bb} d_{cc} + 15d_{ac} d_{cc}^2 + 2d_{bb}^3 + 15d_{bb}^2 d_{bc} + 7d_{bb}^2 d_{cc} + 28d_{bb} d_{bc} d_{cc} + 7d_{bb} d_{cc}^2 + 15d_{bc} d_{cc}^2 + 2d_{cc}^3$

$E_{3,6}$ $= 18d_{aa}^2 d_{ab} + 18d_{aa}^2 d_{ac} + 2d_{aa}^2 d_{bb} + 16d_{aa}^2 d_{bc} + 2d_{aa}^2 d_{cc} - d_{aa}^2 e_{ab} - d_{aa}^2 e_{ac} + d_{aa} d_{ab}^2 + 4d_{aa} d_{ab} d_{ac} + 32d_{aa} d_{ab} d_{bb} +$ kept
$6d_{aa} d_{ab} d_{bc} + 28d_{aa} d_{ab} d_{cc} + d_{aa} d_{ac}^2 + 28d_{aa} d_{ac} d_{bb} + 6d_{aa} d_{ac} d_{bc} + 32d_{aa} d_{ac} d_{cc} + 2d_{aa} d_{bb}^2 + 28d_{aa} d_{bb} d_{bc} +$
$8d_{aa} d_{bb} d_{cc} - 2d_{aa} d_{bb} e_{ac} - 2d_{aa} d_{bb} e_{bc} + 2d_{aa} d_{bc}^2 + 28d_{aa} d_{bc} d_{cc} + 2d_{aa} d_{cc}^2 - 2d_{aa} d_{cc} e_{ab} + 2d_{aa} d_{cc} e_{bc} +$
$d_{ab}^2 d_{bb} + 2d_{ab}^2 d_{cc} + 6d_{ab} d_{ac} d_{bb} + 6d_{ab} d_{ac} d_{cc} + 18d_{ab} d_{bb}^2 + 4d_{ab} d_{bb} d_{bc} + 28d_{ab} d_{bb} d_{cc} + 6d_{ab} d_{bc} d_{cc} + 16d_{ab} d_{cc}^2 +$
$2d_{ac}^2 d_{bb} + d_{ac}^2 d_{cc} + 16d_{ac} d_{bb}^2 + 6d_{ac} d_{bb} d_{bc} + 28d_{ac} d_{bb} d_{cc} + 4d_{ac} d_{bc} d_{cc} + 18d_{ac} d_{cc}^2 + 18d_{bb}^2 d_{bc} + 2d_{bb}^2 d_{cc} +$
$d_{bb}^2 e_{ab} - d_{bb}^2 e_{bc} + d_{bb} d_{bc}^2 + 32d_{bb} d_{bc} d_{cc} + 2d_{bb} d_{cc}^2 + 2d_{bb} d_{cc} e_{ab} + 2d_{bb} d_{cc} e_{ac} + d_{bc}^2 d_{cc} + 18d_{bc} d_{cc}^2 + d_{cc}^2 e_{ac} + d_{cc}^2 e_{bc}$

$E_{3,7}$ $= 15d_{aa}^2 d_{ab} + 15d_{aa}^2 d_{ac} + 10d_{aa}^2 d_{bc} - d_{aa}^2 e_{ab} - d_{aa}^2 e_{ac} + 5d_{aa} d_{ab}^2 + 12d_{aa} d_{ab} d_{ac} + 28d_{aa} d_{ab} d_{bb} +$ kept
$14d_{aa} d_{ab} d_{bc} + 26d_{aa} d_{ab} d_{cc} + 5d_{aa} d_{ac}^2 + 26d_{aa} d_{ac} d_{bb} + 14d_{aa} d_{ac} d_{bc} + 28d_{aa} d_{ac} d_{cc} + 26d_{aa} d_{bb} d_{bc} -$
$2d_{aa} d_{bb} e_{ac} - 2d_{aa} d_{bb} e_{bc} + 10d_{aa} d_{bc}^2 + 26d_{aa} d_{bc} d_{cc} - 2d_{aa} d_{cc} e_{ab} + 2d_{aa} d_{cc} e_{bc} + 5d_{ab}^2 d_{bb} + 10d_{ab}^2 d_{cc} +$
$14d_{ab} d_{ac} d_{bb} + 14d_{ab} d_{ac} d_{cc} + 15d_{ab} d_{bb}^2 + 12d_{ab} d_{bb} d_{bc} + 26d_{ab} d_{bb} d_{cc} + 14d_{ab} d_{bc} d_{cc} + 10d_{ab} d_{cc}^2 + 10d_{ac}^2 d_{bb} +$
$5d_{ac}^2 d_{cc} + 10d_{ac} d_{bb}^2 + 14d_{ac} d_{bb} d_{bc} + 26d_{ac} d_{bb} d_{cc} + 12d_{ac} d_{bc} d_{cc} + 15d_{ac} d_{cc}^2 + 15d_{bb}^2 d_{bc} + d_{bb}^2 e_{ab} - d_{bb}^2 e_{bc} +$
$5d_{bb} d_{bc}^2 + 28d_{bb} d_{bc} d_{cc} + 2d_{bb} d_{cc} e_{ab} + 2d_{bb} d_{cc} e_{ac} + 5d_{bc}^2 d_{cc} + 15d_{bc} d_{cc}^2 + d_{cc}^2 e_{ac} + d_{cc}^2 e_{bc}$

$E_{3,8}$ $= 5d_{aa}^2 d_{ab} + 5d_{aa}^2 d_{ac} + 10d_{aa}^2 d_{bc} - 3d_{aa}^2 e_{ab} - 3d_{aa}^2 e_{ac} + 15d_{aa} d_{ab}^2 + 28d_{aa} d_{ab} d_{ac} + 12d_{aa} d_{ab} d_{bb} +$ kept
$26d_{aa} d_{ab} d_{bc} + 14d_{aa} d_{ab} d_{cc} + 15d_{aa} d_{ac}^2 + 14d_{aa} d_{ac} d_{bb} + 26d_{aa} d_{ac} d_{bc} + 12d_{aa} d_{ac} d_{cc} + 14d_{aa} d_{bb} d_{bc} +$
$2d_{aa} d_{bb} e_{ac} + 2d_{aa} d_{bb} e_{bc} + 10d_{aa} d_{bc}^2 + 14d_{aa} d_{bc} d_{cc} + 2d_{aa} d_{cc} e_{ab} - 2d_{aa} d_{cc} e_{bc} + 15d_{ab}^2 d_{bb} + 10d_{ab}^2 d_{cc} +$
$26d_{ab} d_{ac} d_{bb} + 26d_{ab} d_{ac} d_{cc} + 5d_{ab} d_{bb}^2 + 28d_{ab} d_{bb} d_{bc} + 14d_{ab} d_{bb} d_{cc} + 26d_{ab} d_{bc} d_{cc} + 10d_{ab} d_{cc}^2 + 10d_{ac}^2 d_{bb} +$
$15d_{ac}^2 d_{cc} + 10d_{ac} d_{bb}^2 + 26d_{ac} d_{bb} d_{bc} + 14d_{ac} d_{bb} d_{cc} + 28d_{ac} d_{bc} d_{cc} + 5d_{ac} d_{cc}^2 + 5d_{bb}^2 d_{bc} + 3d_{bb}^2 e_{ab} - 3d_{bb}^2 e_{bc} +$
$15d_{bb} d_{bc}^2 + 12d_{bb} d_{bc} d_{cc} - 2d_{bb} d_{cc} e_{ab} - 2d_{bb} d_{cc} e_{ac} + 15d_{bc}^2 d_{cc} + 5d_{bc} d_{cc}^2 + 3d_{cc}^2 e_{ac} + 3d_{cc}^2 e_{bc}$

$E_{3,9}$ $= d_{aa}^2 d_{ab} + d_{aa}^2 d_{ac} + 2d_{aa}^2 d_{bc} + 18d_{aa} d_{ab}^2 + 32d_{aa} d_{ab} d_{ac} + 4d_{aa} d_{ab} d_{bb} + 28d_{aa} d_{ab} d_{bc} + 6d_{aa} d_{ab} d_{cc} -$ kept
$2d_{aa} d_{ab} e_{ac} + 18d_{aa} d_{ac}^2 + 6d_{aa} d_{ac} d_{bb} + 28d_{aa} d_{ac} d_{bc} + 4d_{aa} d_{ac} d_{cc} - 2d_{aa} d_{ac} e_{ab} + 6d_{aa} d_{bb} d_{bc} + 16d_{aa} d_{bc}^2 +$
$6d_{aa} d_{bc} d_{cc} - 2d_{aa} d_{bc} e_{ab} - 2d_{aa} d_{bc} e_{ac} + 2d_{ab}^2 d_{ac} + 18d_{ab}^2 d_{bb} + 2d_{ab}^2 d_{bc} + 16d_{ab}^2 d_{cc} + 2d_{ab} d_{ac}^2 +$
$28d_{ab} d_{ac} d_{bb} + 8d_{ab} d_{ac} d_{bc} + 28d_{ab} d_{ac} d_{cc} + d_{ab} d_{bb}^2 + 32d_{ab} d_{bb} d_{bc} + 6d_{ab} d_{bb} d_{cc} - 2d_{ab} d_{bb} e_{bc} + 2d_{ab} d_{bc}^2 +$
$28d_{ab} d_{bc} d_{cc} + 2d_{ab} d_{cc}^2 + 2d_{ab} d_{cc} e_{ac} + 2d_{ab} d_{cc} e_{bc} + 16d_{ac}^2 d_{bb} + 2d_{ac}^2 d_{bc} + 18d_{ac}^2 d_{cc} + 2d_{ac} d_{bb}^2 +$
$28d_{ac} d_{bb} d_{bc} + 6d_{ac} d_{bb} d_{cc} + 2d_{ac} d_{bb} e_{ab} - 2d_{ac} d_{bb} e_{bc} + 2d_{ac} d_{bc}^2 + 32d_{ac} d_{bc} d_{cc} + d_{ac} d_{cc}^2 + 2d_{ac} d_{cc} e_{bc} +$
$d_{bb}^2 d_{bc} + 18d_{bb} d_{bc}^2 + 4d_{bb} d_{bc} d_{cc} + 2d_{bb} d_{bc} e_{ab} + 18d_{bc}^2 d_{cc} + d_{bc} d_{cc}^2 + 2d_{bc} d_{cc} e_{ac}$

$E_{3,10}$ $= 3d_{aa}^2 e_{ab} + 3d_{aa}^2 e_{ac} + 15d_{aa} d_{ab}^2 + 28d_{aa} d_{ab} d_{ac} + 26d_{aa} d_{ab} d_{bc} - 4d_{aa} d_{ab} e_{ab} - 2d_{aa} d_{ab} e_{ac} - 4d_{aa} d_{ab} e_{bc} +$ kept
$15d_{aa} d_{ac}^2 + 26d_{aa} d_{ac} d_{bc} - 2d_{aa} d_{ac} e_{ab} - 4d_{aa} d_{ac} e_{ac} + 4d_{aa} d_{ac} e_{bc} - 2d_{aa} d_{bb} e_{ac} - 2d_{aa} d_{bb} e_{bc} + 10d_{aa} d_{bc}^2 +$
$2d_{aa} d_{bc} e_{ab} + 2d_{aa} d_{bc} e_{ac} - 2d_{aa} d_{cc} e_{ab} + 2d_{aa} d_{cc} e_{bc} + 2d_{ab}^3 + 7d_{ab}^2 d_{ac} + 15d_{ab}^2 d_{bb} + 7d_{ab}^2 d_{bc} + 10d_{ab}^2 d_{cc} +$
$7d_{ab} d_{ac}^2 + 26d_{ab} d_{ac} d_{bb} + 12d_{ab} d_{ac} d_{bc} + 26d_{ab} d_{ac} d_{cc} + 28d_{ab} d_{bb} d_{bc} + 4d_{ab} d_{bb} e_{ab} - 4d_{ab} d_{bb} e_{ac} - 2d_{ab} d_{bb} e_{bc} +$
$7d_{ab} d_{bc}^2 + 26d_{ab} d_{bc} d_{cc} - 2d_{ab} d_{cc} e_{ac} - 2d_{ab} d_{cc} e_{bc} + 2d_{ac}^3 + 10d_{ac}^2 d_{bb} + 7d_{ac}^2 d_{bc} + 15d_{ac}^2 d_{cc} + 26d_{ac} d_{bb} d_{bc} -$
$2d_{ac} d_{bb} e_{ab} + 2d_{ac} d_{bb} e_{bc} + 7d_{ac} d_{bc}^2 + 28d_{ac} d_{bc} d_{cc} - 4d_{ac} d_{cc} e_{ab} + 4d_{ac} d_{cc} e_{ac} + 2d_{ac} d_{cc} e_{bc} - 3d_{bb}^2 e_{ab} +$
$3d_{bb}^2 e_{bc} + 15d_{bb} d_{bc}^2 + 2d_{bb} d_{bc} e_{ab} + 4d_{bb} d_{bc} e_{ac} - 4d_{bb} d_{bc} e_{bc} + 2d_{bb} d_{cc} e_{ab} + 2d_{bb} d_{cc} e_{ac} + 2d_{bc}^3 + 15d_{bc}^2 d_{cc} +$
$4d_{bc} d_{cc} e_{ab} + 2d_{bc} d_{cc} e_{ac} + 4d_{bc} d_{cc} e_{bc} - 3d_{cc}^2 e_{ac} - 3d_{cc}^2 e_{bc}$

$E_{3,11}$ $= d_{aa}^2 e_{ab} + d_{aa}^2 e_{ac} + 5d_{aa}d_{ab}^2 + 12d_{aa}d_{ab}d_{ac} + 14d_{aa}d_{ab}d_{bc} - 6d_{aa}d_{ab}e_{ac} + 5d_{aa}d_{ac}^2 + 14d_{aa}d_{ac}d_{bc} -$ kept
$6d_{aa}d_{ac}e_{ab} + 2d_{aa}d_{bb}e_{ac} + 2d_{aa}d_{bb}e_{bc} + 10d_{aa}d_{bc}^2 + 2d_{aa}d_{bc}e_{ab} + 2d_{aa}d_{bc}e_{ac} + 2d_{aa}d_{cc}e_{ab} - 2d_{aa}d_{cc}e_{bc} +$
$4d_{ab}^3 + 14d_{ab}^2 d_{ac} + 5d_{ab}^2 d_{bb} + 14d_{ab}^2 d_{bc} + 10d_{ab}^2 d_{cc} + 14d_{ab}d_{ac}^2 + 14d_{ab}d_{ac}d_{bb} + 24d_{ab}d_{ac}d_{bc} + 14d_{ab}d_{ac}d_{cc} +$
$12d_{ab}d_{bb}d_{bc} - 6d_{ab}d_{bb}e_{bc} + 14d_{ab}d_{bc}^2 + 14d_{ab}d_{bc}d_{cc} - 2d_{ab}d_{cc}e_{ac} - 2d_{ab}d_{cc}e_{bc} + 4d_{ac}^3 + 10d_{ac}^2 d_{bb} + 14d_{ac}^2 d_{bc} +$
$5d_{ac}^2 d_{cc} + 14d_{ac}d_{bb}d_{bc} - 2d_{ac}d_{bb}e_{ab} + 2d_{ac}d_{bb}e_{bc} + 14d_{ac}d_{bc}^2 + 12d_{ac}d_{bc}d_{cc} + 6d_{ac}d_{cc}e_{bc} - d_{bb}^2 e_{ab} + d_{bb}^2 e_{bc} +$
$5d_{bb}d_{bc}^2 + 6d_{bb}d_{bc}e_{ab} - 2d_{bb}d_{cc}e_{ab} - 2d_{bb}d_{cc}e_{ac} + 4d_{bc}^3 + 5d_{bc}^2 d_{cc} + 6d_{bc}d_{cc}e_{ac} - d_{cc}^2 e_{ac} - d_{cc}^2 e_{bc}$

$E_{3,12}$ $= d_{aa}^2 e_{ab} + d_{aa}^2 e_{ac} + d_{aa}d_{ab}^2 + 4d_{aa}d_{ab}d_{ac} + 6d_{aa}d_{ab}d_{bc} + d_{aa}d_{ac}^2 + 6d_{aa}d_{ac}d_{bc} + 2d_{aa}d_{bb}e_{ac} + 2d_{aa}d_{bb}e_{bc} +$ kept
$2d_{aa}d_{bc}^2 + 2d_{aa}d_{cc}e_{ab} - 2d_{aa}d_{cc}e_{bc} - d_{aa}e_{ab}^2 - d_{aa}e_{ac}^2 - 2d_{aa}e_{bc}^2 + 8d_{ab}^3 + 14d_{ab}^2 d_{ac} + d_{ab}^2 d_{bb} + 14d_{ab}^2 d_{bc} +$
$2d_{ab}^2 d_{cc} + 14d_{ab}d_{ac}^2 + 6d_{ab}d_{ac}d_{bb} + 32d_{ab}d_{ac}d_{bc} + 6d_{ab}d_{ac}d_{cc} + 4d_{ab}d_{bb}d_{bc} + 14d_{ab}d_{bc}^2 + 6d_{ab}d_{bc}d_{cc} + 8d_{ac}^3 +$
$2d_{ac}^2 d_{bb} + 14d_{ac}^2 d_{bc} + d_{ac}^2 d_{cc} + 6d_{ac}d_{bb}d_{bc} + 14d_{ac}d_{bc}^2 + 4d_{ac}d_{bc}d_{cc} - d_{bb}^2 e_{ab} + d_{bb}^2 e_{bc} + d_{bb}d_{bc}^2 - 2d_{bb}d_{cc}e_{ab} -$
$2d_{bb}d_{cc}e_{ac} - d_{bb}e_{ab}^2 - 2d_{bb}e_{ac}^2 - d_{bb}e_{bc}^2 + 8d_{bc}^3 + d_{bc}^2 d_{cc} - d_{cc}^2 e_{ac} - d_{cc}^2 e_{bc} - 2d_{cc}e_{ab}^2 - d_{cc}e_{ac}^2 - d_{cc}e_{bc}^2$

$E_{3,13}$ $= 6d_{aa}d_{ab}e_{ac} + 6d_{aa}d_{ac}e_{ab} - 2d_{aa}d_{bc}e_{ab} - 2d_{aa}d_{bc}e_{ac} - d_{aa}e_{ab}^2 + 4d_{aa}e_{ab}e_{bc} - d_{aa}e_{ac}^2 - 4d_{aa}e_{ac}e_{bc} -$ kept
$2d_{aa}e_{bc}^2 + 4d_{ab}^3 + 14d_{ab}^2 d_{ac} + 14d_{ab}^2 d_{bc} - 3d_{ab}^2 e_{ac} - 3d_{ab}^2 e_{bc} + 14d_{ab}d_{ac}^2 + 24d_{ab}d_{ac}d_{bc} - 2d_{ab}d_{ac}e_{ab} -$
$2d_{ab}d_{ac}e_{ac} + 6d_{ab}d_{bb}e_{bc} + 14d_{ab}d_{bc}^2 + 2d_{ab}d_{bc}e_{ab} - 2d_{ab}d_{bc}e_{bc} + 2d_{ab}d_{cc}e_{ac} + 2d_{ab}d_{cc}e_{bc} + 4d_{ac}^3 +$
$14d_{ac}^2 d_{bc} - 3d_{ac}^2 e_{ab} + 3d_{ac}^2 e_{bc} + 2d_{ac}d_{bb}e_{ab} - 2d_{ac}d_{bb}e_{bc} + 14d_{ac}d_{bc}^2 + 2d_{ac}d_{bc}e_{ac} + 2d_{ac}d_{bc}e_{bc} -$
$6d_{ac}d_{cc}e_{bc} - 6d_{bb}d_{bc}e_{ab} - d_{bb}e_{ab}^2 - 4d_{bb}e_{ab}e_{ac} - 2d_{bb}e_{ac}^2 - 4d_{bb}e_{ac}e_{bc} - d_{bb}e_{bc}^2 + 4d_{bc}^3 + 3d_{bc}^2 e_{ab} + 3d_{bc}^2 e_{ac} -$
$6d_{bc}d_{cc}e_{ac} - 2d_{cc}e_{ab}^2 - 4d_{cc}e_{ab}e_{ac} + 4d_{cc}e_{ab}e_{bc} - d_{cc}e_{ac}^2 - d_{cc}e_{bc}^2$

$E_{3,14}$ $= 4d_{aa}d_{ab}e_{ab} + 2d_{aa}d_{ab}e_{ac} + 4d_{aa}d_{ab}e_{bc} + 2d_{aa}d_{ac}e_{ab} + 4d_{aa}d_{ac}e_{ac} - 4d_{aa}d_{ac}e_{bc} - 2d_{aa}d_{bc}e_{ab} - 2d_{aa}d_{bc}e_{ac} -$ kept
$3d_{aa}e_{ab}^2 - 3d_{aa}e_{ac}^2 - 2d_{aa}e_{bc}^2 + 2d_{ab}^3 + 7d_{ab}^2 d_{ac} + 7d_{ab}^2 d_{bc} - 2d_{ab}^2 e_{ac} - 2d_{ab}^2 e_{bc} + 7d_{ab}d_{ac}^2 + 12d_{ab}d_{ac}d_{bc} -$
$4d_{ab}d_{ac}e_{ab} - 4d_{ab}d_{ac}e_{ac} - 4d_{ab}d_{bb}e_{ab} + 4d_{ab}d_{bb}e_{ac} + 2d_{ab}d_{bb}e_{bc} + 7d_{ab}d_{bc}^2 + 4d_{ab}d_{bc}e_{ab} - 4d_{ab}d_{bc}e_{bc} +$
$2d_{ab}d_{cc}e_{ac} + 2d_{ab}d_{cc}e_{bc} + 2d_{ac}^3 + 7d_{ac}^2 d_{bc} - 2d_{ac}^2 e_{ab} + 2d_{ac}^2 e_{bc} + 2d_{ac}d_{bb}e_{ab} - 2d_{ac}d_{bb}e_{bc} + 7d_{ac}d_{bc}^2 + 4d_{ac}d_{bc}e_{ac} +$
$4d_{ac}d_{bc}e_{bc} + 4d_{ac}d_{cc}e_{ab} - 4d_{ac}d_{cc}e_{ac} - 2d_{ac}d_{cc}e_{bc} - 2d_{bb}d_{bc}e_{ab} - 4d_{bb}d_{bc}e_{ac} + 4d_{bb}d_{bc}e_{bc} - 3d_{bb}e_{ab}^2 - 2d_{bb}e_{ac}^2 -$
$3d_{bb}e_{bc}^2 + 2d_{bc}^3 + 2d_{bc}^2 e_{ab} + 2d_{bc}^2 e_{ac} - 4d_{bc}d_{cc}e_{ab} - 2d_{bc}d_{cc}e_{ac} - 4d_{bc}d_{cc}e_{bc} - 2d_{cc}e_{ab}^2 - 3d_{cc}e_{ac}^2 - 3d_{cc}e_{bc}^2$

$E_{3,15}$ $= 2d_{aa}d_{ab}e_{ac} + 2d_{aa}d_{ac}e_{ab} + 2d_{aa}d_{bc}e_{ab} + 2d_{aa}d_{bc}e_{ac} - 2d_{aa}e_{ab}^2 - 8d_{aa}e_{ab}e_{ac} - 2d_{aa}e_{ac}^2 + 2d_{ab}^2 d_{ac} +$ kept
$2d_{ab}^2 d_{bc} + 2d_{ab}d_{ac}^2 + 8d_{ab}d_{ac}d_{bc} + 2d_{ab}d_{bb}e_{bc} + 2d_{ab}d_{bc}^2 - 2d_{ab}d_{cc}e_{ac} - 2d_{ab}d_{cc}e_{bc} - 2d_{ab}e_{ac}^2 - 2d_{ab}e_{bc}^2 +$
$2d_{ac}^2 d_{bc} - 2d_{ac}d_{bb}e_{ab} + 2d_{ac}d_{bb}e_{bc} + 2d_{ac}d_{bc}^2 - 2d_{ac}d_{cc}e_{bc} - 2d_{ac}e_{ab}^2 - 2d_{ac}e_{bc}^2 - 2d_{bb}d_{bc}e_{ab} - 2d_{bb}e_{ab}^2 +$
$8d_{bb}e_{ab}e_{bc} - 2d_{bb}e_{bc}^2 - 2d_{bc}d_{cc}e_{ac} - 2d_{bc}e_{ab}^2 - 2d_{bc}e_{ac}^2 - 2d_{cc}e_{ac}^2 - 8d_{cc}e_{ac}e_{bc} - 2d_{cc}e_{bc}^2$

$E_{3,16}$ $= -3d_{aa}e_{ab}^2 - 3d_{aa}e_{ac}^2 - 2d_{aa}e_{bc}^2 + 2d_{ab}^2 e_{ac} + 2d_{ab}^2 e_{bc} + 4d_{ab}d_{ac}e_{ab} + 4d_{ab}d_{ac}e_{ac} - 4d_{ab}d_{bc}e_{ab} + 4d_{ab}d_{bc}e_{bc} -$ kept
$2d_{ab}e_{ab}^2 - 2d_{ab}e_{ab}e_{ac} + 2d_{ab}e_{ab}e_{bc} - d_{ab}e_{ac}^2 - 4d_{ab}e_{ac}e_{bc} - d_{ab}e_{bc}^2 + 2d_{ac}^2 e_{ab} - 2d_{ac}^2 e_{bc} - 4d_{ac}d_{bc}e_{ac} -$
$4d_{ac}d_{bc}e_{bc} - d_{ac}e_{ab}^2 - 2d_{ac}e_{ab}e_{ac} + 4d_{ac}e_{ab}e_{bc} - 2d_{ac}e_{ac}^2 - 2d_{ac}e_{ac}e_{bc} - d_{ac}e_{bc}^2 - 3d_{bb}e_{ab}^2 - 2d_{bb}e_{ac}^2 - 3d_{bb}e_{bc}^2 -$
$2d_{bc}^2 e_{ab} - 2d_{bc}^2 e_{ac} - d_{bc}e_{ab}^2 - 4d_{bc}e_{ab}e_{ac} + 2d_{bc}e_{ab}e_{bc} - d_{bc}e_{ac}^2 - 2d_{bc}e_{ac}e_{bc} - 2d_{bc}e_{bc}^2 - 2d_{cc}e_{ab}^2 - 3d_{cc}e_{ac}^2 - 3d_{cc}e_{bc}^2$

$E_{3,17}$ $= -d_{aa}e_{ab}^2 + 4d_{aa}e_{ab}e_{bc} - d_{aa}e_{ac}^2 - 4d_{aa}e_{ac}e_{bc} - 2d_{aa}e_{bc}^2 + 3d_{ab}^2 e_{ac} + 3d_{ab}^2 e_{bc} + 2d_{ab}d_{ac}e_{ab} + 2d_{ab}d_{ac}e_{ac} -$ kept
$2d_{ab}d_{bc}e_{ab} + 2d_{ab}d_{bc}e_{bc} - 4d_{ab}e_{ab}^2 - 2d_{ab}e_{ac}^2 - 2d_{ab}e_{bc}^2 + 3d_{ac}^2 e_{ab} - 3d_{ac}^2 e_{bc} - 2d_{ac}d_{bc}e_{ac} - 2d_{ac}d_{bc}e_{bc} -$
$2d_{ac}e_{ab}^2 - 4d_{ac}e_{ac}^2 - 2d_{ac}e_{bc}^2 - d_{bb}e_{ab}^2 - 4d_{bb}e_{ab}e_{ac} - 2d_{bb}e_{ac}^2 - 4d_{bb}e_{ac}e_{bc} - d_{bb}e_{bc}^2 - 3d_{bc}^2 e_{ab} - 3d_{bc}^2 e_{ac} -$
$2d_{bc}e_{ab}^2 - 2d_{bc}e_{ac}^2 - 4d_{bc}e_{bc}^2 - 2d_{cc}e_{ab}^2 - 4d_{cc}e_{ab}e_{ac} + 4d_{cc}e_{ab}e_{bc} - d_{cc}e_{ac}^2 - d_{cc}e_{bc}^2$

$E_{3,18}$ $=$ kept
$-d_{aa}e_{ab}^2 - d_{aa}e_{ac}^2 - 2d_{aa}e_{bc}^2 - 4d_{ab}e_{ab}e_{ac} + 4d_{ab}e_{ab}e_{bc} - 2d_{ab}e_{ac}^2 - 2d_{ab}e_{bc}^2 - 2d_{ac}e_{ab}^2 - 4d_{ac}e_{ab}e_{ac} - 4d_{ac}e_{ac}e_{bc} -$
$2d_{ac}e_{bc}^2 - d_{bb}e_{ab}^2 - 2d_{bb}e_{ac}^2 - d_{bb}e_{bc}^2 - 2d_{bc}e_{ab}^2 + 4d_{bc}e_{ab}e_{bc} - 2d_{bc}e_{ac}^2 - 4d_{bc}e_{ac}e_{bc} - 2d_{cc}e_{ab}^2 - d_{cc}e_{ac}^2 - d_{cc}e_{bc}^2$

$E_{3,19}$ $= -4d_{ab}e_{ab}^2 - 2d_{ab}e_{ac}^2 - 2d_{ab}e_{bc}^2 - 2d_{ac}e_{ab}^2 - 4d_{ac}e_{ac}^2 - 2d_{ac}e_{bc}^2 - 2d_{bc}e_{ab}^2 - 2d_{bc}e_{ac}^2 - 4d_{bc}e_{bc}^2 + e_{ab}^2 e_{ac} +$ kept
$e_{ab}^2 e_{bc} + e_{ab}e_{ac}^2 - e_{ab}e_{bc}^2 - e_{ac}^2 e_{bc} - e_{ac}e_{bc}^2$

$E_{3,20}$	$= -2d_{ab}e_{ab}^2 - 2d_{ab}e_{ab}e_{ac} + 2d_{ab}e_{ab}e_{bc} - d_{ab}e_{ac}^2 - 4d_{ab}e_{ac}e_{bc} - d_{ab}e_{bc}^2 - d_{ac}e_{ab}^2 - 2d_{ac}e_{ab}e_{ac} + 4d_{ac}e_{ab}e_{bc} - 2d_{ac}e_{ac}^2 - 2d_{ac}e_{ac}e_{bc} - d_{ac}e_{bc}^2 - d_{bc}e_{ab}^2 - 4d_{bc}e_{ab}e_{ac} + 2d_{bc}e_{ab}e_{bc} - d_{bc}e_{ac}^2 - 2d_{bc}e_{ac}e_{bc} - 2d_{bc}e_{bc}^2 + 2e_{ab}^2e_{ac} + 2e_{ab}^2e_{bc} + 2e_{ab}e_{ac}^2 - 2e_{ab}e_{bc}^2 - 2e_{ac}^2e_{bc} - 2e_{ac}e_{bc}^2$	kept
$E_{3,21}$	$= -2d_{ab}e_{ac}^2 - 2d_{ab}e_{bc}^2 - 2d_{ac}e_{ab}^2 - 2d_{ac}e_{bc}^2 - 2d_{bc}e_{ab}^2 - 2d_{bc}e_{ac}^2$	kept
$E_{3,22}$	$= -2e_{ab}^2e_{ac} - 2e_{ab}^2e_{bc} - 2e_{ab}e_{ac}^2 + 2e_{ab}e_{bc}^2 + 2e_{ac}^2e_{bc} + 2e_{ac}e_{bc}^2$	kept
$E_{3,23}$	$= -e_{ab}^2e_{ac} - e_{ab}^2e_{bc} - e_{ab}e_{ac}^2 + e_{ab}e_{bc}^2 + e_{ac}^2e_{bc} + e_{ac}e_{bc}^2$	dropped: $= 1/2E_{3,22}$
$E_{3,24}$	$= 0$	dropped: $= 0$

$c = 4$

output	expanded	safe reductions
$E_{4,0}$	$= d_{aa}^2d_{bb}^2 + 4d_{aa}^2d_{bb}d_{cc} + d_{aa}^2d_{cc}^2 + 4d_{aa}d_{bb}^2d_{cc} + 4d_{aa}d_{bb}d_{cc}^2 + d_{bb}^2d_{cc}^2$	kept
$E_{4,1}$	$= 3d_{aa}^3d_{bb} + 3d_{aa}^3d_{cc} + 4d_{aa}^2d_{bb}^2 + 10d_{aa}^2d_{bb}d_{cc} + 4d_{aa}^2d_{cc}^2 + 3d_{aa}d_{bb}^3 + 10d_{aa}d_{bb}^2d_{cc} + 10d_{aa}d_{bb}d_{cc}^2 + 3d_{aa}d_{cc}^3 + 3d_{bb}^3d_{cc} + 4d_{bb}^2d_{cc}^2 + 3d_{bb}d_{cc}^3$	kept
$E_{4,2}$	$= d_{aa}^4 + 7d_{aa}^3d_{bb} + 7d_{aa}^3d_{cc} + 13d_{aa}^2d_{bb}^2 + 22d_{aa}^2d_{bb}d_{cc} + 13d_{aa}^2d_{cc}^2 + 7d_{aa}d_{bb}^3 + 22d_{aa}d_{bb}^2d_{cc} + 22d_{aa}d_{bb}d_{cc}^2 + 7d_{aa}d_{cc}^3 + d_{bb}^4 + 7d_{bb}^3d_{cc} + 13d_{bb}^2d_{cc}^2 + 7d_{bb}d_{cc}^3 + d_{cc}^4$	kept
$E_{4,3}$	$= 4d_{aa}^4 + 13d_{aa}^3d_{bb} + 13d_{aa}^3d_{cc} + d_{aa}^2d_{ab}d_{bb} + 2d_{aa}^2d_{ab}d_{cc} + 2d_{aa}^2d_{ac}d_{bb} + d_{aa}^2d_{ac}d_{cc} + 16d_{aa}^2d_{bb}^2 + 3d_{aa}^2d_{bb}d_{bc} + 34d_{aa}^2d_{bb}d_{cc} + 3d_{aa}^2d_{bc}d_{cc} + 16d_{aa}^2d_{cc}^2 + d_{aa}d_{ab}d_{bb}^2 + 8d_{aa}d_{ab}d_{bb}d_{cc} + 3d_{aa}d_{ab}d_{cc}^2 + 3d_{aa}d_{ac}d_{bb}^2 + 8d_{aa}d_{ac}d_{bb}d_{cc} + d_{aa}d_{ac}d_{cc}^2 + 13d_{aa}d_{bb}^3 + 2d_{aa}d_{bb}^2d_{bc} + 34d_{aa}d_{bb}^2d_{cc} + 8d_{aa}d_{bb}d_{bc}d_{cc} + 34d_{aa}d_{bb}d_{cc}^2 + 2d_{aa}d_{bc}d_{cc}^2 + 13d_{aa}d_{cc}^3 + 2d_{ab}d_{bb}^2d_{cc} + 3d_{ab}d_{bb}d_{cc}^2 + 3d_{ac}d_{bb}^2d_{cc} + 2d_{ac}d_{bb}d_{cc}^2 + 4d_{bb}^4 + 13d_{bb}^3d_{cc} + d_{bb}^2d_{bc}d_{cc} + 16d_{bb}^2d_{cc}^2 + d_{bb}d_{bc}d_{cc}^2 + 13d_{bb}d_{cc}^3 + 4d_{cc}^4$	kept
$E_{4,4}$	$= 5d_{aa}^4 + d_{aa}^3d_{ab} + d_{aa}^3d_{ac} + 14d_{aa}^3d_{bb} + 4d_{aa}^3d_{bc} + 14d_{aa}^3d_{cc} + 7d_{aa}^2d_{ab}d_{bb} + 8d_{aa}^2d_{ab}d_{cc} + 8d_{aa}^2d_{ac}d_{bb} + 7d_{aa}^2d_{ac}d_{cc} + 22d_{aa}^2d_{bb}^2 + 12d_{aa}^2d_{bb}d_{bc} + 40d_{aa}^2d_{bb}d_{cc} + 12d_{aa}^2d_{bc}d_{cc} + 22d_{aa}^2d_{cc}^2 + 7d_{aa}d_{ab}d_{bb}^2 + 20d_{aa}d_{ab}d_{bb}d_{cc} + 12d_{aa}d_{ab}d_{cc}^2 + 12d_{aa}d_{ac}d_{bb}^2 + 20d_{aa}d_{ac}d_{bb}d_{cc} + 7d_{aa}d_{ac}d_{cc}^2 + 14d_{aa}d_{bb}^3 + 8d_{aa}d_{bb}^2d_{bc} + 40d_{aa}d_{bb}^2d_{cc} + 20d_{aa}d_{bb}d_{bc}d_{cc} + 40d_{aa}d_{bb}d_{cc}^2 + 8d_{aa}d_{bc}d_{cc}^2 + 14d_{aa}d_{cc}^3 + d_{ab}d_{bb}^3 + 8d_{ab}d_{bb}^2d_{cc} + 12d_{ab}d_{bb}d_{cc}^2 + 4d_{ab}d_{cc}^3 + 4d_{ac}d_{bb}^3 + 12d_{ac}d_{bb}^2d_{cc} + 8d_{ac}d_{bb}d_{cc}^2 + d_{ac}d_{cc}^3 + 5d_{bb}^4 + d_{bb}^3d_{bc} + 14d_{bb}^3d_{cc} + 7d_{bb}^2d_{bc}d_{cc} + 22d_{bb}^2d_{cc}^2 + 7d_{bb}d_{bc}d_{cc}^2 + 14d_{bb}d_{cc}^3 + d_{bc}d_{cc}^3 + 5d_{cc}^4$	kept
$E_{4,5}$	$= 4d_{aa}^4 + 5d_{aa}^3d_{ab} + 5d_{aa}^3d_{ac} + 13d_{aa}^3d_{bb} + 8d_{aa}^3d_{bc} + 13d_{aa}^3d_{cc} + 22d_{aa}^2d_{ab}d_{bb} + 26d_{aa}^2d_{ab}d_{cc} + 26d_{aa}^2d_{ac}d_{bb} + 22d_{aa}^2d_{ac}d_{cc} + 16d_{aa}^2d_{bb}^2 + 21d_{aa}^2d_{bb}d_{bc} + 34d_{aa}^2d_{bb}d_{cc} + 21d_{aa}^2d_{bc}d_{cc} + 16d_{aa}^2d_{cc}^2 + 22d_{aa}d_{ab}d_{bb}^2 + 44d_{aa}d_{ab}d_{bb}d_{cc} + 21d_{aa}d_{ab}d_{cc}^2 + 21d_{aa}d_{ac}d_{bb}^2 + 44d_{aa}d_{ac}d_{bb}d_{cc} + 22d_{aa}d_{ac}d_{cc}^2 + 13d_{aa}d_{bb}^3 + 26d_{aa}d_{bb}^2d_{bc} + 34d_{aa}d_{bb}^2d_{cc} + 44d_{aa}d_{bb}d_{bc}d_{cc} + 34d_{aa}d_{bb}d_{cc}^2 + 26d_{aa}d_{bc}d_{cc}^2 + 13d_{aa}d_{cc}^3 + 5d_{ab}d_{bb}^3 + 26d_{ab}d_{bb}^2d_{cc} + 21d_{ab}d_{bb}d_{cc}^2 + 8d_{ab}d_{cc}^3 + 8d_{ac}d_{bb}^3 + 21d_{ac}d_{bb}^2d_{cc} + 26d_{ac}d_{bb}d_{cc}^2 + 5d_{ac}d_{cc}^3 + 4d_{bb}^4 + 5d_{bb}^3d_{bc} + 13d_{bb}^3d_{cc} + 22d_{bb}^2d_{bc}d_{cc} + 16d_{bb}^2d_{cc}^2 + 22d_{bb}d_{bc}d_{cc}^2 + 13d_{bb}d_{cc}^3 + 5d_{bc}d_{cc}^3 + 4d_{cc}^4$	kept

$E_{4,6}$ $= d_{aa}^4 + 15d_{aa}^3 d_{ab} + 15d_{aa}^3 d_{ac} + 7d_{aa}^3 d_{bb} + 12d_{aa}^3 d_{bc} + 7d_{aa}^3 d_{cc} + d_{aa}^2 d_{ab} d_{ac} + 37d_{aa}^2 d_{ab} d_{bb} + 2d_{aa}^2 d_{ab} d_{bc} +$ kept
$32d_{aa}^2 d_{ab} d_{cc} + 32d_{aa}^2 d_{ac} d_{bb} + 2d_{aa}^2 d_{ac} d_{bc} + 37d_{aa}^2 d_{ac} d_{cc} + 13d_{aa}^2 d_{bb}^2 + 36d_{aa}^2 d_{bb} d_{bc} + 22d_{aa}^2 d_{bb} d_{cc} -$
$d_{aa}^2 d_{bb} e_{ab} - 2d_{aa}^2 d_{bb} e_{ac} - d_{aa}^2 d_{bb} e_{bc} + d_{aa}^2 d_{bc}^2 + 36d_{aa}^2 d_{bc} d_{cc} + 13d_{aa}^2 d_{cc}^2 - 2d_{aa}^2 d_{cc} e_{ab} - d_{aa}^2 d_{cc} e_{ac} +$
$d_{aa}^2 d_{cc} e_{bc} + d_{aa} d_{ab}^2 d_{bb} + 2d_{aa} d_{ab}^2 d_{cc} + 5d_{aa} d_{ab} d_{ac} d_{bb} + 5d_{aa} d_{ab} d_{ac} d_{cc} + 37d_{aa} d_{ab} d_{bb}^2 + 5d_{aa} d_{ab} d_{bb} d_{bc} +$
$68d_{aa} d_{ab} d_{bb} d_{cc} + 9d_{aa} d_{ab} d_{bc} d_{cc} + 36d_{aa} d_{ab} d_{cc}^2 + 2d_{aa} d_{ac}^2 d_{bb} + d_{aa} d_{ac}^2 d_{cc} + 36d_{aa} d_{ac} d_{bb}^2 + 9d_{aa} d_{ac} d_{bb} d_{bc} +$
$68d_{aa} d_{ac} d_{bb} d_{cc} + 5d_{aa} d_{ac} d_{bc} d_{cc} + 37d_{aa} d_{ac} d_{cc}^2 + 7d_{aa} d_{bb}^3 + 32d_{aa} d_{bb}^2 d_{bc} + 22d_{aa} d_{bb}^2 d_{cc} + d_{aa} d_{bb}^2 e_{ab} -$
$d_{aa} d_{bb}^2 e_{ac} - 2d_{aa} d_{bb}^2 e_{bc} + 2d_{aa} d_{bb} d_{bc}^2 + 68d_{aa} d_{bb} d_{bc} d_{cc} + 22d_{aa} d_{bb} d_{cc}^2 + 2d_{aa} d_{bc}^2 d_{cc} + 32d_{aa} d_{bc} d_{cc}^2 +$
$7d_{aa} d_{cc}^3 - d_{aa} d_{cc}^2 e_{ab} + d_{aa} d_{cc}^2 e_{ac} + 2d_{aa} d_{cc}^2 e_{bc} + 2d_{ab}^2 d_{bb} d_{cc} + d_{ab}^2 d_{cc}^2 + 2d_{ab} d_{ac} d_{bb}^2 + 9d_{ab} d_{ac} d_{bb} d_{cc} +$
$2d_{ab} d_{ac} d_{cc}^2 + 15d_{ab} d_{bb}^3 + d_{ab} d_{bb}^2 d_{bc} + 32d_{ab} d_{bb}^2 d_{cc} + 5d_{ab} d_{bb} d_{bc} d_{cc} + 36d_{ab} d_{bb} d_{cc}^2 + 2d_{ab} d_{bc} d_{cc}^2 + 12d_{ab} d_{cc}^3 +$
$d_{ac}^2 d_{bb}^2 + 2d_{ac}^2 d_{bb} d_{cc} + 12d_{ac} d_{bb}^3 + 2d_{ac} d_{bb}^2 d_{bc} + 36d_{ac} d_{bb}^2 d_{cc} + 5d_{ac} d_{bb} d_{bc} d_{cc} + 32d_{ac} d_{bb} d_{cc}^2 + d_{ac} d_{bc} d_{cc}^2 +$
$15d_{ac} d_{cc}^3 + d_{bb}^4 + 15d_{bb}^3 d_{bc} + 7d_{bb}^3 d_{cc} + 37d_{bb}^2 d_{bc} d_{cc} + 13d_{bb}^2 d_{cc}^2 + 2d_{bb}^2 d_{cc} e_{ab} + d_{bb}^2 d_{cc} e_{ac} - d_{bb}^2 d_{cc} e_{bc} +$
$d_{bb} d_{bc}^2 d_{cc} + 37d_{bb} d_{bc} d_{cc}^2 + 7d_{bb} d_{cc}^3 + d_{bb} d_{cc}^2 e_{ab} + 2d_{bb} d_{cc}^2 e_{ac} + d_{bb} d_{cc}^2 e_{bc} + 15d_{bc} d_{cc}^3 + d_{cc}^4$

$E_{4,7}$ $=$ (102 terms; see the full text file) kept

$E_{4,8}$ $= 15d_{aa}^3 d_{ab} + 15d_{aa}^3 d_{ac} + 12d_{aa}^3 d_{bc} - 3d_{aa}^3 e_{ab} - 3d_{aa}^3 e_{ac} + 9d_{aa}^2 d_{ab}^2 + 22d_{aa}^2 d_{ab} d_{ac} + 37d_{aa}^2 d_{ab} d_{bb} +$ kept
$23d_{aa}^2 d_{ab} d_{bc} + 32d_{aa}^2 d_{ab} d_{cc} + 9d_{aa}^2 d_{ac}^2 + 32d_{aa}^2 d_{ac} d_{bb} + 23d_{aa}^2 d_{ac} d_{bc} + 37d_{aa}^2 d_{ac} d_{cc} + d_{aa}^2 d_{bb}^2 + 36d_{aa}^2 d_{bb} d_{bc} +$
$4d_{aa}^2 d_{bb} d_{cc} - 6d_{aa}^2 d_{bb} e_{ac} + 3d_{aa}^2 d_{bb} e_{bc} + 10d_{aa}^2 d_{bc}^2 + 36d_{aa}^2 d_{bc} d_{cc} + d_{aa}^2 d_{cc}^2 - 6d_{aa}^2 d_{cc} e_{ab} - 3d_{aa}^2 d_{cc} e_{bc} +$
$22d_{aa} d_{ab}^2 d_{bb} + 23d_{aa} d_{ab}^2 d_{cc} + 44d_{aa} d_{ab} d_{ac} d_{bb} + 44d_{aa} d_{ab} d_{ac} d_{cc} + 37d_{aa} d_{ab} d_{bb}^2 + 44d_{aa} d_{ab} d_{bb} d_{bc} +$
$68d_{aa} d_{ab} d_{bb} d_{cc} + 48d_{aa} d_{ab} d_{bc} d_{cc} + 36d_{aa} d_{ab} d_{cc}^2 + 23d_{aa} d_{ac}^2 d_{bb} + 22d_{aa} d_{ac}^2 d_{cc} + 36d_{aa} d_{ac} d_{bb}^2 +$
$48d_{aa} d_{ac} d_{bb} d_{bc} + 68d_{aa} d_{ac} d_{bb} d_{cc} + 44d_{aa} d_{ac} d_{bc} d_{cc} + 37d_{aa} d_{ac} d_{cc}^2 + 32d_{aa} d_{bb}^2 d_{bc} + 4d_{aa} d_{bb}^2 d_{cc} +$
$3d_{aa} d_{bb}^2 e_{ac} - 6d_{aa} d_{bb}^2 e_{bc} + 23d_{aa} d_{bb} d_{bc}^2 + 68d_{aa} d_{bb} d_{bc} d_{cc} + 4d_{aa} d_{bb} d_{cc}^2 + 23d_{aa} d_{bc}^2 d_{cc} + 32d_{aa} d_{bc} d_{cc}^2 +$
$3d_{aa} d_{cc}^2 e_{ab} + 6d_{aa} d_{cc}^2 e_{bc} + 9d_{ab}^2 d_{bb}^2 + 23d_{ab}^2 d_{bb} d_{cc} + 10d_{ab}^2 d_{cc}^2 + 23d_{ab} d_{ac} d_{bb}^2 + 48d_{ab} d_{ac} d_{bb} d_{cc} + 23d_{ab} d_{ac} d_{cc}^2 +$
$15d_{ab} d_{bb}^3 + 22d_{ab} d_{bb}^2 d_{bc} + 32d_{ab} d_{bb}^2 d_{cc} + 44d_{ab} d_{bb} d_{bc} d_{cc} + 36d_{ab} d_{bb} d_{cc}^2 + 23d_{ab} d_{bc} d_{cc}^2 + 12d_{ab} d_{cc}^3 +$
$10d_{ac}^2 d_{bb}^2 + 23d_{ac}^2 d_{bb} d_{cc} + 9d_{ac}^2 d_{cc}^2 + 12d_{ac} d_{bb}^3 + 23d_{ac} d_{bb}^2 d_{bc} + 36d_{ac} d_{bb}^2 d_{cc} + 44d_{ac} d_{bb} d_{bc} d_{cc} + 32d_{ac} d_{bb} d_{cc}^2 +$
$22d_{ac} d_{bc} d_{cc}^2 + 15d_{ac} d_{cc}^3 + 15d_{bb}^3 d_{bc} + 3d_{bb}^3 e_{ab} - 3d_{bb}^3 e_{bc} + 9d_{bb}^2 d_{bc}^2 + 37d_{bb}^2 d_{bc} d_{cc} + d_{bb}^2 d_{cc}^2 + 6d_{bb}^2 d_{cc} e_{ab} -$
$3d_{bb}^2 d_{cc} e_{ac} + 22d_{bb} d_{bc}^2 d_{cc} + 37d_{bb} d_{bc} d_{cc}^2 - 3d_{bb} d_{cc}^2 e_{ab} + 6d_{bb} d_{cc}^2 e_{ac} + 9d_{bc}^2 d_{cc}^2 + 15d_{bc} d_{cc}^3 + 3d_{cc}^3 e_{ac} + 3d_{cc}^3 e_{bc}$

$E_{4,9}$ $=$ (141 terms; see the full text file) kept

$E_{4,10}$ $=$ (144 terms; see the full text file) kept

$E_{4,11}$ $=$ (147 terms; see the full text file) kept

$E_{4,12}$ $=$ (162 terms; see the full text file) kept

$E_{4,13}$ $=$ (165 terms; see the full text file) kept

$E_{4,14}$ $=$ (207 terms; see the full text file) kept

$E_{4,15}$ $=$ (171 terms; see the full text file) kept

$E_{4,16}$ $=$ (207 terms; see the full text file) kept

$E_{4,17}$ $=$ (207 terms; see the full text file) kept

$E_{4,18}$ $=$ (198 terms; see the full text file) kept

$E_{4,19}$ $=$ (171 terms; see the full text file) kept

$E_{4,20}$ $=$ (186 terms; see the full text file) kept

$E_{4,21}$ $=$ (123 terms; see the full text file) kept

$E_{4,22}$ $=$ (117 terms; see the full text file) kept

$E_{4,23}$ $= -d_{aa}d_{ab}e_{ac}^2 - 3d_{aa}d_{ab}e_{bc}^2 - d_{aa}d_{ac}e_{ab}^2 - 3d_{aa}d_{ac}e_{bc}^2 - 2d_{aa}d_{bc}e_{ab}^2 - 2d_{aa}d_{bc}e_{ac}^2 + d_{aa}e_{ab}^3 - 5d_{aa}e_{ab}^2e_{ac} -$ kept
$4d_{aa}e_{ab}^2e_{bc} - 5d_{aa}e_{ab}e_{ac}^2 + d_{aa}e_{ac}^3 + 4d_{aa}e_{ac}^2e_{bc} - 4d_{ab}^2e_{ab}^2 - 3d_{ab}^2e_{ab}e_{ac} + 3d_{ab}^2e_{ab}e_{bc} - 2d_{ab}^2e_{ac}^2 - 2d_{ab}^2e_{ac}e_{bc} -$
$2d_{ab}^2e_{bc}^2 - 5d_{ab}d_{ac}e_{ab}^2 - 8d_{ab}d_{ac}e_{ab}e_{ac} + 4d_{ab}d_{ac}e_{ab}e_{bc} - 5d_{ab}d_{ac}e_{ac}^2 - 4d_{ab}d_{ac}e_{ac}e_{bc} - 6d_{ab}d_{ac}e_{bc}^2 - 3d_{ab}d_{bb}e_{ac}^2 -$
$d_{ab}d_{bb}e_{bc}^2 - 5d_{ab}d_{bc}e_{ab}^2 - 4d_{ab}d_{bc}e_{ab}e_{ac} + 8d_{ab}d_{bc}e_{ab}e_{bc} - 6d_{ab}d_{bc}e_{ac}^2 - 4d_{ab}d_{bc}e_{ac}e_{bc} - 5d_{ab}d_{bc}e_{bc}^2 - 2d_{ab}d_{cc}e_{ac}^2 -$
$2d_{ab}d_{cc}e_{bc}^2 + 3d_{ab}e_{ab}^2e_{ac} + 3d_{ab}e_{ab}^2e_{bc} + 4d_{ab}e_{ab}e_{ac}^2 - 4d_{ab}e_{ab}e_{bc}^2 + d_{ab}e_{ac}^3 - 2d_{ab}e_{ac}^2e_{bc} - 2d_{ab}e_{ac}e_{bc}^2 + d_{ab}e_{bc}^3 -$
$2d_{ac}^2e_{ab}^2 - 3d_{ac}^2e_{ab}e_{ac} + 2d_{ac}^2e_{ab}e_{bc} - 4d_{ac}^2e_{ac}^2 - 3d_{ac}^2e_{ac}e_{bc} - 2d_{ac}^2e_{bc}^2 - 2d_{ac}d_{bb}e_{ab}^2 - 2d_{ac}d_{bb}e_{bc}^2 - 6d_{ac}d_{bc}e_{ab}^2 -$
$4d_{ac}d_{bc}e_{ab}e_{ac} + 4d_{ac}d_{bc}e_{ab}e_{bc} - 5d_{ac}d_{bc}e_{ac}^2 - 8d_{ac}d_{bc}e_{ac}e_{bc} - 5d_{ac}d_{bc}e_{bc}^2 - 3d_{ac}d_{cc}e_{ab}^2 - d_{ac}d_{cc}e_{bc}^2 + d_{ac}e_{ab}^3 +$
$4d_{ac}e_{ab}^2e_{ac} + 2d_{ac}e_{ab}^2e_{bc} + 3d_{ac}e_{ab}e_{ac}^2 - 2d_{ac}e_{ab}e_{bc}^2 - 3d_{ac}e_{ac}^2e_{bc} - 4d_{ac}e_{ac}e_{bc}^2 - d_{ac}e_{bc}^3 - d_{bb}d_{bc}e_{ab}^2 - 3d_{bb}d_{bc}e_{ac}^2 -$
$d_{bb}e_{ab}^3 - 4d_{bb}e_{ab}^2e_{ac} - 5d_{bb}e_{ab}^2e_{bc} + 5d_{bb}e_{ab}e_{bc}^2 + 4d_{bb}e_{ac}e_{bc}^2 + d_{bb}e_{bc}^3 - 2d_{bc}^2e_{ab}^2 - 2d_{bc}^2e_{ab}e_{ac} + 3d_{bc}^2e_{ab}e_{bc} -$
$2d_{bc}^2e_{ac}^2 - 3d_{bc}^2e_{ac}e_{bc} - 4d_{bc}^2e_{bc}^2 - 3d_{bc}d_{cc}e_{ab}^2 - d_{bc}d_{cc}e_{ac}^2 - d_{bc}e_{ab}^3 + 2d_{bc}e_{ab}^2e_{ac} + 4d_{bc}e_{ab}^2e_{bc} + 2d_{bc}e_{ab}e_{ac}^2 -$
$3d_{bc}e_{ab}e_{bc}^2 - d_{bc}e_{ac}^3 - 4d_{bc}e_{ac}^2e_{bc} - 3d_{bc}e_{ac}e_{bc}^2 - 4d_{cc}e_{ab}e_{ac}^2 + 4d_{cc}e_{ab}e_{bc}^2 - d_{cc}e_{ac}^3 + 5d_{cc}e_{ac}^2e_{bc} + 5d_{cc}e_{ac}e_{bc}^2 - d_{cc}e_{bc}^3$

$E_{4,24}$ $= -d_{aa}e_{ab}^3 - 2d_{aa}e_{ab}^2e_{ac} - 2d_{aa}e_{ab}^2e_{bc} - 2d_{aa}e_{ab}e_{ac}^2 + 5d_{aa}e_{ab}e_{bc}^2 - d_{aa}e_{ac}^3 + 2d_{aa}e_{ac}^2e_{bc} + 5d_{aa}e_{ac}e_{bc}^2 -$ kept
$2d_{ab}^2e_{ab}^2 + d_{ab}^2e_{ab}e_{ac} - d_{ab}^2e_{ab}e_{bc} - d_{ab}^2e_{ac}^2 - 2d_{ab}^2e_{ac}e_{bc} - d_{ab}^2e_{bc}^2 - 7d_{ab}d_{ac}e_{ab}^2 - 4d_{ab}d_{ac}e_{ab}e_{ac} + 4d_{ab}d_{ac}e_{ab}e_{bc} -$
$7d_{ab}d_{ac}e_{ac}^2 - 4d_{ab}d_{ac}e_{ac}e_{bc} - 6d_{ab}d_{ac}e_{bc}^2 - 7d_{ab}d_{bc}e_{ab}^2 - 4d_{ab}d_{bc}e_{ab}e_{ac} + 4d_{ab}d_{bc}e_{ab}e_{bc} - 6d_{ab}d_{bc}e_{ac}^2 -$
$4d_{ab}d_{bc}e_{ac}e_{bc} - 7d_{ab}d_{bc}e_{bc}^2 + 3d_{ab}e_{ab}^2e_{ac} + 3d_{ab}e_{ab}^2e_{bc} + 2d_{ab}e_{ab}e_{ac}^2 - 2d_{ab}e_{ab}e_{bc}^2 - d_{ab}e_{ac}^3 - 4d_{ab}e_{ac}^2e_{bc} -$
$4d_{ab}e_{ac}e_{bc}^2 - d_{ab}e_{bc}^3 - d_{ac}^2e_{ab}^2 + d_{ac}^2e_{ab}e_{ac} + 2d_{ac}^2e_{ab}e_{bc} - 2d_{ac}^2e_{ac}^2 + d_{ac}^2e_{ac}e_{bc} - d_{ac}^2e_{bc}^2 - 6d_{ac}d_{bc}e_{ab}^2 -$
$4d_{ac}d_{bc}e_{ab}e_{ac} + 4d_{ac}d_{bc}e_{ab}e_{bc} - 7d_{ac}d_{bc}e_{ac}^2 - 4d_{ac}d_{bc}e_{ac}e_{bc} - 7d_{ac}d_{bc}e_{bc}^2 - d_{ac}e_{ab}^3 + 2d_{ac}e_{ab}^2e_{ac} + 4d_{ac}e_{ab}^2e_{bc} +$
$3d_{ac}e_{ab}e_{ac}^2 - 4d_{ac}e_{ab}e_{bc}^2 - 3d_{ac}e_{ac}^2e_{bc} - 2d_{ac}e_{ac}e_{bc}^2 + d_{ac}e_{bc}^3 + d_{bb}e_{ab}^3 - 2d_{bb}e_{ab}^2e_{ac} - 2d_{bb}e_{ab}^2e_{bc} - 5d_{bb}e_{ab}e_{ac}^2 +$
$2d_{bb}e_{ab}e_{bc}^2 + 5d_{bb}e_{ac}^2e_{bc} + 2d_{bb}e_{ac}e_{bc}^2 - d_{bb}e_{bc}^3 - d_{bc}^2e_{ab}^2 - 2d_{bc}^2e_{ab}e_{ac} - d_{bc}^2e_{ab}e_{bc} - d_{bc}^2e_{ac}^2 + d_{bc}^2e_{ac}e_{bc} - 2d_{bc}^2e_{bc}^2 +$
$d_{bc}e_{ab}^3 + 4d_{bc}e_{ab}^2e_{ac} + 2d_{bc}e_{ab}^2e_{bc} + 4d_{bc}e_{ab}e_{ac}^2 - 3d_{bc}e_{ab}e_{bc}^2 + d_{bc}e_{ac}^3 - 2d_{bc}e_{ac}^2e_{bc} - 3d_{bc}e_{ac}e_{bc}^2 - 5d_{cc}e_{ab}^2e_{ac} -$
$5d_{cc}e_{ab}^2e_{bc} - 2d_{cc}e_{ab}e_{ac}^2 + 2d_{cc}e_{ab}e_{bc}^2 + d_{cc}e_{ac}^3 + 2d_{cc}e_{ac}^2e_{bc} + 2d_{cc}e_{ac}e_{bc}^2 + d_{cc}e_{bc}^3 + e_{ab}^2e_{ac}^2 + e_{ab}^2e_{bc}^2 + e_{ac}^2e_{bc}^2$

$E_{4,25}$ $= -d_{aa}e_{ab}^3 - d_{aa}e_{ab}^2e_{ac} - d_{aa}e_{ab}e_{ac}^2 + 2d_{aa}e_{ab}e_{bc}^2 - d_{aa}e_{ac}^3 + 2d_{aa}e_{ac}e_{bc}^2 - d_{ab}^2e_{ab}e_{ac} + d_{ab}^2e_{ab}e_{bc} -$ kept
$2d_{ab}^2e_{ac}^2 - 2d_{ab}^2e_{ac}e_{bc} - 2d_{ab}^2e_{bc}^2 - 3d_{ab}d_{ac}e_{ab}^2 + 4d_{ab}d_{ac}e_{ab}e_{bc} - 3d_{ab}d_{ac}e_{ac}^2 - 4d_{ab}d_{ac}e_{ac}e_{bc} - 2d_{ab}d_{ac}e_{bc}^2 -$
$3d_{ab}d_{bc}e_{ab}^2 - 4d_{ab}d_{bc}e_{ab}e_{ac} - 2d_{ab}d_{bc}e_{ac}^2 - 4d_{ab}d_{bc}e_{ac}e_{bc} - 3d_{ab}d_{bc}e_{bc}^2 - 2d_{ac}^2e_{ab}^2 - d_{ac}^2e_{ab}e_{ac} + 2d_{ac}^2e_{ab}e_{bc} -$
$d_{ac}^2e_{ac}e_{bc} - 2d_{ac}^2e_{bc}^2 - 2d_{ac}d_{bc}e_{ab}^2 - 4d_{ac}d_{bc}e_{ab}e_{ac} + 4d_{ac}d_{bc}e_{ab}e_{bc} - 3d_{ac}d_{bc}e_{ac}^2 - 3d_{ac}d_{bc}e_{bc}^2 + d_{bb}e_{ab}^3 -$
$d_{bb}e_{ab}^2e_{bc} - 2d_{bb}e_{ab}e_{ac}^2 + d_{bb}e_{ab}e_{bc}^2 + 2d_{bb}e_{ac}^2e_{bc} - d_{bb}e_{bc}^3 - 2d_{bc}^2e_{ab}^2 - 2d_{bc}^2e_{ab}e_{ac} + d_{bc}^2e_{ab}e_{bc} - 2d_{bc}^2e_{ac}^2 -$
$d_{bc}^2e_{ac}e_{bc} - 2d_{cc}e_{ab}^2e_{ac} - 2d_{cc}e_{ab}^2e_{bc} + d_{cc}e_{ac}^3 + d_{cc}e_{ac}^2e_{bc} + d_{cc}e_{ac}e_{bc}^2 + d_{cc}e_{bc}^3 + e_{ab}^3e_{ac} - e_{ab}^3e_{bc} +$
$2e_{ab}^2e_{ac}e_{bc} + e_{ab}e_{ac}^3 - 2e_{ab}e_{ac}^2e_{bc} + 2e_{ab}e_{ac}e_{bc}^2 - e_{ab}e_{bc}^3 + e_{ac}^3e_{bc} + e_{ac}e_{bc}^3$

$E_{4,26}$ $= -d_{aa}e_{ab}^2e_{ac} - d_{aa}e_{ab}e_{ac}^2 - d_{aa}e_{ab}e_{bc}^2 - d_{aa}e_{ac}e_{bc}^2 - d_{ab}^2e_{ac}^2 - d_{ab}^2e_{bc}^2 - 4d_{ab}d_{ac}e_{bc}^2 - 4d_{ab}d_{bc}e_{ac}^2 - 3d_{ab}e_{ab}^2e_{ac} -$ kept
$3d_{ab}e_{ab}^2e_{bc} - 2d_{ab}e_{ab}e_{ac}^2 + 2d_{ab}e_{ab}e_{bc}^2 + d_{ab}e_{ac}^3 + 4d_{ab}e_{ac}^2e_{bc} + 4d_{ab}e_{ac}e_{bc}^2 + d_{ab}e_{bc}^3 - d_{ac}^2e_{ab}^2 - d_{ac}^2e_{bc}^2 - 4d_{ac}d_{bc}e_{ab}^2 +$
$d_{ac}e_{ab}^3 - 2d_{ac}e_{ab}^2e_{ac} - 4d_{ac}e_{ab}^2e_{bc} - 3d_{ac}e_{ab}e_{ac}^2 + 4d_{ac}e_{ab}e_{bc}^2 + 3d_{ac}e_{ac}^2e_{bc} + 2d_{ac}e_{ac}e_{bc}^2 - d_{ac}e_{bc}^3 - d_{bb}e_{ab}^2e_{bc} +$
$d_{bb}e_{ab}e_{ac}^2 + d_{bb}e_{ab}e_{bc}^2 - d_{bb}e_{ac}^2e_{bc} - d_{bc}^2e_{ab}^2 - d_{bc}^2e_{ac}^2 - d_{bc}e_{ab}^3 - 4d_{bc}e_{ab}^2e_{ac} - 2d_{bc}e_{ab}^2e_{bc} - 4d_{bc}e_{ab}e_{ac}^2 + 3d_{bc}e_{ab}e_{bc}^2 -$
$d_{bc}e_{ac}^3 + 2d_{bc}e_{ac}^2e_{bc} + 3d_{bc}e_{ac}e_{bc}^2 + d_{cc}e_{ab}^2e_{ac} + d_{cc}e_{ab}^2e_{bc} + d_{cc}e_{ac}^2e_{bc} + d_{cc}e_{ac}e_{bc}^2 + e_{ab}^4 - e_{ab}^3e_{ac} + e_{ab}^3e_{bc} +$
$e_{ab}^2e_{ac}^2 - 6e_{ab}^2e_{ac}e_{bc} + e_{ab}^2e_{bc}^2 - e_{ab}e_{ac}^3 + 6e_{ab}e_{ac}^2e_{bc} - 6e_{ab}e_{ac}e_{bc}^2 + e_{ab}e_{bc}^3 + e_{ac}^4 - e_{ac}^3e_{bc} + e_{ac}^2e_{bc}^2 - e_{ac}e_{bc}^3 + e_{bc}^4$

$E_{4,27}$ $= -3d_{ab}e_{ab}^2e_{ac} - 3d_{ab}e_{ab}^2e_{bc} - 4d_{ab}e_{ab}e_{ac}^2 + 4d_{ab}e_{ab}e_{bc}^2 - d_{ab}e_{ac}^3 + 2d_{ab}e_{ac}^2e_{bc} + 2d_{ab}e_{ac}e_{bc}^2 - d_{ab}e_{bc}^3 -$ kept
$d_{ac}e_{ab}^3 - 4d_{ac}e_{ab}^2e_{ac} - 2d_{ac}e_{ab}^2e_{bc} - 3d_{ac}e_{ab}e_{ac}^2 + 2d_{ac}e_{ab}e_{bc}^2 + 3d_{ac}e_{ac}^2e_{bc} + 4d_{ac}e_{ac}e_{bc}^2 + d_{ac}e_{bc}^3 + d_{bc}e_{ab}^3 -$
$2d_{bc}e_{ab}^2e_{ac} - 4d_{bc}e_{ab}^2e_{bc} - 2d_{bc}e_{ab}e_{ac}^2 + 3d_{bc}e_{ab}e_{bc}^2 + d_{bc}e_{ac}^3 + 4d_{bc}e_{ac}^2e_{bc} + 3d_{bc}e_{ac}e_{bc}^2 + 3e_{ab}^3e_{ac} - 3e_{ab}^3e_{bc} -$
$4e_{ab}^2e_{ac}^2 + 6e_{ab}^2e_{ac}e_{bc} - 4e_{ab}^2e_{bc}^2 + 3e_{ab}e_{ac}^3 - 6e_{ab}e_{ac}^2e_{bc} + 6e_{ab}e_{ac}e_{bc}^2 - 3e_{ab}e_{bc}^3 + 3e_{ac}^3e_{bc} - 4e_{ac}^2e_{bc}^2 + 3e_{ac}e_{bc}^3$

$E_{4,28}$	$= -d_{ab}e_{ab}^2e_{ac} - d_{ab}e_{ab}^2e_{bc} - 2d_{ab}e_{ab}e_{ac}^2 + 2d_{ab}e_{ab}e_{bc}^2 - d_{ab}e_{ac}^3 - d_{ab}e_{bc}^3 - d_{ac}e_{ab}^3 - 2d_{ac}e_{ab}^2e_{ac} - d_{ac}e_{ab}e_{ac}^2 + d_{ac}e_{ac}^2e_{bc} + 2d_{ac}e_{ac}e_{bc}^2 + d_{ac}e_{bc}^3 + d_{bc}e_{ab}^3 - 2d_{bc}e_{ab}^2e_{bc} + d_{bc}e_{ab}e_{bc}^2 + d_{bc}e_{ac}^3 + 2d_{bc}e_{ac}^2e_{bc} + d_{bc}e_{ac}e_{bc}^2 + e_{ab}^4 - 2e_{ab}^3e_{ac} + 2e_{ab}^3e_{bc} + 6e_{ab}^2e_{ac}^2 - 8e_{ab}^2e_{ac}e_{bc} + 6e_{ab}^2e_{bc}^2 - 2e_{ab}e_{ac}^3 + 8e_{ab}e_{ac}^2e_{bc} - 8e_{ab}e_{ac}e_{bc}^2 + 2e_{ab}e_{bc}^3 + e_{ac}^4 - 2e_{ac}^3e_{bc} + 6e_{ac}^2e_{bc}^2 - 2e_{ac}e_{bc}^3 + e_{bc}^4$	kept
$E_{4,29}$	$= 3e_{ab}^3e_{ac} - 3e_{ab}^3e_{bc} - 4e_{ab}^2e_{ac}^2 + 6e_{ab}^2e_{ac}e_{bc} - 4e_{ab}^2e_{bc}^2 + 3e_{ab}e_{ac}^3 - 6e_{ab}e_{ac}^2e_{bc} + 6e_{ab}e_{ac}e_{bc}^2 - 3e_{ab}e_{bc}^3 + 3e_{ac}^3e_{bc} - 4e_{ac}^2e_{bc}^2 + 3e_{ac}e_{bc}^3$	kept
$E_{4,30}$	$= e_{ab}^4 - e_{ab}^3e_{ac} + e_{ab}^3e_{bc} + e_{ab}^2e_{ac}^2 - 6e_{ab}^2e_{ac}e_{bc} + e_{ab}^2e_{bc}^2 - e_{ab}e_{ac}^3 + 6e_{ab}e_{ac}^2e_{bc} - 6e_{ab}e_{ac}e_{bc}^2 + e_{ab}e_{bc}^3 + e_{ac}^4 - e_{ac}^3e_{bc} + e_{ac}^2e_{bc}^2 - e_{ac}e_{bc}^3 + e_{bc}^4$	kept
$E_{4,31}$	$= e_{ab}^3e_{ac} - e_{ab}^3e_{bc} + 2e_{ab}^2e_{ac}e_{bc} + e_{ab}e_{ac}^3 - 2e_{ab}e_{ac}^2e_{bc} + 2e_{ab}e_{ac}e_{bc}^2 - e_{ab}e_{bc}^3 + e_{ac}^3e_{bc} + e_{ac}e_{bc}^3$	kept
$E_{4,32}$	$= e_{ab}^2e_{ac}^2 + e_{ab}^2e_{bc}^2 + e_{ac}^2e_{bc}^2$	dropped: $= -1/4E_{4,29} + 3/4E_{4,31}$

$c = 5$

output	expanded	safe reductions
$E_{5,0}$	$= 2d_{aa}^2d_{bb}^2d_{cc} + 2d_{aa}^2d_{bb}d_{cc}^2 + 2d_{aa}d_{bb}^2d_{cc}^2$	kept
$E_{5,1}$	$= d_{aa}^3d_{bb}^2 + 4d_{aa}^3d_{bb}d_{cc} + d_{aa}^3d_{cc}^2 + d_{aa}^2d_{bb}^3 + 4d_{aa}^2d_{bb}^2d_{cc} + 4d_{aa}^2d_{bb}d_{cc}^2 + d_{aa}^2d_{cc}^3 + 4d_{aa}d_{bb}^3d_{cc} + 4d_{aa}d_{bb}^2d_{cc}^2 + 4d_{aa}d_{bb}d_{cc}^3 + d_{bb}^3d_{cc}^2 + d_{bb}^2d_{cc}^3$	kept
$E_{5,2}$	$= d_{aa}^4d_{bb} + d_{aa}^4d_{cc} + 4d_{aa}^3d_{bb}^2 + 8d_{aa}^3d_{bb}d_{cc} + 4d_{aa}^3d_{cc}^2 + 4d_{aa}^2d_{bb}^3 + 12d_{aa}^2d_{bb}^2d_{cc} + 12d_{aa}^2d_{bb}d_{cc}^2 + 4d_{aa}^2d_{cc}^3 + d_{aa}d_{bb}^4 + 8d_{aa}d_{bb}^3d_{cc} + 12d_{aa}d_{bb}^2d_{cc}^2 + 8d_{aa}d_{bb}d_{cc}^3 + d_{aa}d_{cc}^4 + d_{bb}^4d_{cc} + 4d_{bb}^3d_{cc}^2 + 4d_{bb}^2d_{cc}^3 + d_{bb}d_{cc}^4$	kept
$E_{5,3}$	$= 5d_{aa}^4d_{bb}+5d_{aa}^4d_{cc}+7d_{aa}^3d_{bb}^2+12d_{aa}^3d_{bb}d_{cc}+7d_{aa}^3d_{cc}^2+2d_{aa}^2d_{ab}d_{bb}d_{cc}+d_{aa}^2d_{ab}d_{cc}^2+d_{aa}^2d_{ac}d_{bb}^2+2d_{aa}^2d_{ac}d_{bb}d_{cc}+7d_{aa}^2d_{bb}^3 + d_{aa}^2d_{bb}^2d_{bc} + 24d_{aa}^2d_{bb}^2d_{cc} + 4d_{aa}^2d_{bb}d_{bc}d_{cc} + 24d_{aa}^2d_{bb}d_{cc}^2 + d_{aa}^2d_{bc}d_{cc}^2 + 7d_{aa}^2d_{cc}^3 + 2d_{aa}d_{ab}d_{bb}^2d_{cc} + 4d_{aa}d_{ab}d_{bb}d_{cc}^2 + 4d_{aa}d_{ac}d_{bb}^2d_{cc} + 2d_{aa}d_{ac}d_{bb}d_{cc}^2 + 5d_{aa}d_{bb}^4 + 12d_{aa}d_{bb}^3d_{cc} + 2d_{aa}d_{bb}^2d_{bc}d_{cc} + 24d_{aa}d_{bb}^2d_{cc}^2 + 2d_{aa}d_{bb}d_{bc}d_{cc}^2 + 12d_{aa}d_{bb}d_{cc}^3 + 5d_{aa}d_{cc}^4 + d_{ab}d_{bb}^2d_{cc}^2 + d_{ac}d_{bb}^2d_{cc}^2 + 5d_{bb}^4d_{cc} + 7d_{bb}^3d_{cc}^2 + 7d_{bb}^2d_{cc}^3 + 5d_{bb}d_{cc}^4$	kept
$E_{5,4}$	$= 2d_{aa}^5 + 5d_{aa}^4d_{bb} + 5d_{aa}^4d_{cc} + d_{aa}^3d_{ab}d_{bb} + d_{aa}^3d_{ab}d_{cc} + d_{aa}^3d_{ac}d_{bb} + d_{aa}^3d_{ac}d_{cc} + 12d_{aa}^3d_{bb}^2 + 4d_{aa}^3d_{bb}d_{bc} + 24d_{aa}^3d_{bb}d_{cc}+4d_{aa}^3d_{bc}d_{cc}+12d_{aa}^3d_{cc}^2+2d_{aa}^2d_{ab}d_{bb}^2+9d_{aa}^2d_{ab}d_{bb}d_{cc}+4d_{aa}^2d_{ab}d_{cc}^2+4d_{aa}^2d_{ac}d_{bb}^2+9d_{aa}^2d_{ac}d_{bb}d_{cc}+2d_{aa}^2d_{ac}d_{cc}^2+12d_{aa}^2d_{bb}^3+4d_{aa}^2d_{bb}^2d_{bc}+30d_{aa}^2d_{bb}^2d_{cc}+10d_{aa}^2d_{bb}d_{bc}d_{cc}+30d_{aa}^2d_{bb}d_{cc}^2+4d_{aa}^2d_{bc}d_{cc}^2+12d_{aa}^2d_{cc}^3+d_{aa}d_{ab}d_{bb}^3+9d_{aa}d_{ab}d_{bb}^2d_{cc}+10d_{aa}d_{ab}d_{bb}d_{cc}^2+4d_{aa}d_{ab}d_{cc}^3+4d_{aa}d_{ac}d_{bb}^3+10d_{aa}d_{ac}d_{bb}^2d_{cc}+9d_{aa}d_{ac}d_{bb}d_{cc}^2+d_{aa}d_{ac}d_{cc}^3+5d_{aa}d_{bb}^4+d_{aa}d_{bb}^3d_{bc}+24d_{aa}d_{bb}^3d_{cc}+9d_{aa}d_{bb}^2d_{bc}d_{cc}+30d_{aa}d_{bb}^2d_{cc}^2+9d_{aa}d_{bb}d_{bc}d_{cc}^2+24d_{aa}d_{bb}d_{cc}^3+d_{aa}d_{bc}d_{cc}^3+5d_{aa}d_{cc}^4+d_{ab}d_{bb}^3d_{cc}+4d_{ab}d_{bb}^2d_{cc}^2+4d_{ab}d_{bb}d_{cc}^3+4d_{ac}d_{bb}^3d_{cc}+4d_{ac}d_{bb}^2d_{cc}^2+d_{ac}d_{bb}d_{cc}^3+2d_{bb}^5+5d_{bb}^4d_{cc}+d_{bb}^3d_{bc}d_{cc}+12d_{bb}^3d_{cc}^2+2d_{bb}^2d_{bc}d_{cc}^2+12d_{bb}^2d_{cc}^3+d_{bb}d_{bc}d_{cc}^3+5d_{bb}d_{cc}^4+2d_{cc}^5$	kept
$E_{5,5}$	$= 2d_{aa}^5 + 8d_{aa}^4d_{bb} + 2d_{aa}^4d_{bc} + 8d_{aa}^4d_{cc} + 4d_{aa}^3d_{ab}d_{bb} + 8d_{aa}^3d_{ab}d_{cc} + 8d_{aa}^3d_{ac}d_{bb} + 4d_{aa}^3d_{ac}d_{cc} + 12d_{aa}^3d_{bb}^2 + 8d_{aa}^3d_{bb}d_{bc} + 24d_{aa}^3d_{bb}d_{cc} + 8d_{aa}^3d_{bc}d_{cc} + 12d_{aa}^3d_{cc}^2 + 12d_{aa}^2d_{ab}d_{bb}^2 + 24d_{aa}^2d_{ab}d_{bb}d_{cc} + 12d_{aa}^2d_{ab}d_{cc}^2 + 12d_{aa}^2d_{ac}d_{bb}^2 + 24d_{aa}^2d_{ac}d_{bb}d_{cc} + 12d_{aa}^2d_{ac}d_{cc}^2 + 12d_{aa}^2d_{bb}^3 + 12d_{aa}^2d_{bb}^2d_{bc} + 36d_{aa}^2d_{bb}^2d_{cc} + 24d_{aa}^2d_{bb}d_{bc}d_{cc} + 36d_{aa}^2d_{bb}d_{cc}^2 + 12d_{aa}^2d_{bc}d_{cc}^2 + 12d_{aa}^2d_{cc}^3 + 4d_{aa}d_{ab}d_{bb}^3 + 24d_{aa}d_{ab}d_{bb}^2d_{cc} + 24d_{aa}d_{ab}d_{bb}d_{cc}^2 + 8d_{aa}d_{ab}d_{cc}^3 + 8d_{aa}d_{ac}d_{bb}^3 + 24d_{aa}d_{ac}d_{bb}^2d_{cc} + 24d_{aa}d_{ac}d_{bb}d_{cc}^2 + 4d_{aa}d_{ac}d_{cc}^3 + 8d_{aa}d_{bb}^4 + 8d_{aa}d_{bb}^3d_{bc} + 24d_{aa}d_{bb}^3d_{cc} + 24d_{aa}d_{bb}^2d_{bc}d_{cc} + 36d_{aa}d_{bb}^2d_{cc}^2 + 24d_{aa}d_{bb}d_{bc}d_{cc}^2 + 24d_{aa}d_{bb}d_{cc}^3 + 8d_{aa}d_{bc}d_{cc}^3 + 8d_{aa}d_{cc}^4 + 8d_{ab}d_{bb}^3d_{cc} + 12d_{ab}d_{bb}^2d_{cc}^2 + 8d_{ab}d_{bb}d_{cc}^3 + 2d_{ab}d_{cc}^4 + 2d_{ac}d_{bb}^4 + 8d_{ac}d_{bb}^3d_{cc} + 12d_{ac}d_{bb}^2d_{cc}^2 + 8d_{ac}d_{bb}d_{cc}^3 + 2d_{bb}^5 + 8d_{bb}^4d_{cc} + 4d_{bb}^3d_{bc}d_{cc} + 12d_{bb}^3d_{cc}^2 + 12d_{bb}^2d_{bc}d_{cc}^2 + 12d_{bb}^2d_{cc}^3 + 4d_{bb}d_{bc}d_{cc}^3 + 8d_{bb}d_{cc}^4 + 2d_{cc}^5$	kept

$E_{5,6}$ = (108 terms; see the full text file) kept

$E_{5,7}$ = (144 terms; see the full text file) kept

$E_{5,8}$ = (147 terms; see the full text file) kept

$E_{5,9}$ = (213 terms; see the full text file) kept

$E_{5,10}$ = (249 terms; see the full text file) kept

$E_{5,11}$ = (243 terms; see the full text file) kept

$E_{5,12}$ = (345 terms; see the full text file) kept

$E_{5,13}$ = (408 terms; see the full text file) kept

$E_{5,14}$ = (327 terms; see the full text file) kept

$E_{5,15}$ = (465 terms; see the full text file) kept

$E_{5,16}$ = (534 terms; see the full text file) kept

$E_{5,17}$ = (438 terms; see the full text file) kept

$E_{5,18}$ = (543 terms; see the full text file) kept

$E_{5,19}$ = (612 terms; see the full text file) kept

$E_{5,20}$ = (456 terms; see the full text file) kept

$E_{5,21}$ = (567 terms; see the full text file) kept

$E_{5,22}$ = (528 terms; see the full text file) kept

$E_{5,23}$ = (378 terms; see the full text file) kept

$E_{5,24}$ = (426 terms; see the full text file) kept

$E_{5,25}$ = (411 terms; see the full text file) kept

$E_{5,26}$ = (267 terms; see the full text file) kept

$E_{5,27}$ = (336 terms; see the full text file) kept

$E_{5,28}$ = (228 terms; see the full text file) kept

$E_{5,29}$ = (135 terms; see the full text file) kept

$E_{5,30}$ = (186 terms; see the full text file) kept

$E_{5,31}$ = (123 terms; see the full text file) kept

$E_{5,32}$ = (102 terms; see the full text file) kept

$E_{5,33}$ = (105 terms; see the full text file) kept

$E_{5,34} = d_{aa}e_{ab}^2e_{bc}^2 + d_{aa}e_{ac}^2e_{bc}^2 - 3d_{ab}e_{ab}^2e_{ac}^2 + 4d_{ab}e_{ab}^2e_{ac}e_{bc} - 3d_{ab}e_{ab}^2e_{bc}^2 + 2d_{ab}e_{ab}e_{ac}^3 - 2d_{ab}e_{ab}e_{bc}^3 + d_{ab}e_{ac}^4 + 4d_{ab}e_{ac}^3e_{bc} + 4d_{ab}e_{ac}e_{bc}^3 + d_{ab}e_{bc}^4 + d_{ac}e_{ab}^4 + 2d_{ac}e_{ab}^3e_{ac} - 4d_{ac}e_{ab}^3e_{bc} - 3d_{ac}e_{ab}^2e_{ac}^2 - 4d_{ac}e_{ab}e_{ac}^2e_{bc} - 4d_{ac}e_{ab}e_{bc}^3 - 3d_{ac}e_{ac}^2e_{bc}^2 + 2d_{ac}e_{ac}e_{bc}^3 + d_{ac}e_{bc}^4 + d_{bb}e_{ab}^2e_{ac}^2 + d_{bb}e_{ac}^2e_{bc}^2 + d_{bc}e_{ab}^4 + 4d_{bc}e_{ab}^3e_{ac} - 2d_{bc}e_{ab}^3e_{bc} - 3d_{bc}e_{ab}^2e_{bc}^2 + 4d_{bc}e_{ab}e_{ac}^3 + 4d_{bc}e_{ab}e_{ac}e_{bc}^2 + d_{bc}e_{ac}^4 + 2d_{bc}e_{ac}^3e_{bc} - 3d_{bc}e_{ac}^2e_{bc}^2 + d_{cc}e_{ab}^2e_{ac}^2 + d_{cc}e_{ab}^2e_{bc}^2 + e_{ab}^4e_{ac} + e_{ab}^4e_{bc} + e_{ab}e_{ac}^4 - e_{ab}e_{bc}^4 - e_{ac}^4e_{bc} - e_{ac}e_{bc}^4$ kept

$E_{5,35} = 4d_{ab}e_{ab}e_{ac}^2e_{bc} - 4d_{ab}e_{ab}e_{ac}e_{bc}^2 + d_{ab}e_{ac}^4 + 4d_{ab}e_{ac}^2e_{bc}^2 + d_{ab}e_{bc}^4 + d_{ac}e_{ab}^4 - 4d_{ac}e_{ab}^2e_{ac}e_{bc} + 4d_{ac}e_{ab}^2e_{bc}^2 - 4d_{ac}e_{ab}e_{ac}e_{bc}^2 + d_{ac}e_{bc}^4 + d_{bc}e_{ab}^4 + 4d_{bc}e_{ab}^2e_{ac}^2 - 4d_{bc}e_{ab}^2e_{ac}e_{bc} + 4d_{bc}e_{ab}e_{ac}^2e_{bc} + d_{bc}e_{ac}^4$ kept

$E_{5,36} = d_{ab}e_{ab}^2e_{ac}^2 + d_{ab}e_{ab}^2e_{bc}^2 - 2d_{ab}e_{ab}e_{ac}^2e_{bc} + 2d_{ab}e_{ab}e_{ac}e_{bc}^2 + 2d_{ab}e_{ac}^3e_{bc} + 2d_{ab}e_{ac}e_{bc}^3 - 2d_{ac}e_{ab}^3e_{bc} + d_{ac}e_{ab}^2e_{ac}^2 + 2d_{ac}e_{ab}^2e_{ac}e_{bc} + 2d_{ac}e_{ab}e_{ac}e_{bc}^2 - 2d_{ac}e_{ab}e_{bc}^3 + d_{ac}e_{ac}^2e_{bc}^2 + 2d_{bc}e_{ab}^3e_{ac} + 2d_{bc}e_{ab}^2e_{ac}e_{bc} + d_{bc}e_{ab}^2e_{bc}^2 + 2d_{bc}e_{ab}e_{ac}^3 - 2d_{bc}e_{ab}e_{ac}^2e_{bc} + d_{bc}e_{ac}^2e_{bc}^2 - e_{ab}^4e_{ac} - e_{ab}^4e_{bc} - e_{ab}e_{ac}^4 + e_{ab}e_{bc}^4 + e_{ac}^4e_{bc} + e_{ac}e_{bc}^4$ kept

$E_{5,37} = 2d_{ab}e_{ac}^2e_{bc}^2 + 2d_{ac}e_{ab}^2e_{bc}^2 + 2d_{bc}e_{ab}^2e_{ac}^2 + e_{ab}^4e_{ac} + e_{ab}^4e_{bc} + e_{ab}^3e_{ac}^2 - e_{ab}^3e_{bc}^2 + e_{ab}^2e_{ac}^3 + e_{ab}^2e_{bc}^3 + e_{ab}e_{ac}^4 - e_{ab}e_{bc}^4 - e_{ac}^4e_{bc} - e_{ac}^3e_{bc}^2 - e_{ac}^2e_{bc}^3 - e_{ac}e_{bc}^4$ kept

$E_{5,38}$	$= e_{ab}^4 e_{ac} + e_{ab}^4 e_{bc} + 2e_{ab}^3 e_{ac}^2 - 2e_{ab}^3 e_{bc}^2 + 2e_{ab}^2 e_{ac}^3 + 2e_{ab}^2 e_{bc}^3 + e_{ab} e_{ac}^4 - e_{ab} e_{bc}^4 - e_{ac}^4 e_{bc} - 2e_{ac}^3 e_{bc}^2 - 2e_{ac}^2 e_{bc}^3 - e_{ac} e_{bc}^4$	kept
$E_{5,39}$	$= e_{ab}^3 e_{ac}^2 - e_{ab}^3 e_{bc}^2 + e_{ab}^2 e_{ac}^3 + e_{ab}^2 e_{bc}^3 - e_{ac}^3 e_{bc}^2 - e_{ac}^2 e_{bc}^3$	kept
$E_{5,40}$	$= 0$	dropped: $= 0$

$c = 6$

output	expanded	safe reductions
$E_{6,0}$	$= d_{aa}^2 d_{bb}^2 d_{cc}^2$	kept
$E_{6,1}$	$= d_{aa}^3 d_{bb}^2 d_{cc} + d_{aa}^3 d_{bb} d_{cc}^2 + d_{aa}^2 d_{bb}^3 d_{cc} + d_{aa}^2 d_{bb} d_{cc}^3 + d_{aa} d_{bb}^3 d_{cc}^2 + d_{aa} d_{bb}^2 d_{cc}^3$	kept
$E_{6,2}$	$= d_{aa}^4 d_{bb} d_{cc} + d_{aa}^3 d_{bb}^3 + 2d_{aa}^3 d_{bb}^2 d_{cc} + 2d_{aa}^3 d_{bb} d_{cc}^2 + d_{aa}^3 d_{cc}^3 + 2d_{aa}^2 d_{bb}^3 d_{cc} + 3d_{aa}^2 d_{bb}^2 d_{cc}^2 + 2d_{aa}^2 d_{bb} d_{cc}^3 + d_{aa} d_{bb}^4 d_{cc} + 2d_{aa} d_{bb}^3 d_{cc}^2 + 2d_{aa} d_{bb}^2 d_{cc}^3 + d_{aa} d_{bb} d_{cc}^4 + d_{bb}^3 d_{cc}^3$	kept
$E_{6,3}$	$= 2d_{aa}^4 d_{bb}^2 + 2d_{aa}^4 d_{bb} d_{cc} + 2d_{aa}^4 d_{cc}^2 + 4d_{aa}^3 d_{bb}^2 d_{cc} + 4d_{aa}^3 d_{bb} d_{cc}^2 + d_{aa}^2 d_{ab} d_{bb} d_{cc}^2 + d_{aa}^2 d_{ac} d_{bb}^2 d_{cc} + 2d_{aa}^2 d_{bb}^4 + 4d_{aa}^2 d_{bb}^3 d_{cc} + d_{aa}^2 d_{bb}^2 d_{bc} d_{cc} + 8d_{aa}^2 d_{bb}^2 d_{cc}^2 + d_{aa}^2 d_{bb} d_{bc} d_{cc}^2 + 4d_{aa}^2 d_{bb} d_{cc}^3 + 2d_{aa}^2 d_{cc}^4 + d_{aa} d_{ab} d_{bb}^2 d_{cc}^2 + d_{aa} d_{ac} d_{bb}^2 d_{cc}^2 + 2d_{aa} d_{bb}^4 d_{cc} + 4d_{aa} d_{bb}^3 d_{cc}^2 + 4d_{aa} d_{bb}^2 d_{cc}^3 + 2d_{aa} d_{bb} d_{cc}^4 + 2d_{bb}^4 d_{cc}^2 + 2d_{bb}^2 d_{cc}^4$	kept
$E_{6,4}$	$= d_{aa}^5 d_{bb} + d_{aa}^5 d_{cc} + d_{aa}^4 d_{bb}^2 + 4d_{aa}^4 d_{bb} d_{cc} + d_{aa}^4 d_{cc}^2 + d_{aa}^3 d_{ab} d_{bb} d_{cc} + d_{aa}^3 d_{ac} d_{bb} d_{cc} + 4d_{aa}^3 d_{bb}^3 + d_{aa}^3 d_{bb}^2 d_{bc} + 7d_{aa}^3 d_{bb}^2 d_{cc} + 2d_{aa}^3 d_{bb} d_{bc} d_{cc} + 7d_{aa}^3 d_{bb} d_{cc}^2 + d_{aa}^3 d_{bc} d_{cc}^2 + 4d_{aa}^3 d_{cc}^3 + 2d_{aa}^2 d_{ab} d_{bb}^2 d_{cc} + 2d_{aa}^2 d_{ab} d_{bb} d_{cc}^2 + d_{aa}^2 d_{ab} d_{cc}^3 + d_{aa}^2 d_{ac} d_{bb}^3 + 2d_{aa}^2 d_{ac} d_{bb}^2 d_{cc} + 2d_{aa}^2 d_{ac} d_{bb} d_{cc}^2 + d_{aa}^2 d_{bb}^4 + 7d_{aa}^2 d_{bb}^3 d_{cc} + 2d_{aa}^2 d_{bb}^2 d_{bc} d_{cc} + 12d_{aa}^2 d_{bb}^2 d_{cc}^2 + 2d_{aa}^2 d_{bb} d_{bc} d_{cc}^2 + 7d_{aa}^2 d_{bb} d_{cc}^3 + d_{aa}^2 d_{cc}^4 + d_{aa} d_{ab} d_{bb}^3 d_{cc} + 2d_{aa} d_{ab} d_{bb}^2 d_{cc}^2 + 2d_{aa} d_{ab} d_{bb} d_{cc}^3 + 2d_{aa} d_{ac} d_{bb}^3 d_{cc} + 2d_{aa} d_{ac} d_{bb}^2 d_{cc}^2 + d_{aa} d_{ac} d_{bb} d_{cc}^3 + d_{aa} d_{bb}^5 + 4d_{aa} d_{bb}^4 d_{cc} + d_{aa} d_{bb}^3 d_{bc} d_{cc} + 7d_{aa} d_{bb}^3 d_{cc}^2 + 2d_{aa} d_{bb}^2 d_{bc} d_{cc}^2 + 7d_{aa} d_{bb}^2 d_{cc}^3 + d_{aa} d_{bb} d_{bc} d_{cc}^3 + 4d_{aa} d_{bb} d_{cc}^4 + d_{aa} d_{cc}^5 + d_{ab} d_{bb}^2 d_{cc}^3 + d_{ac} d_{bb}^3 d_{cc}^2 + d_{bb}^5 d_{cc} + d_{bb}^4 d_{cc}^2 + 4d_{bb}^3 d_{cc}^3 + d_{bb}^2 d_{cc}^4 + d_{bb} d_{cc}^5$	kept
$E_{6,5}$	$= 2d_{aa}^5 d_{bb} + 2d_{aa}^5 d_{cc} + 2d_{aa}^4 d_{bb}^2 + d_{aa}^4 d_{bb} d_{bc} + 4d_{aa}^4 d_{bb} d_{cc} + d_{aa}^4 d_{bc} d_{cc} + 2d_{aa}^4 d_{cc}^2 + d_{aa}^3 d_{ab} d_{bb}^2 + 3d_{aa}^3 d_{ab} d_{bb} d_{cc} + 3d_{aa}^3 d_{ab} d_{cc}^2 + 3d_{aa}^3 d_{ac} d_{bb}^2 + 3d_{aa}^3 d_{ac} d_{bb} d_{cc} + d_{aa}^3 d_{ac} d_{cc}^2 + 4d_{aa}^3 d_{bb}^3 + d_{aa}^3 d_{bb}^2 d_{bc} + 11d_{aa}^3 d_{bb}^2 d_{cc} + 6d_{aa}^3 d_{bb} d_{bc} d_{cc} + 11d_{aa}^3 d_{bb} d_{cc}^2 + d_{aa}^3 d_{bc} d_{cc}^2 + 4d_{aa}^3 d_{cc}^3 + d_{aa}^2 d_{ab} d_{bb}^3 + 8d_{aa}^2 d_{ab} d_{bb}^2 d_{cc} + 5d_{aa}^2 d_{ab} d_{bb} d_{cc}^2 + d_{aa}^2 d_{ab} d_{cc}^3 + d_{aa}^2 d_{ac} d_{bb}^3 + 5d_{aa}^2 d_{ac} d_{bb}^2 d_{cc} + 8d_{aa}^2 d_{ac} d_{bb} d_{cc}^2 + d_{aa}^2 d_{ac} d_{cc}^3 + 2d_{aa}^2 d_{bb}^4 + 3d_{aa}^2 d_{bb}^3 d_{bc} + 11d_{aa}^2 d_{bb}^3 d_{cc} + 5d_{aa}^2 d_{bb}^2 d_{bc} d_{cc} + 12d_{aa}^2 d_{bb}^2 d_{cc}^2 + 5d_{aa}^2 d_{bb} d_{bc} d_{cc}^2 + 11d_{aa}^2 d_{bb} d_{cc}^3 + 3d_{aa}^2 d_{bc} d_{cc}^3 + 2d_{aa}^2 d_{cc}^4 + 3d_{aa} d_{ab} d_{bb}^3 d_{cc} + 5d_{aa} d_{ab} d_{bb}^2 d_{cc}^2 + 6d_{aa} d_{ab} d_{bb} d_{cc}^3 + d_{aa} d_{ab} d_{cc}^4 + d_{aa} d_{ac} d_{bb}^4 + 6d_{aa} d_{ac} d_{bb}^3 d_{cc} + 5d_{aa} d_{ac} d_{bb}^2 d_{cc}^2 + 3d_{aa} d_{ac} d_{bb} d_{cc}^3 + 2d_{aa} d_{bb}^5 + 4d_{aa} d_{bb}^4 d_{cc} + 3d_{aa} d_{bb}^3 d_{bc} d_{cc} + 11d_{aa} d_{bb}^3 d_{cc}^2 + 8d_{aa} d_{bb}^2 d_{bc} d_{cc}^2 + 11d_{aa} d_{bb}^2 d_{cc}^3 + 3d_{aa} d_{bb} d_{bc} d_{cc}^3 + 4d_{aa} d_{bb} d_{cc}^4 + 2d_{aa} d_{cc}^5 + 3d_{ab} d_{bb}^3 d_{cc}^2 + d_{ab} d_{bb}^2 d_{cc}^3 + d_{ab} d_{bb} d_{cc}^4 + d_{ac} d_{bb}^4 d_{cc} + d_{ac} d_{bb}^3 d_{cc}^2 + 3d_{ac} d_{bb}^2 d_{cc}^3 + 2d_{bb}^5 d_{cc} + 2d_{bb}^4 d_{cc}^2 + d_{bb}^3 d_{bc} d_{cc}^2 + 4d_{bb}^3 d_{cc}^3 + d_{bb}^2 d_{bc} d_{cc}^3 + 2d_{bb}^2 d_{cc}^4 + 2d_{bb} d_{cc}^5$	kept

$E_{6,6}$ $= d_{aa}^6 + d_{aa}^4 d_{ab} d_{bb} + 2d_{aa}^4 d_{ab} d_{cc} + 2d_{aa}^4 d_{ac} d_{bb} + d_{aa}^4 d_{ac} d_{cc} + 5d_{aa}^4 d_{bb}^2 + 3d_{aa}^4 d_{bb} d_{bc} + 8d_{aa}^4 d_{bb} d_{cc} + 3d_{aa}^4 d_{bc} d_{cc} +$ kept
$5d_{aa}^4 d_{cc}^2 + 4d_{aa}^3 d_{ab} d_{bb}^2 + 9d_{aa}^3 d_{ab} d_{bb} d_{cc} + 3d_{aa}^3 d_{ab} d_{cc}^2 + 3d_{aa}^3 d_{ac} d_{bb}^2 + 9d_{aa}^3 d_{ac} d_{bb} d_{cc} + 4d_{aa}^3 d_{ac} d_{cc}^2 + 2d_{aa}^3 d_{bb}^3 +$
$5d_{aa}^3 d_{bb}^2 d_{bc} + 10d_{aa}^3 d_{bb}^2 d_{cc} + 6d_{aa}^3 d_{bb} d_{bc} d_{cc} + 10d_{aa}^3 d_{bb} d_{cc}^2 + 5d_{aa}^3 d_{bc} d_{cc}^2 + 2d_{aa}^3 d_{cc}^3 + d_{aa}^2 d_{ab} d_{ac} d_{bb} d_{cc} +$
$4d_{aa}^2 d_{ab} d_{bb}^3 + 10d_{aa}^2 d_{ab} d_{bb}^2 d_{cc} + d_{aa}^2 d_{ab} d_{bb} d_{bc} d_{cc} + 15d_{aa}^2 d_{ab} d_{bb} d_{cc}^2 + d_{aa}^2 d_{ab} d_{bc} d_{cc}^2 + 5d_{aa}^2 d_{ab} d_{cc}^3 + 5d_{aa}^2 d_{ac} d_{bb}^3 +$
$d_{aa}^2 d_{ac} d_{bb}^2 d_{bc} + 15d_{aa}^2 d_{ac} d_{bb}^2 d_{cc} + d_{aa}^2 d_{ac} d_{bb} d_{bc} d_{cc} + 10d_{aa}^2 d_{ac} d_{bb} d_{cc}^2 + 4d_{aa}^2 d_{ac} d_{cc}^3 + 5d_{aa}^2 d_{bb}^4 + 3d_{aa}^2 d_{bb}^3 d_{bc} +$
$10d_{aa}^2 d_{bb}^3 d_{cc} + 15d_{aa}^2 d_{bb}^2 d_{bc} d_{cc} + 18d_{aa}^2 d_{bb}^2 d_{cc}^2 - d_{aa}^2 d_{bb}^2 d_{cc} e_{ac} - d_{aa}^2 d_{bb}^2 d_{cc} e_{bc} + d_{aa}^2 d_{bb} d_{bc}^2 d_{cc} + 15 d_{aa}^2 d_{bb} d_{bc} d_{cc}^2 +$
$10d_{aa}^2 d_{bb} d_{cc}^3 - d_{aa}^2 d_{bb} d_{cc}^2 e_{ab} + d_{aa}^2 d_{bb} d_{cc}^2 e_{bc} + 3d_{aa}^2 d_{bc} d_{cc}^3 + 5d_{aa}^2 d_{cc}^4 + d_{aa} d_{ab}^2 d_{bb} d_{cc}^2 + d_{aa} d_{ab} d_{ac} d_{bb}^2 d_{cc} +$
$d_{aa} d_{ab} d_{ac} d_{bb} d_{cc}^2 + d_{aa} d_{ab} d_{bb}^4 + 9 d_{aa} d_{ab} d_{bb}^3 d_{cc} + d_{aa} d_{ab} d_{bb}^2 d_{bc} d_{cc} + 15 d_{aa} d_{ab} d_{bb}^2 d_{cc}^2 + d_{aa} d_{ab} d_{bb} d_{bc} d_{cc}^2 +$
$6d_{aa} d_{ab} d_{bb} d_{cc}^3 + 3d_{aa} d_{ab} d_{cc}^4 + d_{aa} d_{ac}^2 d_{bb}^2 d_{cc} + 3d_{aa} d_{ac} d_{bb}^4 + 6d_{aa} d_{ac} d_{bb}^3 d_{cc} + d_{aa} d_{ac} d_{bb}^2 d_{bc} d_{cc} + 15 d_{aa} d_{ac} d_{bb}^2 d_{cc}^2 +$
$d_{aa} d_{ac} d_{bb} d_{bc} d_{cc}^2 + 9 d_{aa} d_{ac} d_{bb} d_{cc}^3 + d_{aa} d_{ac} d_{cc}^4 + 2 d_{aa} d_{bb}^4 d_{bc} + 8 d_{aa} d_{bb}^4 d_{cc} + 9 d_{aa} d_{bb}^3 d_{bc} d_{cc} + 10 d_{aa} d_{bb}^3 d_{cc}^2 +$
$10d_{aa} d_{bb}^2 d_{bc} d_{cc}^2 + 10 d_{aa} d_{bb}^2 d_{cc}^3 + d_{aa} d_{bb}^2 d_{cc}^2 e_{ab} + d_{aa} d_{bb}^2 d_{cc}^2 e_{ac} + 9 d_{aa} d_{bb} d_{bc} d_{cc}^3 + 8 d_{aa} d_{bb} d_{cc}^4 + 2 d_{aa} d_{bc} d_{cc}^4 +$
$d_{ab} d_{ac} d_{bb}^2 d_{cc}^2 + 2 d_{ab} d_{bb}^4 d_{cc} + 3 d_{ab} d_{bb}^3 d_{cc}^2 + 5 d_{ab} d_{bb}^2 d_{cc}^3 + 3 d_{ab} d_{bb} d_{cc}^4 + 3 d_{ac} d_{bb}^4 d_{cc} + 5 d_{ac} d_{bb}^3 d_{cc}^2 + 3 d_{ac} d_{bb}^2 d_{cc}^3 +$
$2d_{ac} d_{bb} d_{cc}^4 + d_{bb}^6 + d_{bb}^4 d_{bc} d_{cc} + 5 d_{bb}^4 d_{cc}^2 + 4 d_{bb}^3 d_{bc} d_{cc}^2 + 2 d_{bb}^3 d_{cc}^3 + 4 d_{bb}^2 d_{bc} d_{cc}^3 + 5 d_{bb}^2 d_{cc}^4 + d_{bb} d_{bc} d_{cc}^4 + d_{cc}^6$

$E_{6,7}$ $=$ (145 terms; see the full text file) kept

$E_{6,8}$ $=$ (190 terms; see the full text file) kept

$E_{6,9}$ $=$ (242 terms; see the full text file) kept

$E_{6,10}$ $=$ (302 terms; see the full text file) kept

$E_{6,11}$ $=$ (370 terms; see the full text file) kept

$E_{6,12}$ $=$ (398 terms; see the full text file) kept

$E_{6,13}$ $=$ (556 terms; see the full text file) kept

$E_{6,14}$ $=$ (565 terms; see the full text file) kept

$E_{6,15}$ $=$ (598 terms; see the full text file) kept

$E_{6,16}$ $=$ (784 terms; see the full text file) kept

$E_{6,17}$ $=$ (832 terms; see the full text file) kept

$E_{6,18}$ $=$ (758 terms; see the full text file) kept

$E_{6,19}$ $=$ (1021 terms; see the full text file) kept

$E_{6,20}$ $=$ (1094 terms; see the full text file) kept

$E_{6,21}$ $=$ (977 terms; see the full text file) kept

$E_{6,22}$ $=$ (1150 terms; see the full text file) kept

$E_{6,23}$ $=$ (1219 terms; see the full text file) kept

$E_{6,24}$ $=$ (949 terms; see the full text file) kept

$E_{6,25}$ $=$ (1156 terms; see the full text file) kept

$E_{6,26}$ $=$ (1084 terms; see the full text file) kept

$E_{6,27}$ $=$ (868 terms; see the full text file) kept

$E_{6,28}$ $=$ (982 terms; see the full text file) kept

$E_{6,29}$ $=$ (930 terms; see the full text file) kept

$E_{6,30}$ $=$ (655 terms; see the full text file) kept

$E_{6,31}$ $=$ (726 terms; see the full text file) kept

$E_{6,32}$ $=$ (708 terms; see the full text file) kept

$E_{6,33}$ $=$ (483 terms; see the full text file) kept

$E_{6,34}$ = (492 terms; see the full text file) kept

$E_{6,35}$ = (420 terms; see the full text file) kept

$E_{6,36}$ = (265 terms; see the full text file) kept

$E_{6,37}$ = (276 terms; see the full text file) kept

$E_{6,38}$ = (214 terms; see the full text file) kept

$E_{6,39}$ = (109 terms; see the full text file) kept

$E_{6,40}$ = (133 terms; see the full text file) kept

$E_{6,41}$ $= -d_{aa}e_{ab}^3e_{ac}e_{bc} + d_{aa}e_{ab}^2e_{bc}^3 + d_{aa}e_{ab}e_{ac}^3e_{bc} - d_{aa}e_{ac}^2e_{bc}^3 - d_{ab}^2e_{ab}e_{ac}^2e_{bc} + d_{ab}^2e_{ab}e_{ac}e_{bc}^2 + d_{ab}^2e_{ac}^3e_{bc} +$ kept
$d_{ab}^2e_{ac}e_{bc}^3 + d_{ab}d_{ac}e_{ab}^2e_{bc}^2 + d_{ab}d_{ac}e_{ac}^2e_{bc}^2 + d_{ab}d_{bc}e_{ab}^2e_{ac}^2 + d_{ab}d_{bc}e_{ac}^2e_{bc}^2 + d_{ab}e_{ab}^4e_{ac} + d_{ab}e_{ab}^4e_{bc} + 4d_{ab}e_{ab}^2e_{ac}^3 -$
$d_{ab}e_{ab}^2e_{ac}^2e_{bc} - d_{ab}e_{ab}^2e_{ac}e_{bc}^2 + 4d_{ab}e_{ab}^2e_{bc}^3 + 2d_{ab}e_{ab}e_{ac}^3e_{bc} - 2d_{ab}e_{ab}e_{ac}e_{bc}^3 + d_{ab}e_{ac}^5 + d_{ab}e_{ac}^3e_{bc}^2 + d_{ab}e_{ac}^2e_{bc}^3 +$
$d_{ab}e_{bc}^5 - d_{ac}^2e_{ab}^3e_{bc} + d_{ac}^2e_{ab}^2e_{ac}e_{bc} + d_{ac}^2e_{ab}e_{ac}e_{bc}^2 - d_{ac}^2e_{ab}e_{bc}^3 + d_{ac}d_{bc}e_{ab}^2e_{ac}^2 + d_{ac}d_{bc}e_{ab}^2e_{bc}^2 + d_{ac}e_{ab}^5 +$
$4d_{ac}e_{ab}^3e_{ac}^2 - 2d_{ac}e_{ab}^3e_{ac}e_{bc} + d_{ac}e_{ab}^3e_{bc}^2 + d_{ac}e_{ab}^2e_{ac}^2e_{bc} - d_{ac}e_{ab}^2e_{bc}^3 + d_{ac}e_{ab}e_{ac}^4 - d_{ac}e_{ab}e_{ac}^2e_{bc}^2 + 2d_{ac}e_{ab}e_{ac}e_{bc}^3 -$
$d_{ac}e_{ac}^4e_{bc} - 4d_{ac}e_{ac}^2e_{bc}^3 - d_{ac}e_{bc}^5 + d_{bb}e_{ab}^3e_{ac}e_{bc} + d_{bb}e_{ab}^2e_{ac}^3 - d_{bb}e_{ab}e_{ac}e_{bc}^3 - d_{bb}e_{ac}^3e_{bc}^2 + d_{bc}^2e_{ab}^3e_{ac} +$
$d_{bc}^2e_{ab}^2e_{ac}e_{bc} + d_{bc}^2e_{ab}e_{ac}^3 - d_{bc}^2e_{ab}e_{ac}^2e_{bc} - d_{bc}e_{ab}^5 - d_{bc}e_{ab}^3e_{ac}^2 + 2d_{bc}e_{ab}^3e_{ac}e_{bc} - 4d_{bc}e_{ab}^3e_{bc}^2 - d_{bc}e_{ab}^2e_{ac}^3 +$
$d_{bc}e_{ab}^2e_{ac}e_{bc}^2 - 2d_{bc}e_{ab}e_{ac}^3e_{bc} + d_{bc}e_{ab}e_{ac}^2e_{bc}^2 - d_{bc}e_{ab}e_{ac}^4 - d_{bc}e_{ac}^5 - 4d_{bc}e_{ac}^3e_{bc}^2 - d_{bc}e_{ac}e_{bc}^4 + d_{cc}e_{ab}^3e_{ac}^2 -$
$d_{cc}e_{ab}^3e_{bc}^2 - d_{cc}e_{ab}e_{ac}^3e_{bc} + d_{cc}e_{ab}e_{ac}e_{bc}^3 - 2e_{ab}^5e_{ac} + 2e_{ab}^5e_{bc} + 2e_{ab}^4e_{ac}^2 - 4e_{ab}^4e_{ac}e_{bc} + 2e_{ab}^4e_{bc}^2 - 4e_{ab}^3e_{ac}^3 +$
$11e_{ab}^3e_{ac}^2e_{bc} - 11e_{ab}^3e_{ac}e_{bc}^2 + 4e_{ab}^3e_{bc}^3 + 2e_{ab}^2e_{ac}^4 - 11e_{ab}^2e_{ac}^3e_{bc} + 12e_{ab}^2e_{ac}^2e_{bc}^2 - 11e_{ab}^2e_{ac}e_{bc}^3 + 2e_{ab}^2e_{bc}^4 - 2e_{ab}e_{ac}^5 +$
$4e_{ab}e_{ac}^4e_{bc} - 11e_{ab}e_{ac}^3e_{bc}^2 + 11e_{ab}e_{ac}^2e_{bc}^3 - 4e_{ab}e_{ac}e_{bc}^4 + 2e_{ab}e_{bc}^5 - 2e_{ac}^5e_{bc} + 2e_{ac}^4e_{bc}^2 - 4e_{ac}^3e_{bc}^3 + 2e_{ac}^2e_{bc}^4 - 2e_{ac}e_{bc}^5$

$E_{6,42}$ $= d_{aa}e_{ab}^2e_{ac}e_{bc}^2 + d_{aa}e_{ab}e_{ac}^2e_{bc}^2 + d_{ab}^2e_{ac}^2e_{bc}^2 + 2d_{ab}e_{ab}^3e_{ac}^2 - 2d_{ab}e_{ab}^3e_{bc}^2 + 2d_{ab}e_{ab}e_{ac}^4 - 2d_{ab}e_{ab}e_{ac}^3e_{bc} +$ kept
$2d_{ab}e_{ab}e_{ac}e_{bc}^3 - 2d_{ab}e_{ab}e_{bc}^4 - 2d_{ab}e_{ac}^3e_{bc}^2 - 2d_{ab}e_{ac}^2e_{bc}^3 + d_{ac}^2e_{ab}^2e_{bc}^2 + 2d_{ac}e_{ac}^4e_{ab}e_{ac} + 2d_{ac}e_{ab}^3e_{ac}e_{bc} - 2d_{ac}e_{ab}^3e_{bc}^2 +$
$2d_{ac}e_{ab}^2e_{ac}^3 + 2d_{ac}e_{ab}^2e_{bc}^3 - 2d_{ac}e_{ab}e_{ac}e_{bc}^3 - 2d_{ac}e_{ac}^3e_{bc}^2 - 2d_{ac}e_{ac}e_{bc}^4 + d_{bb}e_{ab}^2e_{ac}^2e_{bc} - d_{bb}e_{ab}e_{ac}^2e_{bc}^2 +$
$d_{bc}^2e_{ab}^2e_{ac}^2 + 2d_{bc}e_{ab}^4e_{bc} + 2d_{bc}e_{ab}^3e_{ac}^2 - 2d_{bc}e_{ab}^3e_{ac}e_{bc} + 2d_{bc}e_{ab}^2e_{ac}^3 + 2d_{bc}e_{ab}^2e_{bc}^3 + 2d_{bc}e_{ab}e_{ac}^3e_{bc} -$
$2d_{bc}e_{ac}^4e_{bc} - 2d_{bc}e_{ac}^2e_{bc}^3 - d_{cc}e_{ab}^2e_{ac}^2e_{bc} - d_{cc}e_{ab}^2e_{ac}e_{bc}^2 - e_{ab}^6 - 5e_{ab}^4e_{ac}^2 + 8e_{ab}^4e_{ac}e_{bc} - 5e_{ab}^4e_{bc}^2 + 2e_{ab}^3e_{ac}^3 -$
$10e_{ab}^3e_{ac}^2e_{bc} + 10e_{ab}^3e_{ac}e_{bc}^2 - 2e_{ab}^3e_{bc}^3 - 5e_{ab}^2e_{ac}^4 + 10e_{ab}^2e_{ac}^3e_{bc} - 18e_{ab}^2e_{ac}^2e_{bc}^2 + 10e_{ab}^2e_{ac}e_{bc}^3 - 5e_{ab}^2e_{bc}^4 -$
$8e_{ab}e_{ac}^4e_{bc} + 10e_{ab}e_{ac}^3e_{bc}^2 - 10e_{ab}e_{ac}^2e_{bc}^3 + 8e_{ab}e_{ac}e_{bc}^4 - e_{ac}^6 - 5e_{ac}^4e_{bc}^2 + 2e_{ac}^3e_{bc}^3 - 5e_{ac}^2e_{bc}^4 - e_{bc}^6$

$E_{6,43}$ $= d_{ab}e_{ab}^2e_{ac}^3 + d_{ab}e_{ab}^2e_{bc}^3 + 2d_{ab}e_{ab}e_{ac}^3e_{bc} - 2d_{ab}e_{ab}e_{ac}e_{bc}^3 - d_{ab}e_{ac}^4e_{bc} - 2d_{ab}e_{ac}^3e_{bc}^2 - 2d_{ab}e_{ac}^2e_{bc}^3 -$ kept
$d_{ab}e_{ac}e_{bc}^4 + d_{ac}e_{ab}^4e_{bc} + d_{ac}e_{ab}^3e_{ac}^2 - 2d_{ac}e_{ab}^3e_{ac}e_{bc} - 2d_{ac}e_{ab}^3e_{bc}^2 + 2d_{ac}e_{ab}^2e_{ac}^3 + 2d_{ac}e_{ab}e_{ac}e_{bc}^3 - d_{ac}e_{ab}e_{bc}^4 -$
$d_{ac}e_{ac}^2e_{bc}^3 + d_{bc}e_{ab}^4e_{ac} + 2d_{bc}e_{ab}^3e_{ac}^2 + 2d_{bc}e_{ab}^3e_{ac}e_{bc} - d_{bc}e_{ab}^3e_{bc}^2 + 2d_{bc}e_{ab}^2e_{ac}^3 + d_{bc}e_{ab}e_{ac}^4 - 2d_{bc}e_{ab}e_{ac}^3e_{bc} -$
$d_{bc}e_{ac}^3e_{bc}^2 - 2e_{ab}^5e_{ac} + 2e_{ab}^5e_{bc} + 2e_{ab}^4e_{ac}^2 - 4e_{ab}^4e_{ac}e_{bc} + 2e_{ab}^4e_{bc}^2 - 4e_{ab}^3e_{ac}^3 + 11e_{ab}^3e_{ac}^2e_{bc} - 11e_{ab}^3e_{ac}e_{bc}^2 +$
$4e_{ab}^3e_{bc}^3 + 2e_{ab}^2e_{ac}^4 - 11e_{ab}^2e_{ac}^3e_{bc} + 12e_{ab}^2e_{ac}^2e_{bc}^2 - 11e_{ab}^2e_{ac}e_{bc}^3 + 2e_{ab}^2e_{bc}^4 - 2e_{ab}e_{ac}^5 + 4e_{ab}e_{ac}^4e_{bc} -$
$11e_{ab}e_{ac}^3e_{bc}^2 + 11e_{ab}e_{ac}^2e_{bc}^3 - 4e_{ab}e_{ac}e_{bc}^4 + 2e_{ab}e_{bc}^5 - 2e_{ac}^5e_{bc} + 2e_{ac}^4e_{bc}^2 - 4e_{ac}^3e_{bc}^3 + 2e_{ac}^2e_{bc}^4 - 2e_{ac}e_{bc}^5$

$E_{6,44}$ $= d_{ab}e_{ab}^2e_{ac}^2e_{bc} + d_{ab}e_{ab}^2e_{ac}e_{bc}^2 - d_{ab}e_{ac}^3e_{bc}^2 - d_{ab}e_{ac}^2e_{bc}^3 - d_{ac}e_{ab}^3e_{bc}^2 - d_{ac}e_{ab}^2e_{ac}^2e_{bc} + d_{ac}e_{ab}^2e_{bc}^3 + d_{ac}e_{ab}e_{ac}^2e_{bc}^2 +$ kept
$d_{bc}e_{ab}^3e_{ac}^2 + d_{bc}e_{ab}^2e_{ac}^3 - d_{bc}e_{ab}^2e_{ac}e_{bc}^2 - d_{bc}e_{ab}e_{ac}^2e_{bc}^2 + e_{ab}^5e_{ac} - e_{ab}^5e_{bc} - e_{ab}^4e_{ac}^2 + 4e_{ab}^4e_{ac}e_{bc} - e_{ab}^4e_{bc}^2 + 4e_{ab}^3e_{ac}^3 -$
$7e_{ab}^3e_{ac}^2e_{bc} + 7e_{ab}^3e_{ac}e_{bc}^2 - 4e_{ab}^3e_{bc}^3 - e_{ab}^2e_{ac}^4 + 7e_{ab}^2e_{ac}^3e_{bc} - 12e_{ab}^2e_{ac}^2e_{bc}^2 + 7e_{ab}^2e_{ac}e_{bc}^3 - e_{ab}^2e_{bc}^4 + e_{ab}e_{ac}^5 -$
$4e_{ab}e_{ac}^4e_{bc} + 7e_{ab}e_{ac}^3e_{bc}^2 - 7e_{ab}e_{ac}^2e_{bc}^3 + 4e_{ab}e_{ac}e_{bc}^4 - e_{ab}e_{bc}^5 + e_{ac}^5e_{bc} - e_{ac}^4e_{bc}^2 + 4e_{ac}^3e_{bc}^3 - e_{ac}^2e_{bc}^4 + e_{ac}e_{bc}^5$

$E_{6,45}$ $= 2e_{ab}^4e_{ac}^2 - 2e_{ab}^4e_{ac}e_{bc} + 2e_{ab}^4e_{bc}^2 + 4e_{ab}^3e_{ac}^2e_{bc} - 4e_{ab}^3e_{ac}e_{bc}^2 + 2e_{ab}^2e_{ac}^4 - 4e_{ab}^2e_{ac}^3e_{bc} + 8e_{ab}^2e_{ac}^2e_{bc}^2 -$ kept
$4e_{ab}^2e_{ac}e_{bc}^3 + 2e_{ab}^2e_{bc}^4 + 2e_{ab}e_{ac}^4e_{bc} - 4e_{ab}e_{ac}^3e_{bc}^2 + 4e_{ab}e_{ac}^2e_{bc}^3 - 2e_{ab}e_{ac}e_{bc}^4 + 2e_{ac}^4e_{bc}^2 + 2e_{ac}^2e_{bc}^4$

$E_{6,46}$ $= e_{ab}^4e_{ac}e_{bc} + e_{ab}^3e_{ac}^3 - 2e_{ab}^3e_{ac}^2e_{bc} + 2e_{ab}^3e_{ac}e_{bc}^2 - e_{ab}^3e_{bc}^3 + 2e_{ab}^2e_{ac}^3e_{bc} - 3e_{ab}^2e_{ac}^2e_{bc}^2 + 2e_{ab}^2e_{ac}e_{bc}^3 -$ kept
$e_{ab}e_{ac}^4e_{bc} + 2e_{ab}e_{ac}^3e_{bc}^2 - 2e_{ab}e_{ac}^2e_{bc}^3 + e_{ab}e_{ac}e_{bc}^4 + e_{ac}^3e_{bc}^3$

$E_{6,47}$ $= e_{ab}^3e_{ac}^2e_{bc} - e_{ab}^3e_{ac}e_{bc}^2 - e_{ab}^2e_{ac}^3e_{bc} - e_{ab}^2e_{ac}e_{bc}^3 - e_{ab}e_{ac}^3e_{bc}^2 + e_{ab}e_{ac}^2e_{bc}^3$ kept

$E_{6,48}$	$= -e_{ab}^2 e_{ac}^2 e_{bc}^2$	kept

3 $S(2) \times S(1)$ acting on (a, b, p), $k = 2$

$n = 3$, $k = 2$, $|G| = 2$, $m = 9$ slots. Bursarial outputs: 26; after safe reductions: 22 (exact linear dependencies checked for $c \leq 2$). Quotient dimension (the minimum number of outputs, for generic inputs): 5.

slot j	w_j	degree in L
0	d_{aa}	2
1	d_{bb}	2
2	d_{pp}	2
3	d_{ab}	2
4	d_{ap}	2
5	d_{bp}	2
6	e_{ab}	2
7	e_{ap}	2
8	e_{bp}	2

g	$g \cdot w$
e	$(d_{aa},\ d_{bb},\ d_{pp},\ d_{ab},\ d_{ap},\ d_{bp},\ e_{ab},\ e_{ap},\ e_{bp})$
(ab)	$(d_{bb},\ d_{aa},\ d_{pp},\ d_{ab},\ d_{bp},\ d_{ap},\ -e_{ab},\ e_{bp},\ e_{ap})$

$c = 1$

output	expanded	safe reductions
$E_{1,0}$	$= d_{aa} + d_{bb}$	kept
$E_{1,1}$	$= d_{aa} + d_{bb}$	dropped: $= E_{1,0}$
$E_{1,2}$	$= 2d_{pp}$	kept
$E_{1,3}$	$= 2d_{ab}$	kept
$E_{1,4}$	$= d_{ap} + d_{bp}$	kept
$E_{1,5}$	$= d_{ap} + d_{bp}$	dropped: $= E_{1,4}$
$E_{1,6}$	$= 0$	dropped: $= 0$
$E_{1,7}$	$= e_{ap} + e_{bp}$	kept
$E_{1,8}$	$= e_{ap} + e_{bp}$	dropped: $= E_{1,7}$

$c = 2$

output	expanded	safe reductions

$E_{2,0}$	$= d_{aa}d_{bb}$	kept
$E_{2,1}$	$= d_{aa}^2 + d_{bb}^2$	kept
$E_{2,2}$	$= d_{aa}d_{bb} + d_{aa}d_{pp} + d_{bb}d_{pp}$	kept
$E_{2,3}$	$= d_{aa}d_{ab} + d_{aa}d_{pp} + d_{ab}d_{bb} + d_{bb}d_{pp}$	kept
$E_{2,4}$	$= d_{aa}d_{ab} + d_{aa}d_{bp} + d_{ab}d_{bb} + d_{ap}d_{bb} + d_{pp}^2$	kept
$E_{2,5}$	$= 2d_{aa}d_{ap} + 2d_{ab}d_{pp} + 2d_{bb}d_{bp}$	kept
$E_{2,6}$	$= d_{aa}d_{bp} - d_{aa}e_{ab} + d_{ab}^2 + d_{ap}d_{bb} + d_{ap}d_{pp} + d_{bb}e_{ab} + d_{bp}d_{pp}$	kept
$E_{2,7}$	$= d_{aa}e_{ab} + d_{aa}e_{bp} + d_{ab}d_{ap} + d_{ab}d_{bp} + d_{ap}d_{pp} - d_{bb}e_{ab} + d_{bb}e_{ap} + d_{bp}d_{pp}$	kept
$E_{2,8}$	$= 2d_{aa}e_{ap} + d_{ab}d_{ap} + d_{ab}d_{bp} + d_{ap}d_{bp} + 2d_{bb}e_{bp}$	kept
$E_{2,9}$	$= d_{aa}e_{bp} + d_{ap}^2 + d_{bb}e_{ap} + d_{bp}^2 + d_{pp}e_{ap} + d_{pp}e_{bp}$	kept
$E_{2,10}$	$= d_{ab}e_{ap} + d_{ab}e_{bp} + d_{ap}d_{bp} - d_{ap}e_{ab} + d_{bp}e_{ab} + d_{pp}e_{ap} + d_{pp}e_{bp}$	kept
$E_{2,11}$	$= d_{ab}e_{ap} + d_{ab}e_{bp} + d_{ap}e_{ab} + d_{ap}e_{bp} - d_{bp}e_{ab} + d_{bp}e_{ap}$	kept
$E_{2,12}$	$= 2d_{ap}e_{ap} + 2d_{bp}e_{bp} - e_{ab}^2$	kept
$E_{2,13}$	$= d_{ap}e_{bp} + d_{bp}e_{ap} - e_{ab}e_{ap} + e_{ab}e_{bp}$	kept
$E_{2,14}$	$= e_{ab}e_{ap} - e_{ab}e_{bp} + e_{ap}e_{bp}$	kept
$E_{2,15}$	$= e_{ap}^2 + e_{bp}^2$	kept
$E_{2,16}$	$= e_{ap}e_{bp}$	kept

4 $S(2)$ acting on (a, b), $k = 1$

$n = 2$, $k = 1$, $|G| = 2$, $m = 5$ slots. Bursarial outputs: 14; after safe reductions: 11 (exact linear dependencies checked for $c \leq 2$). Quotient dimension (the minimum number of outputs, for generic inputs): 2.

slot j	w_j	degree in L
0	d_{aa}	2
1	d_{bb}	2
2	d_{ab}	2
3	e_a	1
4	e_b	1

g	$g \cdot w$
e	$(d_{aa},\ d_{bb},\ d_{ab},\ e_a,\ e_b)$
(ab)	$(d_{bb},\ d_{aa},\ d_{ab},\ e_b,\ e_a)$

$c = 1$

output	expanded	safe reductions
$E_{1,0}$	$= d_{aa} + d_{bb}$	kept
$E_{1,1}$	$= d_{aa} + d_{bb}$	dropped: $= E_{1,0}$
$E_{1,2}$	$= 2d_{ab}$	kept
$E_{1,3}$	$= e_a + e_b$	kept
$E_{1,4}$	$= e_a + e_b$	dropped: $= E_{1,3}$

$c = 2$

output	expanded	safe reductions
$E_{2,0}$	$= d_{aa}d_{bb}$	kept
$E_{2,1}$	$= d_{aa}^2 + d_{bb}^2$	kept
$E_{2,2}$	$= d_{aa}d_{ab} + d_{aa}d_{bb} + d_{ab}d_{bb}$	kept
$E_{2,3}$	$= d_{aa}d_{ab} + d_{aa}e_b + d_{ab}d_{bb} + d_{bb}e_a$	kept
$E_{2,4}$	$= 2d_{aa}e_a + d_{ab}^2 + 2d_{bb}e_b$	kept
$E_{2,5}$	$= d_{aa}e_b + d_{ab}e_a + d_{ab}e_b + d_{bb}e_a$	kept
$E_{2,6}$	$= d_{ab}e_a + d_{ab}e_b + e_ae_b$	kept
$E_{2,7}$	$= e_a^2 + e_b^2$	kept
$E_{2,8}$	$= e_ae_b$	dropped: $= E_{2,0} - E_{2,2} + E_{2,3} - E_{2,5} + E_{2,6}$

5 $S(3)$ acting on (a, b, c), $k = 1$

$n = 3$, $k = 1$, $|G| = 6$, $m = 9$ slots. Bursarial outputs: 174; after safe reductions: 151 (exact linear dependencies checked for $c \leq 6$). Quotient dimension (the minimum number of outputs, for generic inputs): 3.

slot j	w_j	degree in L
0	d_{aa}	2
1	d_{bb}	2
2	d_{cc}	2
3	d_{ab}	2
4	d_{ac}	2
5	d_{bc}	2
6	e_a	1
7	e_b	1
8	e_c	1

g	$g \cdot w$
e	$(d_{aa},\ d_{bb},\ d_{cc},\ d_{ab},\ d_{ac},\ d_{bc},\ e_a,\ e_b,\ e_c)$
(bc)	$(d_{aa},\ d_{cc},\ d_{bb},\ d_{ac},\ d_{ab},\ d_{bc},\ e_a,\ e_c,\ e_b)$
(ab)	$(d_{bb},\ d_{aa},\ d_{cc},\ d_{ab},\ d_{bc},\ d_{ac},\ e_b,\ e_a,\ e_c)$
(abc)	$(d_{bb},\ d_{cc},\ d_{aa},\ d_{bc},\ d_{ab},\ d_{ac},\ e_b,\ e_c,\ e_a)$
(acb)	$(d_{cc},\ d_{aa},\ d_{bb},\ d_{ac},\ d_{bc},\ d_{ab},\ e_c,\ e_a,\ e_b)$
(ac)	$(d_{cc},\ d_{bb},\ d_{aa},\ d_{bc},\ d_{ac},\ d_{ab},\ e_c,\ e_b,\ e_a)$

$c = 1$

output	expanded	safe reductions
$E_{1,0}$	$= 2d_{aa} + 2d_{bb} + 2d_{cc}$	kept
$E_{1,1}$	$= 2d_{aa} + 2d_{bb} + 2d_{cc}$	dropped: $= E_{1,0}$
$E_{1,2}$	$= 2d_{aa} + 2d_{bb} + 2d_{cc}$	dropped: $= E_{1,0}$
$E_{1,3}$	$= 2d_{ab} + 2d_{ac} + 2d_{bc}$	kept
$E_{1,4}$	$= 2d_{ab} + 2d_{ac} + 2d_{bc}$	dropped: $= E_{1,3}$
$E_{1,5}$	$= 2d_{ab} + 2d_{ac} + 2d_{bc}$	dropped: $= E_{1,3}$
$E_{1,6}$	$= 2e_a + 2e_b + 2e_c$	kept
$E_{1,7}$	$= 2e_a + 2e_b + 2e_c$	dropped: $= E_{1,6}$
$E_{1,8}$	$= 2e_a + 2e_b + 2e_c$	dropped: $= E_{1,6}$

$c = 2$

output	expanded	safe reductions
$E_{2,0}$	$= d_{aa}^2 + 4d_{aa}d_{bb} + 4d_{aa}d_{cc} + d_{bb}^2 + 4d_{bb}d_{cc} + d_{cc}^2$	kept
$E_{2,1}$	$= 4d_{aa}^2 + 6d_{aa}d_{bb} + 6d_{aa}d_{cc} + 4d_{bb}^2 + 6d_{bb}d_{cc} + 4d_{cc}^2$	kept
$E_{2,2}$	$= 5d_{aa}^2 + 10d_{aa}d_{bb} + 10d_{aa}d_{cc} + 5d_{bb}^2 + 10d_{bb}d_{cc} + 5d_{cc}^2$	dropped: $= E_{2,0} + E_{2,1}$
$E_{2,3}$	$= 4d_{aa}^2 + 3d_{aa}d_{ab} + 3d_{aa}d_{ac} + 6d_{aa}d_{bb} + 4d_{aa}d_{bc} + 6d_{aa}d_{cc} + 3d_{ab}d_{bb} + 4d_{ab}d_{cc} + 4d_{ac}d_{bb} + 3d_{ac}d_{cc} + 4d_{bb}^2 + 3d_{bb}d_{bc} + 6d_{bb}d_{cc} + 3d_{bc}d_{cc} + 4d_{cc}^2$	kept
$E_{2,4}$	$= d_{aa}^2 + 6d_{aa}d_{ab} + 6d_{aa}d_{ac} + 4d_{aa}d_{bb} + 8d_{aa}d_{bc} + 4d_{aa}d_{cc} + 6d_{ab}d_{bb} + 8d_{ab}d_{cc} + 8d_{ac}d_{bb} + 6d_{ac}d_{cc} + d_{bb}^2 + 6d_{bb}d_{bc} + 4d_{bb}d_{cc} + 6d_{bc}d_{cc} + d_{cc}^2$	dropped: $= E_{2,0} - 2E_{2,1} + 2E_{2,3}$
$E_{2,5}$	$= 12d_{aa}d_{ab} + 12d_{aa}d_{ac} + 6d_{aa}d_{bc} + 12d_{ab}d_{bb} + 6d_{ab}d_{cc} + 6d_{ac}d_{bb} + 12d_{ac}d_{cc} + 12d_{bb}d_{bc} + 12d_{bc}d_{cc}$	kept
$E_{2,6}$	$= 6d_{aa}d_{ab} + 6d_{aa}d_{ac} + 8d_{aa}d_{bc} + 2d_{aa}e_a + 4d_{aa}e_b + 4d_{aa}e_c + d_{ab}^2 + 4d_{ab}d_{ac} + 6d_{ab}d_{bb} + 4d_{ab}d_{bc} + 8d_{ab}d_{cc} + d_{ac}^2 + 8d_{ac}d_{bb} + 4d_{ac}d_{bc} + 6d_{ac}d_{cc} + 6d_{bb}d_{bc} + 4d_{bb}e_a + 2d_{bb}e_b + 4d_{bb}e_c + d_{bc}^2 + 6d_{bc}d_{cc} + 4d_{cc}e_a + 4d_{cc}e_b + 2d_{cc}e_c$	kept
$E_{2,7}$	$= 3d_{aa}d_{ab} + 3d_{aa}d_{ac} + 4d_{aa}d_{bc} + 8d_{aa}e_a + 6d_{aa}e_b + 6d_{aa}e_c + 4d_{ab}^2 + 6d_{ab}d_{ac} + 3d_{ab}d_{bb} + 6d_{ab}d_{bc} + 4d_{ab}d_{cc} + 4d_{ac}^2 + 4d_{ac}d_{bb} + 6d_{ac}d_{bc} + 3d_{ac}d_{cc} + 3d_{bb}d_{bc} + 6d_{bb}e_a + 8d_{bb}e_b + 6d_{bb}e_c + 4d_{bc}^2 + 3d_{bc}d_{cc} + 6d_{cc}e_a + 6d_{cc}e_b + 8d_{cc}e_c$	kept
$E_{2,8}$	$= 10d_{aa}e_a + 10d_{aa}e_b + 10d_{aa}e_c + 5d_{ab}^2 + 10d_{ab}d_{ac} + 10d_{ab}d_{bc} + 5d_{ac}^2 + 10d_{ac}d_{bc} + 10d_{bb}e_a + 10d_{bb}e_b + 10d_{bb}e_c + 5d_{bc}^2 + 10d_{cc}e_a + 10d_{cc}e_b + 10d_{cc}e_c$	dropped: $= 3E_{2,1} - 3E_{2,3} + E_{2,6} + E_{2,7}$
$E_{2,9}$	$= 8d_{aa}e_a + 6d_{aa}e_b + 6d_{aa}e_c + 4d_{ab}^2 + 6d_{ab}d_{ac} + 6d_{ab}d_{bc} + 3d_{ab}e_a + 3d_{ab}e_b + 4d_{ab}e_c + 4d_{ac}^2 + 6d_{ac}d_{bc} + 3d_{ac}e_a + 4d_{ac}e_b + 3d_{ac}e_c + 6d_{bb}e_a + 8d_{bb}e_b + 6d_{bb}e_c + 4d_{bc}^2 + 4d_{bc}e_a + 3d_{bc}e_b + 3d_{bc}e_c + 6d_{cc}e_a + 6d_{cc}e_b + 8d_{cc}e_c$	kept
$E_{2,10}$	$= 2d_{aa}e_a + 4d_{aa}e_b + 4d_{aa}e_c + d_{ab}^2 + 4d_{ab}d_{ac} + 4d_{ab}d_{bc} + 6d_{ab}e_a + 6d_{ab}e_b + 8d_{ab}e_c + d_{ac}^2 + 4d_{ac}d_{bc} + 6d_{ac}e_a + 8d_{ac}e_b + 6d_{ac}e_c + 4d_{bb}e_a + 2d_{bb}e_b + 4d_{bb}e_c + d_{bc}^2 + 8d_{bc}e_a + 6d_{bc}e_b + 6d_{bc}e_c + 4d_{cc}e_a + 4d_{cc}e_b + 2d_{cc}e_c$	dropped: $= E_{2,6} - 2E_{2,7} + 2E_{2,9}$
$E_{2,11}$	$= 12d_{ab}e_a + 12d_{ab}e_b + 6d_{ab}e_c + 12d_{ac}e_a + 6d_{ac}e_b + 12d_{ac}e_c + 6d_{bc}e_a + 12d_{bc}e_b + 12d_{bc}e_c$	kept
$E_{2,12}$	$= 6d_{ab}e_a + 6d_{ab}e_b + 8d_{ab}e_c + 6d_{ac}e_a + 8d_{ac}e_b + 6d_{ac}e_c + 8d_{bc}e_a + 6d_{bc}e_b + 6d_{bc}e_c + e_a^2 + 4e_ae_b + 4e_ae_c + e_b^2 + 4e_be_c + e_c^2$	kept
$E_{2,13}$	$= 3d_{ab}e_a + 3d_{ab}e_b + 4d_{ab}e_c + 3d_{ac}e_a + 4d_{ac}e_b + 3d_{ac}e_c + 4d_{bc}e_a + 3d_{bc}e_b + 3d_{bc}e_c + 4e_a^2 + 6e_ae_b + 6e_ae_c + 4e_b^2 + 6e_be_c + 4e_c^2$	kept
$E_{2,14}$	$= 5e_a^2 + 10e_ae_b + 10e_ae_c + 5e_b^2 + 10e_be_c + 5e_c^2$	dropped: $= 3E_{2,1} - 3E_{2,3} + 3E_{2,7} - 3E_{2,9} + E_{2,12} + E_{2,13}$
$E_{2,15}$	$= 4e_a^2 + 6e_ae_b + 6e_ae_c + 4e_b^2 + 6e_be_c + 4e_c^2$	dropped: $= E_{2,1} - E_{2,3} + E_{2,7} - E_{2,9} + E_{2,13}$
$E_{2,16}$	$= e_a^2 + 4e_ae_b + 4e_ae_c + e_b^2 + 4e_be_c + e_c^2$	dropped: $= 2E_{2,1} - 2E_{2,3} + 2E_{2,7} - 2E_{2,9} + E_{2,12}$

$c = 3$

output	expanded	safe reductions
$E_{3,0}$	$= 2d_{aa}^2d_{bb} + 2d_{aa}^2d_{cc} + 2d_{aa}d_{bb}^2 + 8d_{aa}d_{bb}d_{cc} + 2d_{aa}d_{cc}^2 + 2d_{bb}^2d_{cc} + 2d_{bb}d_{cc}^2$	kept
$E_{3,1}$	$= 2d_{aa}^3 + 7d_{aa}^2d_{bb} + 7d_{aa}^2d_{cc} + 7d_{aa}d_{bb}^2 + 12d_{aa}d_{bb}d_{cc} + 7d_{aa}d_{cc}^2 + 2d_{bb}^3 + 7d_{bb}^2d_{cc} + 7d_{bb}d_{cc}^2 + 2d_{cc}^3$	kept
$E_{3,2}$	$= 4d_{aa}^3 + 14d_{aa}^2d_{bb} + 14d_{aa}^2d_{cc} + 14d_{aa}d_{bb}^2 + 24d_{aa}d_{bb}d_{cc} + 14d_{aa}d_{cc}^2 + 4d_{bb}^3 + 14d_{bb}^2d_{cc} + 14d_{bb}d_{cc}^2 + 4d_{cc}^3$	dropped: $= 2E_{3,1}$

$E_{3,3}$ $= 8d_{aa}^3 + d_{aa}^2 d_{ab} + d_{aa}^2 d_{ac} + 14d_{aa}^2 d_{bb} + 2d_{aa}^2 d_{bc} + 14d_{aa}^2 d_{cc} + 4d_{aa} d_{ab} d_{bb} + 6d_{aa} d_{ab} d_{cc} + 6d_{aa} d_{ac} d_{bb} +$ kept
$4d_{aa} d_{ac} d_{cc} + 14d_{aa} d_{bb}^2 + 6d_{aa} d_{bb} d_{bc} + 32d_{aa} d_{bb} d_{cc} + 6d_{aa} d_{bc} d_{cc} + 14d_{aa} d_{cc}^2 + d_{ab} d_{bb}^2 + 6d_{ab} d_{bb} d_{cc} +$
$2d_{ab} d_{cc}^2 + 2d_{ac} d_{bb}^2 + 6d_{ac} d_{bb} d_{cc} + d_{ac} d_{cc}^2 + 8d_{bb}^3 + d_{bb}^2 d_{bc} + 14d_{bb}^2 d_{cc} + 4d_{bb} d_{bc} d_{cc} + 14d_{bb} d_{cc}^2 + d_{bc} d_{cc}^2 + 8d_{cc}^3$

$E_{3,4}$ $= 4d_{aa}^3 + 5d_{aa}^2 d_{ab} + 5d_{aa}^2 d_{ac} + 14d_{aa}^2 d_{bb} + 10d_{aa}^2 d_{bc} + 14d_{aa}^2 d_{cc} + 12d_{aa} d_{ab} d_{bb} + 14d_{aa} d_{ab} d_{cc} + 14d_{aa} d_{ac} d_{bb} +$ kept
$12d_{aa} d_{ac} d_{cc} + 14d_{aa} d_{bb}^2 + 14d_{aa} d_{bb} d_{bc} + 24d_{aa} d_{bb} d_{cc} + 14d_{aa} d_{bc} d_{cc} + 14d_{aa} d_{cc}^2 + 5d_{ab} d_{bb}^2 + 14d_{ab} d_{bb} d_{cc} +$
$10d_{ab} d_{cc}^2 + 10d_{ac} d_{bb}^2 + 14d_{ac} d_{bb} d_{cc} + 5d_{ac} d_{cc}^2 + 4d_{bb}^3 + 5d_{bb}^2 d_{bc} + 14d_{bb}^2 d_{cc} + 12d_{bb} d_{bc} d_{cc} + 14d_{bb} d_{cc}^2 + 5d_{bc} d_{cc}^2 + 4d_{cc}^3$

$E_{3,5}$ $= 2d_{aa}^3 + 15d_{aa}^2 d_{ab} + 15d_{aa}^2 d_{ac} + 7d_{aa}^2 d_{bb} + 10d_{aa}^2 d_{bc} + 7d_{aa}^2 d_{cc} + 28d_{aa} d_{ab} d_{bb} + 26d_{aa} d_{ab} d_{cc} + 26d_{aa} d_{ac} d_{bb} +$ kept
$28d_{aa} d_{ac} d_{cc} + 7d_{aa} d_{bb}^2 + 26d_{aa} d_{bb} d_{bc} + 12d_{aa} d_{bb} d_{cc} + 26d_{aa} d_{bc} d_{cc} + 7d_{aa} d_{cc}^2 + 15d_{ab} d_{bb}^2 + 26d_{ab} d_{bb} d_{cc} +$
$10d_{ab} d_{cc}^2 + 10d_{ac} d_{bb}^2 + 26d_{ac} d_{bb} d_{cc} + 15d_{ac} d_{cc}^2 + 2d_{bb}^3 + 15d_{bb}^2 d_{bc} + 7d_{bb}^2 d_{cc} + 28d_{bb} d_{bc} d_{cc} + 7d_{bb} d_{cc}^2 + 15d_{bc} d_{cc}^2 + 2d_{cc}^3$

$E_{3,6}$ $= 18d_{aa}^2 d_{ab} + 18d_{aa}^2 d_{ac} + 2d_{aa}^2 d_{bb} + 16d_{aa}^2 d_{bc} + 2d_{aa}^2 d_{cc} + 2d_{aa}^2 e_b + 2d_{aa}^2 e_c + d_{aa} d_{ab}^2 + 4d_{aa} d_{ab} d_{ac} +$ kept
$32d_{aa} d_{ab} d_{bb} + 6d_{aa} d_{ab} d_{bc} + 28d_{aa} d_{ab} d_{cc} + d_{aa} d_{ac}^2 + 28d_{aa} d_{ac} d_{bb} + 6d_{aa} d_{ac} d_{bc} + 32d_{aa} d_{ac} d_{cc} + 2d_{aa} d_{bb}^2 +$
$28d_{aa} d_{bb} d_{bc} + 8d_{aa} d_{bb} d_{cc} + 4d_{aa} d_{bb} e_a + 4d_{aa} d_{bb} e_b + 8d_{aa} d_{bb} e_c + 2d_{aa} d_{bc}^2 + 28d_{aa} d_{bc} d_{cc} + 2d_{aa} d_{cc}^2 +$
$4d_{aa} d_{cc} e_a + 8d_{aa} d_{cc} e_b + 4d_{aa} d_{cc} e_c + d_{ab}^2 d_{bb} + 2d_{ab}^2 d_{cc} + 6d_{ab} d_{ac} d_{bb} + 6d_{ab} d_{ac} d_{cc} + 18d_{ab} d_{bb}^2 + 4d_{ab} d_{bb} d_{bc} +$
$28d_{ab} d_{bb} d_{cc} + 6d_{ab} d_{bc} d_{cc} + 16d_{ab} d_{cc}^2 + 2d_{ac}^2 d_{bb} + d_{ac}^2 d_{cc} + 16d_{ac} d_{bb}^2 + 6d_{ac} d_{bb} d_{bc} + 28d_{ac} d_{bb} d_{cc} +$
$4d_{ac} d_{bc} d_{cc} + 18d_{ac} d_{cc}^2 + 18d_{bb}^2 d_{bc} + 2d_{bb}^2 d_{cc} + 2d_{bb}^2 e_a + 2d_{bb}^2 e_c + d_{bb} d_{bc}^2 + 32d_{bb} d_{bc} d_{cc} + 2d_{bb} d_{cc}^2 +$
$8d_{bb} d_{cc} e_a + 4d_{bb} d_{cc} e_b + 4d_{bb} d_{cc} e_c + d_{bc}^2 d_{cc} + 18d_{bc} d_{cc}^2 + 2d_{cc}^2 e_a + 2d_{cc}^2 e_b$

$E_{3,7}$ $= 15d_{aa}^2 d_{ab} + 15d_{aa}^2 d_{ac} + 10d_{aa}^2 d_{bc} + 6d_{aa}^2 e_a + 7d_{aa}^2 e_b + 7d_{aa}^2 e_c + 5d_{aa} d_{ab}^2 + 12d_{aa} d_{ab} d_{ac} + 28d_{aa} d_{ab} d_{bb} +$ kept
$14d_{aa} d_{ab} d_{bc} + 26d_{aa} d_{ab} d_{cc} + 5d_{aa} d_{ac}^2 + 26d_{aa} d_{ac} d_{bb} + 14d_{aa} d_{ac} d_{bc} + 28d_{aa} d_{ac} d_{cc} + 26d_{aa} d_{bb} d_{bc} +$
$14d_{aa} d_{bb} e_a + 14d_{aa} d_{bb} e_b + 12d_{aa} d_{bb} e_c + 10d_{aa} d_{bc}^2 + 26d_{aa} d_{bc} d_{cc} + 14d_{aa} d_{cc} e_a + 12d_{aa} d_{cc} e_b +$
$14d_{aa} d_{cc} e_c + 5d_{ab}^2 d_{bb} + 10d_{ab}^2 d_{cc} + 14d_{ab} d_{ac} d_{bb} + 14d_{ab} d_{ac} d_{cc} + 15d_{ab} d_{bb}^2 + 12d_{ab} d_{bb} d_{bc} + 26d_{ab} d_{bb} d_{cc} +$
$14d_{ab} d_{bc} d_{cc} + 10d_{ab} d_{cc}^2 + 10d_{ac}^2 d_{bb} + 5d_{ac}^2 d_{cc} + 10d_{ac} d_{bb}^2 + 14d_{ac} d_{bb} d_{bc} + 26d_{ac} d_{bb} d_{cc} + 12d_{ac} d_{bc} d_{cc} +$
$15d_{ac} d_{cc}^2 + 15d_{bb}^2 d_{bc} + 7d_{bb}^2 e_a + 6d_{bb}^2 e_b + 7d_{bb}^2 e_c + 5d_{bb} d_{bc}^2 + 28d_{bb} d_{bc} d_{cc} + 12d_{bb} d_{cc} e_a + 14d_{bb} d_{cc} e_b +$
$14d_{bb} d_{cc} e_c + 5d_{bc}^2 d_{cc} + 15d_{bc} d_{cc}^2 + 7d_{cc}^2 e_a + 7d_{cc}^2 e_b + 6d_{cc}^2 e_c$

$E_{3,8}$ $= 5d_{aa}^2 d_{ab} + 5d_{aa}^2 d_{ac} + 10d_{aa}^2 d_{bc} + 12d_{aa}^2 e_a + 14d_{aa}^2 e_b + 14d_{aa}^2 e_c + 15d_{aa} d_{ab}^2 + 28d_{aa} d_{ab} d_{ac} +$ kept
$12d_{aa} d_{ab} d_{bb} + 26d_{aa} d_{ab} d_{bc} + 14d_{aa} d_{ab} d_{cc} + 15d_{aa} d_{ac}^2 + 14d_{aa} d_{ac} d_{bb} + 26d_{aa} d_{ac} d_{bc} + 12d_{aa} d_{ac} d_{cc} +$
$14d_{aa} d_{bb} d_{bc} + 28d_{aa} d_{bb} e_a + 28d_{aa} d_{bb} e_b + 24d_{aa} d_{bb} e_c + 10d_{aa} d_{bc}^2 + 14d_{aa} d_{bc} d_{cc} + 28d_{aa} d_{cc} e_a +$
$24d_{aa} d_{cc} e_b + 28d_{aa} d_{cc} e_c + 15d_{ab}^2 d_{bb} + 10d_{ab}^2 d_{cc} + 26d_{ab} d_{ac} d_{bb} + 26d_{ab} d_{ac} d_{cc} + 5d_{ab} d_{bb}^2 + 28d_{ab} d_{bb} d_{bc} +$
$14d_{ab} d_{bb} d_{cc} + 26d_{ab} d_{bc} d_{cc} + 10d_{ab} d_{cc}^2 + 10d_{ac}^2 d_{bb} + 15d_{ac}^2 d_{cc} + 10d_{ac} d_{bb}^2 + 26d_{ac} d_{bb} d_{bc} + 14d_{ac} d_{bb} d_{cc} +$
$28d_{ac} d_{bc} d_{cc} + 5d_{ac} d_{cc}^2 + 5d_{bb}^2 d_{bc} + 14d_{bb}^2 e_a + 12d_{bb}^2 e_b + 14d_{bb}^2 e_c + 15d_{bb} d_{bc}^2 + 12d_{bb} d_{bc} d_{cc} + 24d_{bb} d_{cc} e_a +$
$28d_{bb} d_{cc} e_b + 28d_{bb} d_{cc} e_c + 15d_{bc}^2 d_{cc} + 5d_{bc} d_{cc}^2 + 14d_{cc}^2 e_a + 14d_{cc}^2 e_b + 12d_{cc}^2 e_c$

$E_{3,9}$ $= d_{aa}^2 d_{ab} + d_{aa}^2 d_{ac} + 2d_{aa}^2 d_{bc} + 24d_{aa}^2 e_a + 14d_{aa}^2 e_b + 14d_{aa}^2 e_c + 18d_{aa} d_{ab}^2 + 32d_{aa} d_{ab} d_{ac} + 4d_{aa} d_{ab} d_{bb} +$ kept
$28d_{aa} d_{ab} d_{bc} + 6d_{aa} d_{ab} d_{cc} + 2d_{aa} d_{ab} e_a + 4d_{aa} d_{ab} e_b + 6d_{aa} d_{ab} e_c + 18d_{aa} d_{ac}^2 + 6d_{aa} d_{ac} d_{bb} + 28d_{aa} d_{ac} d_{bc} +$
$4d_{aa} d_{ac} d_{cc} + 2d_{aa} d_{ac} e_a + 6d_{aa} d_{ac} e_b + 4d_{aa} d_{ac} e_c + 6d_{aa} d_{bb} d_{bc} + 28d_{aa} d_{bb} e_a + 28d_{aa} d_{bb} e_b + 32d_{aa} d_{bb} e_c +$
$16d_{aa} d_{bc}^2 + 6d_{aa} d_{bc} d_{cc} + 4d_{aa} d_{bc} e_a + 6d_{aa} d_{bc} e_b + 6d_{aa} d_{bc} e_c + 28d_{aa} d_{cc} e_a + 32d_{aa} d_{cc} e_b + 28d_{aa} d_{cc} e_c +$
$2d_{ab}^2 d_{ac} + 18d_{ab}^2 d_{bb} + 2d_{ab}^2 d_{bc} + 16d_{ab}^2 d_{cc} + 2d_{ab} d_{ac}^2 + 28d_{ab} d_{ac} d_{bb} + 8d_{ab} d_{ac} d_{bc} + 28d_{ab} d_{ac} d_{cc} + d_{ab} d_{bb}^2 +$
$32d_{ab} d_{bb} d_{bc} + 6d_{ab} d_{bb} d_{cc} + 4d_{ab} d_{bb} e_a + 2d_{ab} d_{bb} e_b + 6d_{ab} d_{bb} e_c + 2d_{ab} d_{bc}^2 + 28d_{ab} d_{bc} d_{cc} + 2d_{ab} d_{cc}^2 + 6d_{ab} d_{cc} e_a +$
$6d_{ab} d_{cc} e_b + 4d_{ab} d_{cc} e_c + 16d_{ac}^2 d_{bb} + 2d_{ac}^2 d_{bc} + 18d_{ac}^2 d_{cc} + 2d_{ac} d_{bb}^2 + 28d_{ac} d_{bb} d_{bc} + 6d_{ac} d_{bb} d_{cc} + 6d_{ac} d_{bb} e_a +$
$4d_{ac} d_{bb} e_b + 6d_{ac} d_{bb} e_c + 2d_{ac} d_{bc}^2 + 32d_{ac} d_{bc} d_{cc} + d_{ac} d_{cc}^2 + 4d_{ac} d_{cc} e_a + 6d_{ac} d_{cc} e_b + 2d_{ac} d_{cc} e_c + d_{bb}^2 d_{bc} +$
$14d_{bb}^2 e_a + 24d_{bb}^2 e_b + 14d_{bb}^2 e_c + 18d_{bb} d_{bc}^2 + 4d_{bb} d_{bc} d_{cc} + 6d_{bb} d_{bc} e_a + 2d_{bb} d_{bc} e_b + 4d_{bb} d_{bc} e_c + 32d_{bb} d_{cc} e_a +$
$28d_{bb} d_{cc} e_b + 28d_{bb} d_{cc} e_c + 18d_{bc}^2 d_{cc} + d_{bc} d_{cc}^2 + 6d_{bc} d_{cc} e_a + 4d_{bc} d_{cc} e_b + 2d_{bc} d_{cc} e_c + 14d_{cc}^2 e_a + 14d_{cc}^2 e_b + 24d_{cc}^2 e_c$

$E_{3,10}$ $= 12d_{aa}^2e_a + 14d_{aa}^2e_b + 14d_{aa}^2e_c + 15d_{aa}d_{ab}^2 + 28d_{aa}d_{ab}d_{ac} + 26d_{aa}d_{ab}d_{bc} + 10d_{aa}d_{ab}e_a + 12d_{aa}d_{ab}e_b +$ kept
$14d_{aa}d_{ab}e_c + 15d_{aa}d_{ac}^2 + 26d_{aa}d_{ac}d_{bc} + 10d_{aa}d_{ac}e_a + 14d_{aa}d_{ac}e_b + 12d_{aa}d_{ac}e_c + 28d_{aa}d_{bb}e_a +$
$28d_{aa}d_{bb}e_b + 24d_{aa}d_{bb}e_c + 10d_{aa}d_{bc}^2 + 20d_{aa}d_{bc}e_a + 14d_{aa}d_{bc}e_b + 14d_{aa}d_{bc}e_c + 28d_{aa}d_{cc}e_a + 24d_{aa}d_{cc}e_b +$
$28d_{aa}d_{cc}e_c + 2d_{ab}^3 + 7d_{ab}^2d_{ac} + 15d_{ab}^2d_{bb} + 7d_{ab}^2d_{bc} + 10d_{ab}^2d_{cc} + 7d_{ab}d_{ac}^2 + 26d_{ab}d_{ac}d_{bb} + 12d_{ab}d_{ac}d_{bc} +$
$26d_{ab}d_{ac}d_{cc} + 28d_{ab}d_{bb}d_{bc} + 12d_{ab}d_{bb}e_a + 10d_{ab}d_{bb}e_b + 14d_{ab}d_{bb}e_c + 7d_{ab}d_{bc}^2 + 26d_{ab}d_{bc}d_{cc} + 14d_{ab}d_{cc}e_a +$
$14d_{ab}d_{cc}e_b + 20d_{ab}d_{cc}e_c + 2d_{ac}^3 + 10d_{ac}^2d_{bb} + 7d_{ac}^2d_{bc} + 15d_{ac}^2d_{cc} + 26d_{ac}d_{bb}d_{bc} + 14d_{ac}d_{bb}e_a +$
$20d_{ac}d_{bb}e_b + 14d_{ac}d_{bb}e_c + 7d_{ac}d_{bc}^2 + 28d_{ac}d_{bc}d_{cc} + 12d_{ac}d_{cc}e_a + 14d_{ac}d_{cc}e_b + 10d_{ac}d_{cc}e_c + 14d_{bb}^2e_a +$
$12d_{bb}^2e_b + 14d_{bb}^2e_c + 15d_{bb}d_{bc}^2 + 14d_{bb}d_{bc}e_a + 10d_{bb}d_{bc}e_b + 12d_{bb}d_{bc}e_c + 24d_{bb}d_{cc}e_a + 28d_{bb}d_{cc}e_b +$
$28d_{bb}d_{cc}e_c + 2d_{bc}^3 + 15d_{bc}^2d_{cc} + 14d_{bc}d_{cc}e_a + 12d_{bc}d_{cc}e_b + 10d_{bc}d_{cc}e_c + 14d_{cc}^2e_a + 14d_{cc}^2e_b + 12d_{cc}^2e_c$

$E_{3,11}$ $= 6d_{aa}^2e_a + 7d_{aa}^2e_b + 7d_{aa}^2e_c + 5d_{aa}d_{ab}^2 + 12d_{aa}d_{ab}d_{ac} + 14d_{aa}d_{ab}d_{bc} + 30d_{aa}d_{ab}e_a + 28d_{aa}d_{ab}e_b +$ kept
$26d_{aa}d_{ab}e_c + 5d_{aa}d_{ac}^2 + 14d_{aa}d_{ac}d_{bc} + 30d_{aa}d_{ac}e_a + 26d_{aa}d_{ac}e_b + 28d_{aa}d_{ac}e_c + 14d_{aa}d_{bb}e_a +$
$14d_{aa}d_{bb}e_b + 12d_{aa}d_{bb}e_c + 10d_{aa}d_{bc}^2 + 20d_{aa}d_{bc}e_a + 26d_{aa}d_{bc}e_b + 26d_{aa}d_{bc}e_c + 14d_{aa}d_{cc}e_a + 12d_{aa}d_{cc}e_b +$
$14d_{aa}d_{cc}e_c + 4d_{ab}^3 + 14d_{ab}^2d_{ac} + 5d_{ab}^2d_{bb} + 14d_{ab}^2d_{bc} + 10d_{ab}^2d_{cc} + 14d_{ab}d_{ac}^2 + 14d_{ab}d_{ac}d_{bb} + 24d_{ab}d_{ac}d_{bc} +$
$14d_{ab}d_{ac}d_{cc} + 12d_{ab}d_{bb}d_{bc} + 28d_{ab}d_{bb}e_a + 30d_{ab}d_{bb}e_b + 26d_{ab}d_{bb}e_c + 14d_{ab}d_{bc}^2 + 14d_{ab}d_{bc}d_{cc} +$
$26d_{ab}d_{cc}e_a + 26d_{ab}d_{cc}e_b + 20d_{ab}d_{cc}e_c + 4d_{ac}^3 + 10d_{ac}^2d_{bb} + 14d_{ac}^2d_{bc} + 5d_{ac}^2d_{cc} + 14d_{ac}d_{bb}d_{bc} +$
$26d_{ac}d_{bb}e_a + 20d_{ac}d_{bb}e_b + 26d_{ac}d_{bb}e_c + 14d_{ac}d_{bc}^2 + 12d_{ac}d_{bc}d_{cc} + 28d_{ac}d_{cc}e_a + 26d_{ac}d_{cc}e_b + 30d_{ac}d_{cc}e_c +$
$7d_{bb}^2e_a + 6d_{bb}^2e_b + 7d_{bb}^2e_c + 5d_{bb}d_{bc}^2 + 26d_{bb}d_{bc}e_a + 30d_{bb}d_{bc}e_b + 28d_{bb}d_{bc}e_c + 12d_{bb}d_{cc}e_a + 14d_{bb}d_{cc}e_b +$
$14d_{bb}d_{cc}e_c + 4d_{bc}^3 + 5d_{bc}^2d_{cc} + 26d_{bc}d_{cc}e_a + 28d_{bc}d_{cc}e_b + 30d_{bc}d_{cc}e_c + 7d_{cc}^2e_a + 7d_{cc}^2e_b + 6d_{cc}^2e_c$

$E_{3,12}$ $=$ (103 terms; see the full text file) kept

$E_{3,13}$ $= 30d_{aa}d_{ab}e_a + 28d_{aa}d_{ab}e_b + 26d_{aa}d_{ab}e_c + 30d_{aa}d_{ac}e_a + 26d_{aa}d_{ac}e_b + 28d_{aa}d_{ac}e_c + 20d_{aa}d_{bc}e_a + 26d_{aa}d_{bc}e_b +$ kept
$26d_{aa}d_{bc}e_c + 6d_{aa}e_a^2 + 14d_{aa}e_ae_b + 14d_{aa}e_ae_c + 7d_{aa}e_b^2 + 12d_{aa}e_be_c + 7d_{aa}e_c^2 + 4d_{ab}^3 + 14d_{ab}^2d_{ac} + 14d_{ab}^2d_{bc} +$
$5d_{ab}^2e_a + 5d_{ab}^2e_b + 10d_{ab}^2e_c + 14d_{ab}d_{ac}^2 + 24d_{ab}d_{ac}d_{bc} + 12d_{ab}d_{ac}e_a + 14d_{ab}d_{ac}e_b + 14d_{ab}d_{ac}e_c + 28d_{ab}d_{bb}e_a +$
$30d_{ab}d_{bb}e_b + 26d_{ab}d_{bb}e_c + 14d_{ab}d_{bc}^2 + 14d_{ab}d_{bc}e_a + 12d_{ab}d_{bc}e_b + 14d_{ab}d_{bc}e_c + 26d_{ab}d_{cc}e_a + 26d_{ab}d_{cc}e_b +$
$20d_{ab}d_{cc}e_c + 4d_{ac}^3 + 14d_{ac}^2d_{bc} + 5d_{ac}^2e_a + 10d_{ac}^2e_b + 5d_{ac}^2e_c + 26d_{ac}d_{bb}e_a + 20d_{ac}d_{bb}e_b + 26d_{ac}d_{bb}e_c + 14d_{ac}d_{bc}^2 +$
$14d_{ac}d_{bc}e_a + 14d_{ac}d_{bc}e_b + 12d_{ac}d_{bc}e_c + 28d_{ac}d_{cc}e_a + 26d_{ac}d_{cc}e_b + 30d_{ac}d_{cc}e_c + 26d_{bb}d_{bc}e_a + 30d_{bb}d_{bc}e_b +$
$28d_{bb}d_{bc}e_c + 7d_{bb}e_a^2 + 14d_{bb}e_ae_b + 12d_{bb}e_ae_c + 6d_{bb}e_b^2 + 14d_{bb}e_be_c + 7d_{bb}e_c^2 + 4d_{bc}^3 + 10d_{bc}^2e_a + 5d_{bc}^2e_b +$
$5d_{bc}^2e_c + 26d_{bc}d_{cc}e_a + 28d_{bc}d_{cc}e_b + 30d_{bc}d_{cc}e_c + 7d_{cc}e_a^2 + 12d_{cc}e_ae_b + 14d_{cc}e_ae_c + 7d_{cc}e_b^2 + 14d_{cc}e_be_c + 6d_{cc}e_c^2$

$E_{3,14}$ $= 10d_{aa}d_{ab}e_a + 12d_{aa}d_{ab}e_b + 14d_{aa}d_{ab}e_c + 10d_{aa}d_{ac}e_a + 14d_{aa}d_{ac}e_b + 12d_{aa}d_{ac}e_c + 20d_{aa}d_{bc}e_a +$ kept
$14d_{aa}d_{bc}e_b + 14d_{aa}d_{bc}e_c + 12d_{aa}e_a^2 + 28d_{aa}e_ae_b + 28d_{aa}e_ae_c + 14d_{aa}e_b^2 + 24d_{aa}e_be_c + 14d_{aa}e_c^2 + 2d_{ab}^3 +$
$7d_{ab}^2d_{ac} + 7d_{ab}^2d_{bc} + 15d_{ab}^2e_a + 15d_{ab}^2e_b + 10d_{ab}^2e_c + 7d_{ab}d_{ac}^2 + 12d_{ab}d_{ac}d_{bc} + 28d_{ab}d_{ac}e_a + 26d_{ab}d_{ac}e_b +$
$26d_{ab}d_{ac}e_c + 12d_{ab}d_{bb}e_a + 10d_{ab}d_{bb}e_b + 14d_{ab}d_{bb}e_c + 7d_{ab}d_{bc}^2 + 26d_{ab}d_{bc}e_a + 28d_{ab}d_{bc}e_b + 26d_{ab}d_{bc}e_c +$
$14d_{ab}d_{cc}e_a + 14d_{ab}d_{cc}e_b + 20d_{ab}d_{cc}e_c + 2d_{ac}^3 + 7d_{ac}^2d_{bc} + 15d_{ac}^2e_a + 10d_{ac}^2e_b + 15d_{ac}^2e_c + 14d_{ac}d_{bb}e_a +$
$20d_{ac}d_{bb}e_b + 14d_{ac}d_{bb}e_c + 7d_{ac}d_{bc}^2 + 26d_{ac}d_{bc}e_a + 26d_{ac}d_{bc}e_b + 28d_{ac}d_{bc}e_c + 12d_{ac}d_{cc}e_a + 14d_{ac}d_{cc}e_b +$
$10d_{ac}d_{cc}e_c + 14d_{bb}d_{bc}e_a + 10d_{bb}d_{bc}e_b + 12d_{bb}d_{bc}e_c + 14d_{bb}e_a^2 + 28d_{bb}e_ae_b + 24d_{bb}e_ae_c + 12d_{bb}e_b^2 +$
$28d_{bb}e_be_c + 14d_{bb}e_c^2 + 2d_{bc}^3 + 10d_{bc}^2e_a + 15d_{bc}^2e_b + 15d_{bc}^2e_c + 14d_{bc}d_{cc}e_a + 12d_{bc}d_{cc}e_b + 10d_{bc}d_{cc}e_c +$
$14d_{cc}e_a^2 + 24d_{cc}e_ae_b + 28d_{cc}e_ae_c + 14d_{cc}e_b^2 + 28d_{cc}e_be_c + 12d_{cc}e_c^2$

$E_{3,15}$	$= 2d_{aa}d_{ab}e_a + 4d_{aa}d_{ab}e_b + 6d_{aa}d_{ab}e_c + 2d_{aa}d_{ac}e_a + 6d_{aa}d_{ac}e_b + 4d_{aa}d_{ac}e_c + 4d_{aa}d_{bc}e_a + 6d_{aa}d_{bc}e_b +$ $6d_{aa}d_{bc}e_c + 24d_{aa}e_a^2 + 28d_{aa}e_ae_b + 28d_{aa}e_ae_c + 14d_{aa}e_b^2 + 32d_{aa}e_be_c + 14d_{aa}e_c^2 + 2d_{ab}^2d_{ac} + 2d_{ab}^2d_{bc} +$ $18d_{ab}^2e_a + 18d_{ab}^2e_b + 16d_{ab}^2e_c + 2d_{ab}d_{ac}^2 + 8d_{ab}d_{ac}d_{bc} + 32d_{ab}d_{ac}e_a + 28d_{ab}d_{ac}e_b + 28d_{ab}d_{ac}e_c +$ $4d_{ab}d_{bb}e_a + 2d_{ab}d_{bb}e_b + 6d_{ab}d_{bb}e_c + 2d_{ab}d_{bc}^2 + 28d_{ab}d_{bc}e_a + 32d_{ab}d_{bc}e_b + 28d_{ab}d_{bc}e_c + 6d_{ab}d_{cc}e_a +$ $6d_{ab}d_{cc}e_b + 4d_{ab}d_{cc}e_c + d_{ab}e_a^2 + 4d_{ab}e_ae_b + 6d_{ab}e_ae_c + d_{ab}e_b^2 + 6d_{ab}e_be_c + 2d_{ab}e_c^2 + 2d_{ac}^2d_{bc} + 18d_{ac}^2e_a +$ $16d_{ac}^2e_b + 18d_{ac}^2e_c + 6d_{ac}d_{bb}e_a + 4d_{ac}d_{bb}e_b + 6d_{ac}d_{bb}e_c + 2d_{ac}d_{bc}^2 + 28d_{ac}d_{bc}e_a + 28d_{ac}d_{bc}e_b +$ $32d_{ac}d_{bc}e_c + 4d_{ac}d_{cc}e_a + 6d_{ac}d_{cc}e_b + 2d_{ac}d_{cc}e_c + d_{ac}e_a^2 + 6d_{ac}e_ae_b + 4d_{ac}e_ae_c + 2d_{ac}e_b^2 + 6d_{ac}e_be_c +$ $d_{ac}e_c^2 + 6d_{bb}d_{bc}e_a + 2d_{bb}d_{bc}e_b + 4d_{bb}d_{bc}e_c + 14d_{bb}e_a^2 + 28d_{bb}e_ae_b + 32d_{bb}e_ae_c + 24d_{bb}e_b^2 + 28d_{bb}e_be_c +$ $14d_{bb}e_c^2 + 16d_{bc}^2e_a + 18d_{bc}^2e_b + 18d_{bc}^2e_c + 6d_{bc}d_{cc}e_a + 4d_{bc}d_{cc}e_b + 2d_{bc}d_{cc}e_c + 2d_{bc}e_a^2 + 6d_{bc}e_ae_b +$ $6d_{bc}e_ae_c + d_{bc}e_b^2 + 4d_{bc}e_be_c + d_{bc}e_c^2 + 14d_{cc}e_a^2 + 32d_{cc}e_ae_b + 28d_{cc}e_ae_c + 14d_{cc}e_b^2 + 28d_{cc}e_be_c + 24d_{cc}e_c^2$	kept
$E_{3,16}$	$= 12d_{aa}e_a^2 + 28d_{aa}e_ae_b + 28d_{aa}e_ae_c + 14d_{aa}e_b^2 + 24d_{aa}e_be_c + 14d_{aa}e_c^2 + 15d_{ab}^2e_a + 15d_{ab}^2e_b + 10d_{ab}^2e_c +$ $28d_{ab}d_{ac}e_a + 26d_{ab}d_{ac}e_b + 26d_{ab}d_{ac}e_c + 26d_{ab}d_{bc}e_a + 28d_{ab}d_{bc}e_b + 26d_{ab}d_{bc}e_c + 5d_{ab}e_a^2 + 12d_{ab}e_ae_b +$ $14d_{ab}e_ae_c + 5d_{ab}e_b^2 + 14d_{ab}e_be_c + 10d_{ab}e_c^2 + 15d_{ac}^2e_a + 10d_{ac}^2e_b + 15d_{ac}^2e_c + 26d_{ac}d_{bc}e_a + 26d_{ac}d_{bc}e_b +$ $28d_{ac}d_{bc}e_c + 5d_{ac}e_a^2 + 14d_{ac}e_ae_b + 12d_{ac}e_ae_c + 10d_{ac}e_b^2 + 14d_{ac}e_be_c + 5d_{ac}e_c^2 + 14d_{bb}e_a^2 + 28d_{bb}e_ae_b +$ $24d_{bb}e_ae_c + 12d_{bb}e_b^2 + 28d_{bb}e_be_c + 14d_{bb}e_c^2 + 10d_{bc}^2e_a + 15d_{bc}^2e_b + 15d_{bc}^2e_c + 10d_{bc}e_a^2 + 14d_{bc}e_ae_b +$ $14d_{bc}e_ae_c + 5d_{bc}e_b^2 + 12d_{bc}e_be_c + 5d_{bc}e_c^2 + 14d_{cc}e_a^2 + 24d_{cc}e_ae_b + 28d_{cc}e_ae_c + 14d_{cc}e_b^2 + 28d_{cc}e_be_c + 12d_{cc}e_c^2$	kept
$E_{3,17}$	$= 6d_{aa}e_a^2 + 14d_{aa}e_ae_b + 14d_{aa}e_ae_c + 7d_{aa}e_b^2 + 12d_{aa}e_be_c + 7d_{aa}e_c^2 + 5d_{ab}^2e_a + 5d_{ab}^2e_b + 10d_{ab}^2e_c +$ $12d_{ab}d_{ac}e_a + 14d_{ab}d_{ac}e_b + 14d_{ab}d_{ac}e_c + 14d_{ab}d_{bc}e_a + 12d_{ab}d_{bc}e_b + 14d_{ab}d_{bc}e_c + 15d_{ab}e_a^2 + 28d_{ab}e_ae_b +$ $26d_{ab}e_ae_c + 15d_{ab}e_b^2 + 26d_{ab}e_be_c + 10d_{ab}e_c^2 + 5d_{ac}^2e_a + 10d_{ac}^2e_b + 5d_{ac}^2e_c + 14d_{ac}d_{bc}e_a + 14d_{ac}d_{bc}e_b +$ $12d_{ac}d_{bc}e_c + 15d_{ac}e_a^2 + 26d_{ac}e_ae_b + 28d_{ac}e_ae_c + 10d_{ac}e_b^2 + 26d_{ac}e_be_c + 15d_{ac}e_c^2 + 7d_{bb}e_a^2 + 14d_{bb}e_ae_b +$ $12d_{bb}e_ae_c + 6d_{bb}e_b^2 + 14d_{bb}e_be_c + 7d_{bb}e_c^2 + 10d_{bc}^2e_a + 5d_{bc}^2e_b + 5d_{bc}^2e_c + 10d_{bc}e_a^2 + 26d_{bc}e_ae_b + 26d_{bc}e_ae_c +$ $15d_{bc}e_b^2 + 28d_{bc}e_be_c + 15d_{bc}e_c^2 + 7d_{cc}e_a^2 + 12d_{cc}e_ae_b + 14d_{cc}e_ae_c + 7d_{cc}e_b^2 + 14d_{cc}e_be_c + 6d_{cc}e_c^2$	kept
$E_{3,18}$	$= 4d_{aa}e_ae_b + 4d_{aa}e_ae_c + 2d_{aa}e_b^2 + 8d_{aa}e_be_c + 2d_{aa}e_c^2 + d_{ab}^2e_a + d_{ab}^2e_b + 2d_{ab}^2e_c + 4d_{ab}d_{ac}e_a + 6d_{ab}d_{ac}e_b +$ $6d_{ab}d_{ac}e_c + 6d_{ab}d_{bc}e_a + 4d_{ab}d_{bc}e_b + 6d_{ab}d_{bc}e_c + 18d_{ab}e_a^2 + 32d_{ab}e_ae_b + 28d_{ab}e_ae_c + 18d_{ab}e_b^2 + 28d_{ab}e_be_c +$ $16d_{ab}e_c^2 + d_{ac}^2e_a + 2d_{ac}^2e_b + d_{ac}^2e_c + 6d_{ac}d_{bc}e_a + 6d_{ac}d_{bc}e_b + 4d_{ac}d_{bc}e_c + 18d_{ac}e_a^2 + 28d_{ac}e_ae_b +$ $32d_{ac}e_ae_c + 16d_{ac}e_b^2 + 28d_{ac}e_be_c + 18d_{ac}e_c^2 + 2d_{bb}e_a^2 + 4d_{bb}e_ae_b + 8d_{bb}e_ae_c + 4d_{bb}e_be_c + 2d_{bb}e_c^2 + 2d_{bc}^2e_a +$ $d_{bc}^2e_b + d_{bc}^2e_c + 16d_{bc}e_a^2 + 28d_{bc}e_ae_b + 28d_{bc}e_ae_c + 18d_{bc}e_b^2 + 32d_{bc}e_be_c + 18d_{bc}e_c^2 + 2d_{cc}e_a^2 + 8d_{cc}e_ae_b +$ $4d_{cc}e_ae_c + 2d_{cc}e_b^2 + 4d_{cc}e_be_c + 2e_a^2e_b + 2e_a^2e_c + 2e_ae_b^2 + 8e_ae_be_c + 2e_ae_c^2 + 2e_b^2e_c + 2e_be_c^2$	kept
$E_{3,19}$	$= 15d_{ab}e_a^2 + 28d_{ab}e_ae_b + 26d_{ab}e_ae_c + 15d_{ab}e_b^2 + 26d_{ab}e_be_c + 10d_{ab}e_c^2 + 15d_{ac}e_a^2 + 26d_{ac}e_ae_b + 28d_{ac}e_ae_c +$ $10d_{ac}e_b^2 + 26d_{ac}e_be_c + 15d_{ac}e_c^2 + 10d_{bc}e_a^2 + 26d_{bc}e_ae_b + 26d_{bc}e_ae_c + 15d_{bc}e_b^2 + 28d_{bc}e_be_c + 15d_{bc}e_c^2 + 2e_a^3 +$ $7e_a^2e_b + 7e_a^2e_c + 7e_ae_b^2 + 12e_ae_be_c + 7e_ae_c^2 + 2e_b^3 + 7e_b^2e_c + 7e_be_c^2 + 2e_c^3$	kept
$E_{3,20}$	$= 5d_{ab}e_a^2 + 12d_{ab}e_ae_b + 14d_{ab}e_ae_c + 5d_{ab}e_b^2 + 14d_{ab}e_be_c + 10d_{ab}e_c^2 + 5d_{ac}e_a^2 + 14d_{ac}e_ae_b + 12d_{ac}e_ae_c +$ $10d_{ac}e_b^2 + 14d_{ac}e_be_c + 5d_{ac}e_c^2 + 10d_{bc}e_a^2 + 14d_{bc}e_ae_b + 14d_{bc}e_ae_c + 5d_{bc}e_b^2 + 12d_{bc}e_be_c + 5d_{bc}e_c^2 + 4e_a^3 +$ $14e_a^2e_b + 14e_a^2e_c + 14e_ae_b^2 + 24e_ae_be_c + 14e_ae_c^2 + 4e_b^3 + 14e_b^2e_c + 14e_be_c^2 + 4e_c^3$	dropped: $= E_{3,4} - 2E_{3,5} + 2E_{3,7} - E_{3,8} + E_{3,10} - 2E_{3,11} + 2E_{3,13} - E_{3,14} + E_{3,16} - 2E_{3,17} + 2E_{3,19}$
$E_{3,21}$	$= d_{ab}e_a^2 + 4d_{ab}e_ae_b + 6d_{ab}e_ae_c + d_{ab}e_b^2 + 6d_{ab}e_be_c + 2d_{ab}e_c^2 + d_{ac}e_a^2 + 6d_{ac}e_ae_b + 4d_{ac}e_ae_c + 2d_{ac}e_b^2 +$ $6d_{ac}e_be_c + d_{ac}e_c^2 + 2d_{bc}e_a^2 + 6d_{bc}e_ae_b + 6d_{bc}e_ae_c + d_{bc}e_b^2 + 4d_{bc}e_be_c + d_{bc}e_c^2 + 8e_a^3 + 14e_a^2e_b + 14e_a^2e_c +$ $14e_ae_b^2 + 32e_ae_be_c + 14e_ae_c^2 + 8e_b^3 + 14e_b^2e_c + 14e_be_c^2 + 8e_c^3$	kept
$E_{3,22}$	$= 4e_a^3 + 14e_a^2e_b + 14e_a^2e_c + 14e_ae_b^2 + 24e_ae_be_c + 14e_ae_c^2 + 4e_b^3 + 14e_b^2e_c + 14e_be_c^2 + 4e_c^3$	dropped: $= 2E_{3,1} - 2E_{3,5} + 2E_{3,7} - 2E_{3,11} + 2E_{3,13} - 2E_{3,17} + 2E_{3,19}$

$E_{3,23}$	$= 2e_a^3 + 7e_a^2e_b + 7e_a^2e_c + 7e_ae_b^2 + 12e_ae_be_c + 7e_ae_c^2 + 2e_b^3 + 7e_b^2e_c + 7e_be_c^2 + 2e_c^3$	dropped: $= E_{3,1} - E_{3,5} + E_{3,7} - E_{3,11} + E_{3,13} - E_{3,17} + E_{3,19}$
$E_{3,24}$	$= 2e_a^2e_b + 2e_a^2e_c + 2e_ae_b^2 + 8e_ae_be_c + 2e_ae_c^2 + 2e_b^2e_c + 2e_be_c^2$	kept

$c = 4$

output	expanded	safe reductions
$E_{4,0}$	$= d_{aa}^2d_{bb}^2 + 4d_{aa}^2d_{bb}d_{cc} + d_{aa}^2d_{cc}^2 + 4d_{aa}d_{bb}^2d_{cc} + 4d_{aa}d_{bb}d_{cc}^2 + d_{bb}^2d_{cc}^2$	kept
$E_{4,1}$	$= 3d_{aa}^3d_{bb} + 3d_{aa}^3d_{cc} + 4d_{aa}^2d_{bb}^2 + 10d_{aa}^2d_{bb}d_{cc} + 4d_{aa}^2d_{cc}^2 + 3d_{aa}d_{bb}^3 + 10d_{aa}d_{bb}^2d_{cc} + 10d_{aa}d_{bb}d_{cc}^2 + 3d_{aa}d_{cc}^3 + 3d_{bb}^3d_{cc} + 4d_{bb}^2d_{cc}^2 + 3d_{bb}d_{cc}^3$	kept
$E_{4,2}$	$= d_{aa}^4 + 7d_{aa}^3d_{bb} + 7d_{aa}^3d_{cc} + 13d_{aa}^2d_{bb}^2 + 22d_{aa}^2d_{bb}d_{cc} + 13d_{aa}^2d_{cc}^2 + 7d_{aa}d_{bb}^3 + 22d_{aa}d_{bb}^2d_{cc} + 22d_{aa}d_{bb}d_{cc}^2 + 7d_{aa}d_{cc}^3 + d_{bb}^4 + 7d_{bb}^3d_{cc} + 13d_{bb}^2d_{cc}^2 + 7d_{bb}d_{cc}^3 + d_{cc}^4$	kept
$E_{4,3}$	$= 4d_{aa}^4 + 13d_{aa}^3d_{bb} + 13d_{aa}^3d_{cc} + d_{aa}^2d_{ab}d_{bb} + 2d_{aa}^2d_{ab}d_{cc} + 2d_{aa}^2d_{ac}d_{bb} + d_{aa}^2d_{ac}d_{cc} + 16d_{aa}^2d_{bb}^2 + 3d_{aa}^2d_{bb}d_{bc} + 34d_{aa}^2d_{bb}d_{cc} + 3d_{aa}^2d_{bc}d_{cc} + 16d_{aa}^2d_{cc}^2 + d_{aa}d_{ab}d_{bb}^2 + 8d_{aa}d_{ab}d_{bb}d_{cc} + 3d_{aa}d_{ab}d_{cc}^2 + 3d_{aa}d_{ac}d_{bb}^2 + 8d_{aa}d_{ac}d_{bb}d_{cc} + d_{aa}d_{ac}d_{cc}^2 + 13d_{aa}d_{bb}^3 + 2d_{aa}d_{bb}^2d_{bc} + 34d_{aa}d_{bb}^2d_{cc} + 8d_{aa}d_{bb}d_{bc}d_{cc} + 34d_{aa}d_{bb}d_{cc}^2 + 2d_{aa}d_{bc}d_{cc}^2 + 13d_{aa}d_{cc}^3 + 2d_{ab}d_{bb}^2d_{cc} + 3d_{ab}d_{bb}d_{cc}^2 + 3d_{ac}d_{bb}^2d_{cc} + 2d_{ac}d_{bb}d_{cc}^2 + 4d_{bb}^4 + 13d_{bb}^3d_{cc} + d_{bb}^2d_{bc}d_{cc} + 16d_{bb}^2d_{cc}^2 + d_{bb}d_{bc}d_{cc}^2 + 13d_{bb}d_{cc}^3 + 4d_{cc}^4$	kept
$E_{4,4}$	$= 5d_{aa}^4 + d_{aa}^3d_{ab} + d_{aa}^3d_{ac} + 14d_{aa}^3d_{bb} + 4d_{aa}^3d_{bc} + 14d_{aa}^3d_{cc} + 7d_{aa}^2d_{ab}d_{bb} + 8d_{aa}^2d_{ab}d_{cc} + 8d_{aa}^2d_{ac}d_{bb} + 7d_{aa}^2d_{ac}d_{cc} + 22d_{aa}^2d_{bb}^2 + 12d_{aa}^2d_{bb}d_{bc} + 40d_{aa}^2d_{bb}d_{cc} + 12d_{aa}^2d_{bc}d_{cc} + 22d_{aa}^2d_{cc}^2 + 7d_{aa}d_{ab}d_{bb}^2 + 20d_{aa}d_{ab}d_{bb}d_{cc} + 12d_{aa}d_{ab}d_{cc}^2 + 12d_{aa}d_{ac}d_{bb}^2 + 20d_{aa}d_{ac}d_{bb}d_{cc} + 7d_{aa}d_{ac}d_{cc}^2 + 14d_{aa}d_{bb}^3 + 8d_{aa}d_{bb}^2d_{bc} + 40d_{aa}d_{bb}^2d_{cc} + 20d_{aa}d_{bb}d_{bc}d_{cc} + 40d_{aa}d_{bb}d_{cc}^2 + 8d_{aa}d_{bc}d_{cc}^2 + 14d_{aa}d_{cc}^3 + d_{ab}d_{bb}^3 + 8d_{ab}d_{bb}^2d_{cc} + 12d_{ab}d_{bb}d_{cc}^2 + 4d_{ab}d_{cc}^3 + 4d_{ac}d_{bb}^3 + 12d_{ac}d_{bb}^2d_{cc} + 8d_{ac}d_{bb}d_{cc}^2 + d_{ac}d_{cc}^3 + 5d_{bb}^4 + d_{bb}^3d_{bc} + 14d_{bb}^3d_{cc} + 7d_{bb}^2d_{bc}d_{cc} + 22d_{bb}^2d_{cc}^2 + 7d_{bb}d_{bc}d_{cc}^2 + 14d_{bb}d_{cc}^3 + d_{bc}d_{cc}^3 + 5d_{cc}^4$	kept
$E_{4,5}$	$= 4d_{aa}^4 + 5d_{aa}^3d_{ab} + 5d_{aa}^3d_{ac} + 13d_{aa}^3d_{bb} + 8d_{aa}^3d_{bc} + 13d_{aa}^3d_{cc} + 22d_{aa}^2d_{ab}d_{bb} + 26d_{aa}^2d_{ab}d_{cc} + 26d_{aa}^2d_{ac}d_{bb} + 22d_{aa}^2d_{ac}d_{cc} + 16d_{aa}^2d_{bb}^2 + 21d_{aa}^2d_{bb}d_{bc} + 34d_{aa}^2d_{bb}d_{cc} + 21d_{aa}^2d_{bc}d_{cc} + 16d_{aa}^2d_{cc}^2 + 22d_{aa}d_{ab}d_{bb}^2 + 44d_{aa}d_{ab}d_{bb}d_{cc} + 21d_{aa}d_{ab}d_{cc}^2 + 21d_{aa}d_{ac}d_{bb}^2 + 44d_{aa}d_{ac}d_{bb}d_{cc} + 22d_{aa}d_{ac}d_{cc}^2 + 13d_{aa}d_{bb}^3 + 26d_{aa}d_{bb}^2d_{bc} + 34d_{aa}d_{bb}^2d_{cc} + 44d_{aa}d_{bb}d_{bc}d_{cc} + 34d_{aa}d_{bb}d_{cc}^2 + 26d_{aa}d_{bc}d_{cc}^2 + 13d_{aa}d_{cc}^3 + 5d_{ab}d_{bb}^3 + 26d_{ab}d_{bb}^2d_{cc} + 21d_{ab}d_{bb}d_{cc}^2 + 8d_{ab}d_{cc}^3 + 8d_{ac}d_{bb}^3 + 21d_{ac}d_{bb}^2d_{cc} + 26d_{ac}d_{bb}d_{cc}^2 + 5d_{ac}d_{cc}^3 + 4d_{bb}^4 + 5d_{bb}^3d_{bc} + 13d_{bb}^3d_{cc} + 22d_{bb}^2d_{bc}d_{cc} + 16d_{bb}^2d_{cc}^2 + 22d_{bb}d_{bc}d_{cc}^2 + 13d_{bb}d_{cc}^3 + 5d_{bc}d_{cc}^3 + 4d_{cc}^4$	kept
$E_{4,6}$	$= d_{aa}^4 + 15d_{aa}^3d_{ab} + 15d_{aa}^3d_{ac} + 7d_{aa}^3d_{bb} + 12d_{aa}^3d_{bc} + 7d_{aa}^3d_{cc} + d_{aa}^2d_{ab}d_{ac} + 37d_{aa}^2d_{ab}d_{bb} + 2d_{aa}^2d_{ab}d_{bc} + 32d_{aa}^2d_{ab}d_{cc} + 32d_{aa}^2d_{ac}d_{bb} + 2d_{aa}^2d_{ac}d_{bc} + 37d_{aa}^2d_{ac}d_{cc} + 13d_{aa}^2d_{bb}^2 + 36d_{aa}^2d_{bb}d_{bc} + 22d_{aa}^2d_{bb}d_{cc} + 2d_{aa}^2d_{bb}e_b + 4d_{aa}^2d_{bb}e_c + d_{aa}^2d_{bc}^2 + 36d_{aa}^2d_{bc}d_{cc} + 13d_{aa}^2d_{cc}^2 + 4d_{aa}^2d_{cc}e_b + 2d_{aa}^2d_{cc}e_c + d_{aa}d_{ab}^2d_{bb} + 2d_{aa}d_{ab}^2d_{cc} + 5d_{aa}d_{ab}d_{ac}d_{bb} + 5d_{aa}d_{ab}d_{ac}d_{cc} + 37d_{aa}d_{ab}d_{bb}^2 + 5d_{aa}d_{ab}d_{bb}d_{bc} + 68d_{aa}d_{ab}d_{bb}d_{cc} + 9d_{aa}d_{ab}d_{bc}d_{cc} + 36d_{aa}d_{ab}d_{cc}^2 + 2d_{aa}d_{ac}^2d_{bb} + d_{aa}d_{ac}^2d_{cc} + 36d_{aa}d_{ac}d_{bb}^2 + 9d_{aa}d_{ac}d_{bb}d_{bc} + 68d_{aa}d_{ac}d_{bb}d_{cc} + 5d_{aa}d_{ac}d_{bc}d_{cc} + 37d_{aa}d_{ac}d_{cc}^2 + 7d_{aa}d_{bb}^3 + 32d_{aa}d_{bb}^2d_{bc} + 22d_{aa}d_{bb}^2d_{cc} + 2d_{aa}d_{bb}^2e_a + 4d_{aa}d_{bb}^2e_c + 2d_{aa}d_{bb}d_{bc}^2 + 68d_{aa}d_{bb}d_{bc}d_{cc} + 22d_{aa}d_{bb}d_{cc}^2 + 8d_{aa}d_{bb}d_{cc}e_a + 8d_{aa}d_{bb}d_{cc}e_b + 8d_{aa}d_{bb}d_{cc}e_c + 2d_{aa}d_{bc}^2d_{cc} + 32d_{aa}d_{bc}d_{cc}^2 + 7d_{aa}d_{cc}^3 + 2d_{aa}d_{cc}^2e_a + 4d_{aa}d_{cc}^2e_b + 2d_{ab}^2d_{bb}d_{cc} + d_{ab}^2d_{cc}^2 + 2d_{ab}d_{ac}d_{bb}^2 + 9d_{ab}d_{ac}d_{bb}d_{cc} + 2d_{ab}d_{ac}d_{cc}^2 + 15d_{ab}d_{bb}^3 + d_{ab}d_{bb}^2d_{bc} + 32d_{ab}d_{bb}^2d_{cc} + 5d_{ab}d_{bb}d_{bc}d_{cc} + 36d_{ab}d_{bb}d_{cc}^2 + 2d_{ab}d_{bc}d_{cc}^2 + 12d_{ab}d_{cc}^3 + d_{ac}^2d_{bb}^2 + 2d_{ac}^2d_{bb}d_{cc} + 12d_{ac}d_{bb}^3 + 2d_{ac}d_{bb}^2d_{bc} + 36d_{ac}d_{bb}^2d_{cc} + 5d_{ac}d_{bb}d_{bc}d_{cc} + 32d_{ac}d_{bb}d_{cc}^2 + d_{ac}d_{bc}d_{cc}^2 + 15d_{ac}d_{cc}^3 + d_{bb}^4 + 15d_{bb}^3d_{bc} + 7d_{bb}^3d_{cc} + 37d_{bb}^2d_{bc}d_{cc} + 13d_{bb}^2d_{cc}^2 + 4d_{bb}^2d_{cc}e_a + 2d_{bb}^2d_{cc}e_c + d_{bb}d_{bc}^2d_{cc} + 37d_{bb}d_{bc}d_{cc}^2 + 7d_{bb}d_{cc}^3 + 4d_{bb}d_{cc}^2e_a + 2d_{bb}d_{cc}^2e_b + 15d_{bc}d_{cc}^3 + d_{cc}^4$	kept

$E_{4,7}$ = (105 terms; see the full text file) kept

$E_{4,8}$ = (102 terms; see the full text file) kept

$E_{4,9}$ = (162 terms; see the full text file) kept

$E_{4,10}$ = (180 terms; see the full text file) kept

$E_{4,11}$ = (171 terms; see the full text file) kept

$E_{4,12}$ = (225 terms; see the full text file) kept

$E_{4,13}$ = (246 terms; see the full text file) kept

$E_{4,14}$ = (234 terms; see the full text file) kept

$E_{4,15}$ = (255 terms; see the full text file) kept

$E_{4,16}$ = (273 terms; see the full text file) kept

$E_{4,17}$ = (255 terms; see the full text file) kept

$E_{4,18}$ = (234 terms; see the full text file) kept

$E_{4,19}$ = (246 terms; see the full text file) kept

$E_{4,20}$ = (225 terms; see the full text file) kept

$E_{4,21}$ = (171 terms; see the full text file) kept

$E_{4,22}$ = (180 terms; see the full text file) kept

$E_{4,23}$ = (162 terms; see the full text file) kept

$E_{4,24}$ = (102 terms; see the full text file) kept

$E_{4,25}$ = (105 terms; see the full text file) kept

$E_{4,26}$ = $2d_{aa}e_ae_b^2 + 8d_{aa}e_ae_be_c + 2d_{aa}e_ae_c^2 + 4d_{aa}e_b^2e_c + 4d_{aa}e_be_c^2 + d_{ab}^2e_ae_b + 2d_{ab}^2e_ae_c + 2d_{ab}^2e_be_c + d_{ab}^2e_c^2 + d_{ab}d_{ac}e_a^2 + 5d_{ab}d_{ac}e_ae_b + 5d_{ab}d_{ac}e_ae_c + 2d_{ab}d_{ac}e_b^2 + 9d_{ab}d_{ac}e_be_c + 2d_{ab}d_{ac}e_c^2 + 2d_{ab}d_{bc}e_a^2 + 5d_{ab}d_{bc}e_ae_b + 9d_{ab}d_{bc}e_ae_c + d_{ab}d_{bc}e_b^2 + 5d_{ab}d_{bc}e_be_c + 2d_{ab}d_{bc}e_c^2 + 15d_{ab}e_a^3 + 37d_{ab}e_a^2e_b + 32d_{ab}e_a^2e_c + 37d_{ab}e_ae_b^2 + 68d_{ab}e_ae_be_c + 36d_{ab}e_ae_c^2 + 15d_{ab}e_b^3 + 32d_{ab}e_b^2e_c + 36d_{ab}e_be_c^2 + 12d_{ab}e_c^3 + 2d_{ac}^2e_ae_b + d_{ac}^2e_ae_c + d_{ac}^2e_b^2 + 2d_{ac}^2e_be_c + 2d_{ac}d_{bc}e_a^2 + 9d_{ac}d_{bc}e_ae_b + 5d_{ac}d_{bc}e_ae_c + 2d_{ac}d_{bc}e_b^2 + 5d_{ac}d_{bc}e_be_c + d_{ac}d_{bc}e_c^2 + 15d_{ac}e_a^3 + 32d_{ac}e_a^2e_b + 37d_{ac}e_a^2e_c + 36d_{ac}e_ae_b^2 + 68d_{ac}e_ae_be_c + 37d_{ac}e_ae_c^2 + 12d_{ac}e_b^3 + 36d_{ac}e_b^2e_c + 32d_{ac}e_be_c^2 + 15d_{ac}e_c^3 + 2d_{bb}e_a^2e_b + 4d_{bb}e_a^2e_c + 8d_{bb}e_ae_be_c + 4d_{bb}e_ae_c^2 + 2d_{bb}e_be_c^2 + d_{bc}^2e_a^2 + 2d_{bc}^2e_ae_b + 2d_{bc}^2e_ae_c + d_{bc}^2e_be_c + 12d_{bc}e_a^3 + 36d_{bc}e_a^2e_b + 36d_{bc}e_a^2e_c + 32d_{bc}e_ae_b^2 + 68d_{bc}e_ae_be_c + 32d_{bc}e_ae_c^2 + 15d_{bc}e_b^3 + 37d_{bc}e_b^2e_c + 37d_{bc}e_be_c^2 + 15d_{bc}e_c^3 + 4d_{cc}e_a^2e_b + 2d_{cc}e_a^2e_c + 4d_{cc}e_ae_b^2 + 8d_{cc}e_ae_be_c + 2d_{cc}e_b^2e_c + e_a^4 + 7e_a^3e_b + 7e_a^3e_c + 13e_a^2e_b^2 + 22e_a^2e_be_c + 13e_a^2e_c^2 + 7e_ae_b^3 + 22e_ae_b^2e_c + 22e_ae_be_c^2 + 7e_ae_c^3 + e_b^4 + 7e_b^3e_c + 13e_b^2e_c^2 + 7e_be_c^3 + e_c^4$ kept

$E_{4,27}$ = $5d_{ab}e_a^3 + 22d_{ab}e_a^2e_b + 26d_{ab}e_a^2e_c + 22d_{ab}e_ae_b^2 + 44d_{ab}e_ae_be_c + 21d_{ab}e_ae_c^2 + 5d_{ab}e_b^3 + 26d_{ab}e_b^2e_c + 21d_{ab}e_be_c^2 + 8d_{ab}e_c^3 + 5d_{ac}e_a^3 + 26d_{ac}e_a^2e_b + 22d_{ac}e_a^2e_c + 21d_{ac}e_ae_b^2 + 44d_{ac}e_ae_be_c + 22d_{ac}e_ae_c^2 + 8d_{ac}e_b^3 + 21d_{ac}e_b^2e_c + 26d_{ac}e_be_c^2 + 5d_{ac}e_c^3 + 8d_{bc}e_a^3 + 21d_{bc}e_a^2e_b + 21d_{bc}e_a^2e_c + 26d_{bc}e_ae_b^2 + 44d_{bc}e_ae_be_c + 26d_{bc}e_ae_c^2 + 5d_{bc}e_b^3 + 22d_{bc}e_b^2e_c + 22d_{bc}e_be_c^2 + 5d_{bc}e_c^3 + 4e_a^4 + 13e_a^3e_b + 13e_a^3e_c + 16e_a^2e_b^2 + 34e_a^2e_be_c + 16e_a^2e_c^2 + 13e_ae_b^3 + 34e_ae_b^2e_c + 34e_ae_be_c^2 + 13e_ae_c^3 + 4e_b^4 + 13e_b^3e_c + 16e_b^2e_c^2 + 13e_be_c^3 + 4e_c^4$ kept

$E_{4,28}$ = $d_{ab}e_a^3 + 7d_{ab}e_a^2e_b + 8d_{ab}e_a^2e_c + 7d_{ab}e_ae_b^2 + 20d_{ab}e_ae_be_c + 12d_{ab}e_ae_c^2 + d_{ab}e_b^3 + 8d_{ab}e_b^2e_c + 12d_{ab}e_be_c^2 + 4d_{ab}e_c^3 + d_{ac}e_a^3 + 8d_{ac}e_a^2e_b + 7d_{ac}e_a^2e_c + 12d_{ac}e_ae_b^2 + 20d_{ac}e_ae_be_c + 7d_{ac}e_ae_c^2 + 4d_{ac}e_b^3 + 12d_{ac}e_b^2e_c + 8d_{ac}e_be_c^2 + d_{ac}e_c^3 + 4d_{bc}e_a^3 + 12d_{bc}e_a^2e_b + 12d_{bc}e_a^2e_c + 8d_{bc}e_ae_b^2 + 20d_{bc}e_ae_be_c + 8d_{bc}e_ae_c^2 + d_{bc}e_b^3 + 7d_{bc}e_b^2e_c + 7d_{bc}e_be_c^2 + d_{bc}e_c^3 + 5e_a^4 + 14e_a^3e_b + 14e_a^3e_c + 22e_a^2e_b^2 + 40e_a^2e_be_c + 22e_a^2e_c^2 + 14e_ae_b^3 + 40e_ae_b^2e_c + 40e_ae_be_c^2 + 14e_ae_c^3 + 5e_b^4 + 14e_b^3e_c + 22e_b^2e_c^2 + 14e_be_c^3 + 5e_c^4$ kept

$E_{4,29}$	$= d_{ab}e_a^2e_b + 2d_{ab}e_a^2e_c + d_{ab}e_ae_b^2 + 8d_{ab}e_ae_be_c + 3d_{ab}e_ae_c^2 + 2d_{ab}e_b^2e_c + 3d_{ab}e_be_c^2 + 2d_{ac}e_a^2e_b + d_{ac}e_a^2e_c + 3d_{ac}e_ae_b^2 + 8d_{ac}e_ae_be_c + d_{ac}e_ae_c^2 + 3d_{ac}e_b^2e_c + 2d_{ac}e_be_c^2 + 3d_{bc}e_a^2e_b + 3d_{bc}e_a^2e_c + 2d_{bc}e_ae_b^2 + 8d_{bc}e_ae_be_c + 2d_{bc}e_ae_c^2 + d_{bc}e_b^2e_c + d_{bc}e_be_c^2 + 4e_a^4 + 13e_a^3e_b + 13e_a^3e_c + 16e_a^2e_b^2 + 34e_a^2e_be_c + 16e_a^2e_c^2 + 13e_ae_b^3 + 34e_ae_b^2e_c + 34e_ae_be_c^2 + 13e_ae_c^3 + 4e_b^4 + 13e_b^3e_c + 16e_b^2e_c^2 + 13e_be_c^3 + 4e_c^4$	dropped: $= E_{4,3} - E_{4,5} + E_{4,9} - E_{4,11} + E_{4,15} - E_{4,17} + E_{4,21} - E_{4,23} + E_{4,27}$
$E_{4,30}$	$= e_a^4 + 7e_a^3e_b + 7e_a^3e_c + 13e_a^2e_b^2 + 22e_a^2e_be_c + 13e_a^2e_c^2 + 7e_ae_b^3 + 22e_ae_b^2e_c + 22e_ae_be_c^2 + 7e_ae_c^3 + e_b^4 + 7e_b^3e_c + 13e_b^2e_c^2 + 7e_be_c^3 + e_c^4$	kept
$E_{4,31}$	$= 3e_a^3e_b + 3e_a^3e_c + 4e_a^2e_b^2 + 10e_a^2e_be_c + 4e_a^2e_c^2 + 3e_ae_b^3 + 10e_ae_b^2e_c + 10e_ae_be_c^2 + 3e_ae_c^3 + 3e_b^3e_c + 4e_b^2e_c^2 + 3e_be_c^3$	kept
$E_{4,32}$	$= e_a^2e_b^2 + 4e_a^2e_be_c + e_a^2e_c^2 + 4e_ae_b^2e_c + 4e_ae_be_c^2 + e_b^2e_c^2$	dropped: $= E_{4,0} - E_{4,2} + E_{4,6} - E_{4,8} + E_{4,12} - E_{4,14} + E_{4,18} - E_{4,20} + E_{4,24} - E_{4,26} + E_{4,30}$

$c = 5$

output	expanded	safe reductions
$E_{5,0}$	$= 2d_{aa}^2d_{bb}^2d_{cc} + 2d_{aa}^2d_{bb}d_{cc}^2 + 2d_{aa}d_{bb}^2d_{cc}^2$	kept
$E_{5,1}$	$= d_{aa}^3d_{bb}^2 + 4d_{aa}^3d_{bb}d_{cc} + d_{aa}^3d_{cc}^2 + d_{aa}^2d_{bb}^3 + 4d_{aa}^2d_{bb}^2d_{cc} + 4d_{aa}^2d_{bb}d_{cc}^2 + d_{aa}^2d_{cc}^3 + 4d_{aa}d_{bb}^3d_{cc} + 4d_{aa}d_{bb}^2d_{cc}^2 + 4d_{aa}d_{bb}d_{cc}^3 + d_{bb}^3d_{cc}^2 + d_{bb}^2d_{cc}^3$	kept
$E_{5,2}$	$= d_{aa}^4d_{bb} + d_{aa}^4d_{cc} + 4d_{aa}^3d_{bb}^2 + 8d_{aa}^3d_{bb}d_{cc} + 4d_{aa}^3d_{cc}^2 + 4d_{aa}^2d_{bb}^3 + 12d_{aa}^2d_{bb}^2d_{cc} + 12d_{aa}^2d_{bb}d_{cc}^2 + 4d_{aa}^2d_{cc}^3 + d_{aa}d_{bb}^4 + 8d_{aa}d_{bb}^3d_{cc} + 12d_{aa}d_{bb}^2d_{cc}^2 + 8d_{aa}d_{bb}d_{cc}^3 + d_{aa}d_{cc}^4 + d_{bb}^4d_{cc} + 4d_{bb}^3d_{cc}^2 + 4d_{bb}^2d_{cc}^3 + d_{bb}d_{cc}^4$	kept
$E_{5,3}$	$= 5d_{aa}^4d_{bb}+5d_{aa}^4d_{cc}+7d_{aa}^3d_{bb}^2+12d_{aa}^3d_{bb}d_{cc}+7d_{aa}^3d_{cc}^2+2d_{aa}^2d_{ab}d_{bb}d_{cc}+d_{aa}^2d_{ab}d_{cc}^2+d_{aa}^2d_{ac}d_{bb}^2+2d_{aa}^2d_{ac}d_{bb}d_{cc}+7d_{aa}^2d_{bb}^3 + d_{aa}^2d_{bb}^2d_{bc} + 24d_{aa}^2d_{bb}^2d_{cc} + 4d_{aa}^2d_{bb}d_{bc}d_{cc} + 24d_{aa}^2d_{bb}d_{cc}^2 + d_{aa}^2d_{bc}d_{cc}^2 + 7d_{aa}^2d_{cc}^3 + 2d_{aa}d_{ab}d_{bb}^2d_{cc} + 4d_{aa}d_{ab}d_{bb}d_{cc}^2 + 4d_{aa}d_{ac}d_{bb}^2d_{cc} + 2d_{aa}d_{ac}d_{bb}d_{cc}^2 + 5d_{aa}d_{bb}^4 + 12d_{aa}d_{bb}^3d_{cc} + 2d_{aa}d_{bb}^2d_{bc}d_{cc} + 24d_{aa}d_{bb}^2d_{cc}^2 + 2d_{aa}d_{bb}d_{bc}d_{cc}^2 + 12d_{aa}d_{bb}d_{cc}^3 + 5d_{aa}d_{cc}^4 + d_{ab}d_{bb}^2d_{cc}^2 + d_{ac}d_{bb}^2d_{cc}^2 + 5d_{bb}^4d_{cc} + 7d_{bb}^3d_{cc}^2 + 7d_{bb}^2d_{cc}^3 + 5d_{bb}d_{cc}^4$	kept
$E_{5,4}$	$= 2d_{aa}^5 + 5d_{aa}^4d_{bb} + 5d_{aa}^4d_{cc} + d_{aa}^3d_{ab}d_{bb} + d_{aa}^3d_{ab}d_{cc} + d_{aa}^3d_{ac}d_{bb} + d_{aa}^3d_{ac}d_{cc} + 12d_{aa}^3d_{bb}^2 + 4d_{aa}^3d_{bb}d_{bc} + 24d_{aa}^3d_{bb}d_{cc} + 4d_{aa}^3d_{bc}d_{cc} + 12d_{aa}^3d_{cc}^2 + 2d_{aa}^2d_{ab}d_{bb}^2 + 9d_{aa}^2d_{ab}d_{bb}d_{cc} + 4d_{aa}^2d_{ab}d_{cc}^2 + 4d_{aa}^2d_{ac}d_{bb}^2 + 9d_{aa}^2d_{ac}d_{bb}d_{cc} + 2d_{aa}^2d_{ac}d_{cc}^2 + 12d_{aa}^2d_{bb}^3 + 4d_{aa}^2d_{bb}^2d_{bc} + 30d_{aa}^2d_{bb}^2d_{cc} + 10d_{aa}^2d_{bb}d_{bc}d_{cc} + 30d_{aa}^2d_{bb}d_{cc}^2 + 4d_{aa}^2d_{bc}d_{cc}^2 + 12d_{aa}^2d_{cc}^3 + d_{aa}d_{ab}d_{bb}^3 + 9d_{aa}d_{ab}d_{bb}^2d_{cc} + 10d_{aa}d_{ab}d_{bb}d_{cc}^2 + 4d_{aa}d_{ab}d_{cc}^3 + 4d_{aa}d_{ac}d_{bb}^3 + 10d_{aa}d_{ac}d_{bb}^2d_{cc} + 9d_{aa}d_{ac}d_{bb}d_{cc}^2 + d_{aa}d_{ac}d_{cc}^3 + 5d_{aa}d_{bb}^4 + d_{aa}d_{bb}^3d_{bc} + 24d_{aa}d_{bb}^3d_{cc} + 9d_{aa}d_{bb}^2d_{bc}d_{cc} + 30d_{aa}d_{bb}^2d_{cc}^2 + 9d_{aa}d_{bb}d_{bc}d_{cc}^2 + 24d_{aa}d_{bb}d_{cc}^3 + d_{aa}d_{bc}d_{cc}^3 + 5d_{aa}d_{cc}^4 + d_{ab}d_{bb}^3d_{cc} + 4d_{ab}d_{bb}^2d_{cc}^2 + 4d_{ab}d_{bb}d_{cc}^3 + 4d_{ac}d_{bb}^3d_{cc} + 4d_{ac}d_{bb}^2d_{cc}^2 + d_{ac}d_{bb}d_{cc}^3 + 2d_{bb}^5 + 5d_{bb}^4d_{cc} + d_{bb}^3d_{bc}d_{cc} + 12d_{bb}^3d_{cc}^2 + 2d_{bb}^2d_{bc}d_{cc}^2 + 12d_{bb}^2d_{cc}^3 + d_{bb}d_{bc}d_{cc}^3 + 5d_{bb}d_{cc}^4 + 2d_{cc}^5$	kept
$E_{5,5}$	$= 2d_{aa}^5 + 8d_{aa}^4d_{bb} + 2d_{aa}^4d_{bc} + 8d_{aa}^4d_{cc} + 4d_{aa}^3d_{ab}d_{bb} + 8d_{aa}^3d_{ab}d_{cc} + 8d_{aa}^3d_{ac}d_{bb} + 4d_{aa}^3d_{ac}d_{cc} + 12d_{aa}^3d_{bb}^2 + 8d_{aa}^3d_{bb}d_{bc} + 24d_{aa}^3d_{bb}d_{cc} + 8d_{aa}^3d_{bc}d_{cc} + 12d_{aa}^3d_{cc}^2 + 12d_{aa}^2d_{ab}d_{bb}^2 + 24d_{aa}^2d_{ab}d_{bb}d_{cc} + 12d_{aa}^2d_{ab}d_{cc}^2 + 12d_{aa}^2d_{ac}d_{bb}^2 + 24d_{aa}^2d_{ac}d_{bb}d_{cc} + 12d_{aa}^2d_{ac}d_{cc}^2 + 12d_{aa}^2d_{bb}^3 + 12d_{aa}^2d_{bb}^2d_{bc} + 36d_{aa}^2d_{bb}^2d_{cc} + 24d_{aa}^2d_{bb}d_{bc}d_{cc} + 36d_{aa}^2d_{bb}d_{cc}^2 + 12d_{aa}^2d_{bc}d_{cc}^2 + 12d_{aa}^2d_{cc}^3 + 4d_{aa}d_{ab}d_{bb}^3 + 24d_{aa}d_{ab}d_{bb}^2d_{cc} + 24d_{aa}d_{ab}d_{bb}d_{cc}^2 + 8d_{aa}d_{ab}d_{cc}^3 + 8d_{aa}d_{ac}d_{bb}^3 + 24d_{aa}d_{ac}d_{bb}^2d_{cc} + 24d_{aa}d_{ac}d_{bb}d_{cc}^2 + 4d_{aa}d_{ac}d_{cc}^3 + 8d_{aa}d_{bb}^4 + 8d_{aa}d_{bb}^3d_{bc} + 24d_{aa}d_{bb}^3d_{cc} + 24d_{aa}d_{bb}^2d_{bc}d_{cc} + 36d_{aa}d_{bb}^2d_{cc}^2 + 24d_{aa}d_{bb}d_{bc}d_{cc}^2 + 24d_{aa}d_{bb}d_{cc}^3 + 8d_{aa}d_{bc}d_{cc}^3 + 8d_{aa}d_{cc}^4 + 8d_{ab}d_{bb}^3d_{cc} + 12d_{ab}d_{bb}^2d_{cc}^2 + 8d_{ab}d_{bb}d_{cc}^3 + 2d_{ab}d_{cc}^4 + 2d_{ac}d_{bb}^4 + 8d_{ac}d_{bb}^3d_{cc} + 12d_{ac}d_{bb}^2d_{cc}^2 + 8d_{ac}d_{bb}d_{cc}^3 + 2d_{bb}^5 + 8d_{bb}^4d_{cc} + 4d_{bb}^3d_{bc}d_{cc} + 12d_{bb}^3d_{cc}^2 + 12d_{bb}^2d_{bc}d_{cc}^2 + 12d_{bb}^2d_{cc}^3 + 4d_{bb}d_{bc}d_{cc}^3 + 8d_{bb}d_{cc}^4 + 2d_{cc}^5$	kept

$E_{5,6}$ = (105 terms; see the full text file) kept

$E_{5,7}$ = (144 terms; see the full text file) kept

$E_{5,8}$ = (165 terms; see the full text file) kept

$E_{5,9}$ = (228 terms; see the full text file) kept

$E_{5,10}$ = (285 terms; see the full text file) kept

$E_{5,11}$ = (288 terms; see the full text file) kept

$E_{5,12}$ = (387 terms; see the full text file) kept

$E_{5,13}$ = (447 terms; see the full text file) kept

$E_{5,14}$ = (456 terms; see the full text file) kept

$E_{5,15}$ = (549 terms; see the full text file) kept

$E_{5,16}$ = (591 terms; see the full text file) kept

$E_{5,17}$ = (567 terms; see the full text file) kept

$E_{5,18}$ = (657 terms; see the full text file) kept

$E_{5,19}$ = (678 terms; see the full text file) kept

$E_{5,20}$ = (627 terms; see the full text file) kept

$E_{5,21}$ = (678 terms; see the full text file) kept

$E_{5,22}$ = (657 terms; see the full text file) kept

$E_{5,23}$ = (567 terms; see the full text file) kept

$E_{5,24}$ = (591 terms; see the full text file) kept

$E_{5,25}$ = (549 terms; see the full text file) kept

$E_{5,26}$ = (456 terms; see the full text file) kept

$E_{5,27}$ = (447 terms; see the full text file) kept

$E_{5,28}$ = (387 terms; see the full text file) kept

$E_{5,29}$ = (288 terms; see the full text file) kept

$E_{5,30}$ = (285 terms; see the full text file) kept

$E_{5,31}$ = (228 terms; see the full text file) kept

$E_{5,32}$ = (165 terms; see the full text file) kept

$E_{5,33}$ = (144 terms; see the full text file) kept

$E_{5,34}$ = (105 terms; see the full text file) kept

$E_{5,35}$ $= 4d_{ab}e_a^3e_b + 8d_{ab}e_a^3e_c + 12d_{ab}e_a^2e_b^2 + 24d_{ab}e_a^2e_be_c + 12d_{ab}e_a^2e_c^2 + 4d_{ab}e_ae_b^3 + 24d_{ab}e_ae_b^2e_c + 24d_{ab}e_ae_be_c^2 +$ kept
$8d_{ab}e_ae_c^3 + 8d_{ab}e_b^3e_c + 12d_{ab}e_b^2e_c^2 + 8d_{ab}e_be_c^3 + 2d_{ab}e_c^4 + 8d_{ac}e_a^3e_b + 4d_{ac}e_a^3e_c + 12d_{ac}e_a^2e_b^2 + 24d_{ac}e_a^2e_be_c +$
$12d_{ac}e_a^2e_c^2 + 8d_{ac}e_ae_b^3 + 24d_{ac}e_ae_b^2e_c + 24d_{ac}e_ae_be_c^2 + 4d_{ac}e_ae_c^3 + 2d_{ac}e_b^4 + 8d_{ac}e_b^3e_c + 12d_{ac}e_b^2e_c^2 +$
$8d_{ac}e_be_c^3 + 2d_{bc}e_a^4 + 8d_{bc}e_a^3e_b + 8d_{bc}e_a^3e_c + 12d_{bc}e_a^2e_b^2 + 24d_{bc}e_a^2e_be_c + 12d_{bc}e_a^2e_c^2 + 8d_{bc}e_ae_b^3 +$
$24d_{bc}e_ae_b^2e_c + 24d_{bc}e_ae_be_c^2 + 8d_{bc}e_ae_c^3 + 4d_{bc}e_b^3e_c + 12d_{bc}e_b^2e_c^2 + 4d_{bc}e_be_c^3 + 2e_a^5 + 8e_a^4e_b + 8e_a^4e_c + 12e_a^3e_b^2 +$
$24e_a^3e_be_c + 12e_a^3e_c^2 + 12e_a^2e_b^3 + 36e_a^2e_b^2e_c + 36e_a^2e_be_c^2 + 12e_a^2e_c^3 + 8e_ae_b^4 + 24e_ae_b^3e_c + 36e_ae_b^2e_c^2 + 24e_ae_be_c^3 +$
$8e_ae_c^4 + 2e_b^5 + 8e_b^4e_c + 12e_b^3e_c^2 + 12e_b^2e_c^3 + 8e_be_c^4 + 2e_c^5$

$E_{5,36}$	$= d_{ab}e_a^3e_b + d_{ab}e_a^3e_c + 2d_{ab}e_a^2e_b^2 + 9d_{ab}e_a^2e_be_c + 4d_{ab}e_a^2e_c^2 + d_{ab}e_ae_b^3 + 9d_{ab}e_ae_b^2e_c + 10d_{ab}e_ae_be_c^2 + 4d_{ab}e_ae_c^3 + d_{ab}e_b^3e_c + 4d_{ab}e_b^2e_c^2 + 4d_{ab}e_be_c^3 + d_{ac}e_a^3e_b + d_{ac}e_a^3e_c + 4d_{ac}e_a^2e_b^2 + 9d_{ac}e_a^2e_be_c + 2d_{ac}e_a^2e_c^2 + 4d_{ac}e_ae_b^3 + 10d_{ac}e_ae_b^2e_c + 9d_{ac}e_ae_be_c^2 + d_{ac}e_ae_c^3 + 4d_{ac}e_b^3e_c + 4d_{ac}e_b^2e_c^2 + d_{ac}e_be_c^3 + 4d_{bc}e_a^3e_b + 4d_{bc}e_a^3e_c + 4d_{bc}e_a^2e_b^2 + 10d_{bc}e_a^2e_be_c + 4d_{bc}e_a^2e_c^2 + d_{bc}e_ae_b^3 + 9d_{bc}e_ae_b^2e_c + 9d_{bc}e_ae_be_c^2 + d_{bc}e_ae_c^3 + d_{bc}e_b^3e_c + 2d_{bc}e_b^2e_c^2 + d_{bc}e_be_c^3 + 2e_a^5 + 5e_a^4e_b + 5e_a^4e_c + 12e_a^3e_b^2 + 24e_a^3e_be_c + 12e_a^3e_c^2 + 12e_a^2e_b^3 + 30e_a^2e_b^2e_c + 30e_a^2e_be_c^2 + 12e_a^2e_c^3 + 5e_ae_b^4 + 24e_ae_b^3e_c + 30e_ae_b^2e_c^2 + 24e_ae_be_c^3 + 5e_ae_c^4 + 2e_b^5 + 5e_b^4e_c + 12e_b^3e_c^2 + 12e_b^2e_c^3 + 5e_be_c^4 + 2e_c^5$	kept
$E_{5,37}$	$= 2d_{ab}e_a^2e_be_c + d_{ab}e_a^2e_c^2 + 2d_{ab}e_ae_b^2e_c + 4d_{ab}e_ae_be_c^2 + d_{ab}e_b^2e_c^2 + d_{ac}e_a^2e_b^2 + 2d_{ac}e_a^2e_be_c + 4d_{ac}e_ae_b^2e_c + 2d_{ac}e_ae_be_c^2 + d_{ac}e_b^2e_c^2 + d_{bc}e_a^2e_b^2 + 4d_{bc}e_a^2e_be_c + d_{bc}e_a^2e_c^2 + 2d_{bc}e_ae_b^2e_c + 2d_{bc}e_ae_be_c^2 + 5e_a^4e_b + 5e_a^4e_c + 7e_a^3e_b^2 + 12e_a^3e_be_c + 7e_a^3e_c^2 + 7e_a^2e_b^3 + 24e_a^2e_b^2e_c + 24e_a^2e_be_c^2 + 7e_a^2e_c^3 + 5e_ae_b^4 + 12e_ae_b^3e_c + 24e_ae_b^2e_c^2 + 12e_ae_be_c^3 + 5e_ae_c^4 + 5e_b^4e_c + 7e_b^3e_c^2 + 7e_b^2e_c^3 + 5e_be_c^4$	kept
$E_{5,38}$	$= e_a^4e_b + e_a^4e_c + 4e_a^3e_b^2 + 8e_a^3e_be_c + 4e_a^3e_c^2 + 4e_a^2e_b^3 + 12e_a^2e_b^2e_c + 12e_a^2e_be_c^2 + 4e_a^2e_c^3 + e_ae_b^4 + 8e_ae_b^3e_c + 12e_ae_b^2e_c^2 + 8e_ae_be_c^3 + e_ae_c^4 + e_b^4e_c + 4e_b^3e_c^2 + 4e_b^2e_c^3 + e_be_c^4$	kept
$E_{5,39}$	$= e_a^3e_b^2 + 4e_a^3e_be_c + e_a^3e_c^2 + e_a^2e_b^3 + 4e_a^2e_b^2e_c + 4e_a^2e_be_c^2 + e_a^2e_c^3 + 4e_ae_b^3e_c + 4e_ae_b^2e_c^2 + 4e_ae_be_c^3 + e_b^3e_c^2 + e_b^2e_c^3$	dropped: $= E_{5,1} - E_{5,3} + E_{5,7} - E_{5,9} + E_{5,13} - E_{5,15} + E_{5,19} - E_{5,21} + E_{5,25} - E_{5,27} + E_{5,31} - E_{5,33} + E_{5,37}$
$E_{5,40}$	$= 2e_a^2e_b^2e_c + 2e_a^2e_be_c^2 + 2e_ae_b^2e_c^2$	dropped: $= E_{5,0} - E_{5,4} + E_{5,6} - E_{5,10} + E_{5,12} - E_{5,16} + E_{5,18} - E_{5,22} + E_{5,24} - E_{5,28} + E_{5,30} - E_{5,34} + E_{5,36}$

$c = 6$

output	expanded	safe reductions
$E_{6,0}$	$= d_{aa}^2d_{bb}^2d_{cc}^2$	kept
$E_{6,1}$	$= d_{aa}^3d_{bb}^2d_{cc} + d_{aa}^3d_{bb}d_{cc}^2 + d_{aa}^2d_{bb}^3d_{cc} + d_{aa}^2d_{bb}d_{cc}^3 + d_{aa}d_{bb}^3d_{cc}^2 + d_{aa}d_{bb}^2d_{cc}^3$	kept
$E_{6,2}$	$= d_{aa}^4d_{bb}d_{cc} + d_{aa}^3d_{bb}^3 + 2d_{aa}^3d_{bb}^2d_{cc} + 2d_{aa}^3d_{bb}d_{cc}^2 + d_{aa}^3d_{cc}^3 + 2d_{aa}^2d_{bb}^3d_{cc} + 3d_{aa}^2d_{bb}^2d_{cc}^2 + 2d_{aa}^2d_{bb}d_{cc}^3 + d_{aa}d_{bb}^4d_{cc} + 2d_{aa}d_{bb}^3d_{cc}^2 + 2d_{aa}d_{bb}^2d_{cc}^3 + d_{aa}d_{bb}d_{cc}^4 + d_{bb}^3d_{cc}^3$	kept
$E_{6,3}$	$= 2d_{aa}^4d_{bb}^2 + 2d_{aa}^4d_{bb}d_{cc} + 2d_{aa}^4d_{cc}^2 + 4d_{aa}^3d_{bb}^2d_{cc} + 4d_{aa}^3d_{bb}d_{cc}^2 + d_{aa}^2d_{ab}d_{bb}d_{cc}^2 + d_{aa}^2d_{ac}d_{bb}^2d_{cc} + 2d_{aa}^2d_{bb}^4 + 4d_{aa}^2d_{bb}^3d_{cc} + d_{aa}^2d_{bb}^2d_{bc}d_{cc} + 8d_{aa}^2d_{bb}^2d_{cc}^2 + d_{aa}^2d_{bb}d_{bc}d_{cc}^2 + 4d_{aa}^2d_{bb}d_{cc}^3 + 2d_{aa}^2d_{cc}^4 + d_{aa}d_{ab}d_{bb}^2d_{cc}^2 + d_{aa}d_{ac}d_{bb}^2d_{cc}^2 + 2d_{aa}d_{bb}^4d_{cc} + 4d_{aa}d_{bb}^3d_{cc}^2 + 4d_{aa}d_{bb}^2d_{cc}^3 + 2d_{aa}d_{bb}d_{cc}^4 + 2d_{bb}^4d_{cc}^2 + 2d_{bb}^2d_{cc}^4$	kept
$E_{6,4}$	$= d_{aa}^5d_{bb} + d_{aa}^5d_{cc} + d_{aa}^4d_{bb}^2 + 4d_{aa}^4d_{bb}d_{cc} + d_{aa}^4d_{cc}^2 + d_{aa}^3d_{ab}d_{bb}d_{cc} + d_{aa}^3d_{ac}d_{bb}d_{cc} + 4d_{aa}^3d_{bb}^3 + d_{aa}^3d_{bb}^2d_{bc} + 7d_{aa}^3d_{bb}^2d_{cc} + 2d_{aa}^3d_{bb}d_{bc}d_{cc} + 7d_{aa}^3d_{bb}d_{cc}^2 + d_{aa}^3d_{bc}d_{cc}^2 + 4d_{aa}^3d_{cc}^3 + 2d_{aa}^2d_{ab}d_{bb}^2d_{cc} + 2d_{aa}^2d_{ab}d_{bb}d_{cc}^2 + d_{aa}^2d_{ab}d_{cc}^3 + d_{aa}^2d_{ac}d_{bb}^3 + 2d_{aa}^2d_{ac}d_{bb}^2d_{cc} + 2d_{aa}^2d_{ac}d_{bb}d_{cc}^2 + d_{aa}^2d_{bb}^4 + 7d_{aa}^2d_{bb}^3d_{cc} + 2d_{aa}^2d_{bb}^2d_{bc}d_{cc} + 12d_{aa}^2d_{bb}^2d_{cc}^2 + 2d_{aa}^2d_{bb}d_{bc}d_{cc}^2 + 7d_{aa}^2d_{bb}d_{cc}^3 + d_{aa}^2d_{cc}^4 + d_{aa}d_{ab}d_{bb}^3d_{cc} + 2d_{aa}d_{ab}d_{bb}^2d_{cc}^2 + 2d_{aa}d_{ab}d_{bb}d_{cc}^3 + 2d_{aa}d_{ac}d_{bb}^3d_{cc} + 2d_{aa}d_{ac}d_{bb}^2d_{cc}^2 + d_{aa}d_{ac}d_{bb}d_{cc}^3 + d_{aa}d_{bb}^5 + 4d_{aa}d_{bb}^4d_{cc} + d_{aa}d_{bb}^3d_{bc}d_{cc} + 7d_{aa}d_{bb}^3d_{cc}^2 + 2d_{aa}d_{bb}^2d_{bc}d_{cc}^2 + 7d_{aa}d_{bb}^2d_{cc}^3 + d_{aa}d_{bb}d_{bc}d_{cc}^3 + 4d_{aa}d_{bb}d_{cc}^4 + d_{aa}d_{cc}^5 + d_{ab}d_{bb}^2d_{cc}^3 + d_{ac}d_{bb}^3d_{cc}^2 + d_{bb}^5d_{cc} + d_{bb}^4d_{cc}^2 + 4d_{bb}^3d_{cc}^3 + d_{bb}^2d_{cc}^4 + d_{bb}d_{cc}^5$	kept

$E_{6,5}$ $= 2d_{aa}^5d_{bb} + 2d_{aa}^5d_{cc} + 2d_{aa}^4d_{bb}^2 + d_{aa}^4d_{bb}d_{bc} + 4d_{aa}^4d_{bb}d_{cc} + d_{aa}^4d_{bc}d_{cc} + 2d_{aa}^4d_{cc}^2 + d_{aa}^3d_{ab}d_{bb}^2 +$ kept
$3d_{aa}^3d_{ab}d_{bb}d_{cc} + 3d_{aa}^3d_{ab}d_{cc}^2 + 3d_{aa}^3d_{ac}d_{bb}^2 + 3d_{aa}^3d_{ac}d_{bb}d_{cc} + d_{aa}^3d_{ac}d_{cc}^2 + 4d_{aa}^3d_{bb}^3 + d_{aa}^3d_{bb}^2d_{bc} +$
$11d_{aa}^3d_{bb}^2d_{cc} + 6d_{aa}^2d_{bb}d_{bc}d_{cc} + 11d_{aa}^3d_{bb}d_{cc}^2 + d_{aa}^3d_{bc}d_{cc}^2 + 4d_{aa}^3d_{cc}^3 + d_{aa}^2d_{ab}d_{bb}^3 + 8d_{aa}^2d_{ab}d_{bb}^2d_{cc} +$
$5d_{aa}^2d_{ab}d_{bb}d_{cc}^2 + d_{aa}^2d_{ab}d_{cc}^3 + d_{aa}^2d_{ac}d_{bb}^3 + 5d_{aa}^2d_{ac}d_{bb}^2d_{cc} + 8d_{aa}^2d_{ac}d_{bb}d_{cc}^2 + d_{aa}^2d_{ac}d_{cc}^3 + 2d_{aa}^2d_{bb}^4 +$
$3d_{aa}^2d_{bb}^3d_{bc} + 11d_{aa}^2d_{bb}^3d_{cc} + 5d_{aa}^2d_{bb}^2d_{bc}d_{cc} + 12d_{aa}^2d_{bb}^2d_{cc}^2 + 5d_{aa}^2d_{bb}d_{bc}d_{cc}^2 + 11d_{aa}^2d_{bb}d_{cc}^3 + 3d_{aa}^2d_{bc}d_{cc}^3 +$
$2d_{aa}^2d_{cc}^4 + 3d_{aa}d_{ab}d_{bb}^3d_{cc} + 5d_{aa}d_{ab}d_{bb}^2d_{cc}^2 + 6d_{aa}d_{ab}d_{bb}d_{cc}^3 + d_{aa}d_{ab}d_{cc}^4 + d_{aa}d_{ac}d_{bb}^4 + 6d_{aa}d_{ac}d_{bb}^3d_{cc} +$
$5d_{aa}d_{ac}d_{bb}^2d_{cc}^2 + 3d_{aa}d_{ac}d_{bb}d_{cc}^3 + 2d_{aa}d_{bb}^5 + 4d_{aa}d_{bb}^4d_{cc} + 3d_{aa}d_{bb}^3d_{bc}d_{cc} + 11d_{aa}d_{bb}^3d_{cc}^2 + 8d_{aa}d_{bb}^2d_{bc}d_{cc}^2 +$
$11d_{aa}d_{bb}^2d_{cc}^3 + 3d_{aa}d_{bb}d_{bc}d_{cc}^3 + 4d_{aa}d_{bb}d_{cc}^4 + 2d_{aa}d_{cc}^5 + 3d_{ab}d_{bb}^3d_{cc}^2 + d_{ab}d_{bb}^2d_{cc}^3 + d_{ab}d_{bb}d_{cc}^4 + d_{ac}d_{bb}^4d_{cc} +$
$d_{ac}d_{bb}^3d_{cc}^2 + 3d_{ac}d_{bb}^2d_{cc}^3 + 2d_{bb}^5d_{cc} + 2d_{bb}^4d_{cc}^2 + d_{bb}^3d_{bc}d_{cc}^2 + 4d_{bb}^3d_{cc}^3 + d_{bb}^2d_{bc}d_{cc}^3 + 2d_{bb}^2d_{cc}^4 + 2d_{bb}d_{cc}^5$

$E_{6,6}$ $= d_{aa}^6 + d_{aa}^4d_{ab}d_{bb} + 2d_{aa}^4d_{ab}d_{cc} + 2d_{aa}^4d_{ac}d_{bb} + d_{aa}^4d_{ac}d_{cc} + 5d_{aa}^4d_{bb}^2 + 3d_{aa}^4d_{bb}d_{bc} + 8d_{aa}^4d_{bb}d_{cc} +$ kept
$3d_{aa}^4d_{bc}d_{cc} + 5d_{aa}^4d_{cc}^2 + 4d_{aa}^3d_{ab}d_{bb}^2 + 9d_{aa}^3d_{ab}d_{bb}d_{cc} + 3d_{aa}^3d_{ab}d_{cc}^2 + 3d_{aa}^3d_{ac}d_{bb}^2 + 9d_{aa}^3d_{ac}d_{bb}d_{cc} +$
$4d_{aa}^3d_{ac}d_{cc}^2 + 2d_{aa}^3d_{bb}^3 + 5d_{aa}^3d_{bb}^2d_{bc} + 10d_{aa}^3d_{bb}^2d_{cc} + 6d_{aa}^3d_{bb}d_{bc}d_{cc} + 10d_{aa}^3d_{bb}d_{cc}^2 + 5d_{aa}^3d_{bc}d_{cc}^2 + 2d_{aa}^3d_{cc}^3 +$
$d_{aa}^2d_{ab}d_{ac}d_{bb}d_{cc} + 4d_{aa}^2d_{ab}d_{bb}^3 + 10d_{aa}^2d_{ab}d_{bb}^2d_{cc} + d_{aa}^2d_{ab}d_{bb}d_{bc}d_{cc} + 15d_{aa}^2d_{ab}d_{bb}d_{cc}^2 + d_{aa}^2d_{ab}d_{bc}d_{cc}^2 +$
$5d_{aa}^2d_{ab}d_{cc}^3 + 5d_{aa}^2d_{ac}d_{bb}^3 + d_{aa}^2d_{ac}d_{bb}^2d_{bc} + 15d_{aa}^2d_{ac}d_{bb}^2d_{cc} + d_{aa}^2d_{ac}d_{bb}d_{bc}d_{cc} + 10d_{aa}^2d_{ac}d_{bb}d_{cc}^2 + 4d_{aa}^2d_{ac}d_{cc}^3 +$
$5d_{aa}^2d_{bb}^4 + 3d_{aa}^2d_{bb}^3d_{bc} + 10d_{aa}^2d_{bb}^3d_{cc} + 15d_{aa}^2d_{bb}^2d_{bc}d_{cc} + 18d_{aa}^2d_{bb}^2d_{cc}^2 + 2d_{aa}^2d_{bb}^2d_{cc}e_c + d_{aa}^2d_{bb}d_{bc}^2d_{cc} +$
$15d_{aa}^2d_{bb}d_{bc}d_{cc}^2 + 10d_{aa}^2d_{bb}d_{cc}^3 + 2d_{aa}^2d_{bb}d_{cc}^2e_b + 3d_{aa}^2d_{bc}d_{cc}^3 + 5d_{aa}^2d_{cc}^4 + d_{aa}d_{ab}^2d_{bb}d_{cc}^2 + d_{aa}d_{ab}d_{ac}d_{bb}^2d_{cc} +$
$d_{aa}d_{ab}d_{ac}d_{bb}d_{cc}^2 + d_{aa}d_{ab}d_{bb}^4 + 9d_{aa}d_{ab}d_{bb}^3d_{cc} + d_{aa}d_{ab}d_{bb}^2d_{bc}d_{cc} + 15d_{aa}d_{ab}d_{bb}^2d_{cc}^2 + d_{aa}d_{ab}d_{bb}d_{bc}d_{cc}^2 +$
$6d_{aa}d_{ab}d_{bb}d_{cc}^3 + 3d_{aa}d_{ab}d_{cc}^4 + d_{aa}d_{ac}^2d_{bb}^2d_{cc} + 3d_{aa}d_{ac}d_{bb}^4 + 6d_{aa}d_{ac}d_{bb}^3d_{cc} + d_{aa}d_{ac}d_{bb}^2d_{bc}d_{cc} +$
$15d_{aa}d_{ac}d_{bb}^2d_{cc}^2 + d_{aa}d_{ac}d_{bb}d_{bc}d_{cc}^2 + 9d_{aa}d_{ac}d_{bb}d_{cc}^3 + d_{aa}d_{ac}d_{cc}^4 + 2d_{aa}d_{bb}^4d_{bc} + 8d_{aa}d_{bb}^4d_{cc} + 9d_{aa}d_{bb}^3d_{bc}d_{cc} +$
$10d_{aa}d_{bb}^3d_{cc}^2 + 10d_{aa}d_{bb}^2d_{bc}d_{cc}^2 + 10d_{aa}d_{bb}^2d_{cc}^3 + 2d_{aa}d_{bb}^2d_{cc}^2e_a + 9d_{aa}d_{bb}d_{bc}d_{cc}^3 + 8d_{aa}d_{bb}d_{cc}^4 + 2d_{aa}d_{bc}d_{cc}^4 +$
$d_{ab}d_{ac}d_{bb}^2d_{cc}^2 + 2d_{ab}d_{bb}^4d_{cc} + 3d_{ab}d_{bb}^3d_{cc}^2 + 5d_{ab}d_{bb}^2d_{cc}^3 + 3d_{ab}d_{bb}d_{cc}^4 + 3d_{ac}d_{bb}^4d_{cc} + 5d_{ac}d_{bb}^3d_{cc}^2 + 3d_{ac}d_{bb}^2d_{cc}^3 +$
$2d_{ac}d_{bb}d_{cc}^4 + d_{bb}^6 + d_{bb}^4d_{bc}d_{cc} + 5d_{bb}^4d_{cc}^2 + 4d_{bb}^3d_{bc}d_{cc}^2 + 2d_{bb}^3d_{cc}^3 + 4d_{bb}^2d_{bc}d_{cc}^3 + 5d_{bb}^2d_{cc}^4 + d_{bb}d_{bc}d_{cc}^4 + d_{cc}^6$

$E_{6,7}$ $=$ (151 terms; see the full text file) kept

$E_{6,8}$ $=$ (190 terms; see the full text file) kept

$E_{6,9}$ $=$ (236 terms; see the full text file) kept

$E_{6,10}$ $=$ (344 terms; see the full text file) kept

$E_{6,11}$ $=$ (394 terms; see the full text file) kept

$E_{6,12}$ $=$ (476 terms; see the full text file) kept

$E_{6,13}$ $=$ (619 terms; see the full text file) kept

$E_{6,14}$ $=$ (682 terms; see the full text file) kept

$E_{6,15}$ $=$ (754 terms; see the full text file) kept

$E_{6,16}$ $=$ (925 terms; see the full text file) kept

$E_{6,17}$ $=$ (994 terms; see the full text file) kept

$E_{6,18}$ $=$ (1071 terms; see the full text file) kept

$E_{6,19}$ $=$ (1222 terms; see the full text file) kept

$E_{6,20}$ $=$ (1266 terms; see the full text file) kept

$E_{6,21}$ $=$ (1266 terms; see the full text file) kept

$E_{6,22}$ $=$ (1400 terms; see the full text file) kept

$E_{6,23}$ $=$ (1424 terms; see the full text file) kept

$E_{6,24}$ $=$ (1379 terms; see the full text file) kept

$E_{6,25}$ = (1424 terms; see the full text file) kept

$E_{6,26}$ = (1400 terms; see the full text file) kept

$E_{6,27}$ = (1266 terms; see the full text file) kept

$E_{6,28}$ = (1266 terms; see the full text file) kept

$E_{6,29}$ = (1222 terms; see the full text file) kept

$E_{6,30}$ = (1071 terms; see the full text file) kept

$E_{6,31}$ = (994 terms; see the full text file) kept

$E_{6,32}$ = (925 terms; see the full text file) kept

$E_{6,33}$ = (754 terms; see the full text file) kept

$E_{6,34}$ = (682 terms; see the full text file) kept

$E_{6,35}$ = (619 terms; see the full text file) kept

$E_{6,36}$ = (476 terms; see the full text file) kept

$E_{6,37}$ = (394 terms; see the full text file) kept

$E_{6,38}$ = (344 terms; see the full text file) kept

$E_{6,39}$ = (236 terms; see the full text file) kept

$E_{6,40}$ = (190 terms; see the full text file) kept

$E_{6,41}$ = (151 terms; see the full text file) kept

$E_{6,42}$ = $2d_{aa}e_ae_b^2e_c^2 + d_{ab}^2e_ae_be_c^2 + d_{ab}d_{ac}e_a^2e_be_c + d_{ab}d_{ac}e_ae_b^2e_c + d_{ab}d_{ac}e_ae_be_c^2 + d_{ab}d_{ac}e_b^2e_c^2 + d_{ab}d_{bc}e_a^2e_be_c +$ kept
$d_{ab}d_{bc}e_a^2e_c^2 + d_{ab}d_{bc}e_ae_b^2e_c + d_{ab}d_{bc}e_ae_be_c^2 + d_{ab}e_a^4e_b + 2d_{ab}e_a^4e_c + 4d_{ab}e_a^3e_b^2 + 9d_{ab}e_a^3e_be_c + 3d_{ab}e_a^3e_c^2 +$
$4d_{ab}e_a^2e_b^3 + 10d_{ab}e_a^2e_b^2e_c + 15d_{ab}e_a^2e_be_c^2 + 5d_{ab}e_a^2e_c^3 + d_{ab}e_ae_b^4 + 9d_{ab}e_ae_b^3e_c + 15d_{ab}e_ae_b^2e_c^2 + 6d_{ab}e_ae_be_c^3 +$
$3d_{ab}e_ae_c^4 + 2d_{ab}e_b^4e_c + 3d_{ab}e_b^3e_c^2 + 5d_{ab}e_b^2e_c^3 + 3d_{ab}e_be_c^4 + d_{ac}^2e_ae_b^2e_c + d_{ac}d_{bc}e_a^2e_b^2 + d_{ac}d_{bc}e_a^2e_be_c +$
$d_{ac}d_{bc}e_ae_b^2e_c + d_{ac}d_{bc}e_ae_be_c^2 + 2d_{ac}e_a^4e_b + d_{ac}e_a^4e_c + 3d_{ac}e_a^3e_b^2 + 9d_{ac}e_a^3e_be_c + 4d_{ac}e_a^3e_c^2 + 5d_{ac}e_a^2e_b^3 +$
$15d_{ac}e_a^2e_b^2e_c + 10d_{ac}e_a^2e_be_c^2 + 4d_{ac}e_a^2e_c^3 + 3d_{ac}e_ae_b^4 + 6d_{ac}e_ae_b^3e_c + 15d_{ac}e_ae_b^2e_c^2 + 9d_{ac}e_ae_be_c^3 + d_{ac}e_ae_c^4 +$
$3d_{ac}e_b^4e_c + 5d_{ac}e_b^3e_c^2 + 3d_{ac}e_b^2e_c^3 + 2d_{ac}e_be_c^4 + 2d_{bb}e_a^2e_be_c^2 + d_{bc}^2e_a^2e_be_c + 3d_{bc}e_a^4e_b + 3d_{bc}e_a^4e_c + 5d_{bc}e_a^3e_b^2 +$
$6d_{bc}e_a^3e_be_c + 5d_{bc}e_a^3e_c^2 + 3d_{bc}e_a^2e_b^3 + 15d_{bc}e_a^2e_b^2e_c + 15d_{bc}e_a^2e_be_c^2 + 3d_{bc}e_a^2e_c^3 + 2d_{bc}e_ae_b^4 + 9d_{bc}e_ae_b^3e_c +$
$10d_{bc}e_ae_b^2e_c^2 + 9d_{bc}e_ae_be_c^3 + 2d_{bc}e_ae_c^4 + d_{bc}e_b^4e_c + 4d_{bc}e_b^3e_c^2 + 4d_{bc}e_b^2e_c^3 + d_{bc}e_be_c^4 + 2d_{cc}e_a^2e_b^2e_c + e_a^6 +$
$5e_a^4e_b^2 + 8e_a^4e_be_c + 5e_a^4e_c^2 + 2e_a^3e_b^3 + 10e_a^3e_b^2e_c + 10e_a^3e_be_c^2 + 2e_a^3e_c^3 + 5e_a^2e_b^4 + 10e_a^2e_b^3e_c + 18e_a^2e_b^2e_c^2 +$
$10e_a^2e_be_c^3 + 5e_a^2e_c^4 + 8e_ae_b^4e_c + 10e_ae_b^3e_c^2 + 10e_ae_b^2e_c^3 + 8e_ae_be_c^4 + e_b^6 + 5e_b^4e_c^2 + 2e_b^3e_c^3 + 5e_b^2e_c^4 + e_c^6$

$E_{6,43}$ = $d_{ab}e_a^3e_b^2 + 3d_{ab}e_a^3e_be_c + 3d_{ab}e_a^3e_c^2 + d_{ab}e_a^2e_b^3 + 8d_{ab}e_a^2e_b^2e_c + 5d_{ab}e_a^2e_be_c^2 + d_{ab}e_a^2e_c^3 + 3d_{ab}e_ae_b^3e_c +$ kept
$5d_{ab}e_ae_b^2e_c^2 + 6d_{ab}e_ae_be_c^3 + d_{ab}e_ae_c^4 + 3d_{ab}e_b^3e_c^2 + d_{ab}e_b^2e_c^3 + d_{ab}e_be_c^4 + 3d_{ac}e_a^3e_b^2 + 3d_{ac}e_a^3e_be_c + d_{ac}e_a^3e_c^2 +$
$d_{ac}e_a^2e_b^3 + 5d_{ac}e_a^2e_b^2e_c + 8d_{ac}e_a^2e_be_c^2 + d_{ac}e_a^2e_c^3 + d_{ac}e_ae_b^4 + 6d_{ac}e_ae_b^3e_c + 5d_{ac}e_ae_b^2e_c^2 + 3d_{ac}e_ae_be_c^3 + d_{ac}e_ae_c^4 +$
$d_{ac}e_b^3e_c^2 + 3d_{ac}e_b^2e_c^3 + d_{bc}e_a^4e_b + d_{bc}e_a^4e_c + d_{bc}e_a^3e_b^2 + 6d_{bc}e_a^3e_be_c + d_{bc}e_a^3e_c^2 + 3d_{bc}e_a^2e_b^3 + 5d_{bc}e_a^2e_b^2e_c +$
$5d_{bc}e_a^2e_be_c^2 + 3d_{bc}e_a^2e_c^3 + 3d_{bc}e_ae_b^3e_c + 8d_{bc}e_ae_b^2e_c^2 + 3d_{bc}e_ae_be_c^3 + d_{bc}e_b^3e_c^2 + d_{bc}e_b^2e_c^3 + 2e_a^5e_b + 2e_a^5e_c +$
$2e_a^4e_b^2 + 4e_a^4e_be_c + 2e_a^4e_c^2 + 4e_a^3e_b^3 + 11e_a^3e_b^2e_c + 11e_a^3e_be_c^2 + 4e_a^3e_c^3 + 2e_a^2e_b^4 + 11e_a^2e_b^3e_c + 12e_a^2e_b^2e_c^2 + 11e_a^2e_be_c^3 +$
$2e_a^2e_c^4 + 2e_ae_b^5 + 4e_ae_b^4e_c + 11e_ae_b^3e_c^2 + 11e_ae_b^2e_c^3 + 4e_ae_be_c^4 + 2e_ae_c^5 + 2e_b^5e_c + 2e_b^4e_c^2 + 4e_b^3e_c^3 + 2e_b^2e_c^4 + 2e_be_c^5$

$E_{6,44}$ = $d_{ab}e_a^3e_be_c + 2d_{ab}e_a^2e_b^2e_c + 2d_{ab}e_a^2e_be_c^2 + d_{ab}e_a^2e_c^3 + d_{ab}e_ae_b^3e_c + 2d_{ab}e_ae_b^2e_c^2 + 2d_{ab}e_ae_be_c^3 + d_{ab}e_b^2e_c^3 +$ kept
$d_{ac}e_a^3e_be_c + d_{ac}e_a^2e_b^3 + 2d_{ac}e_a^2e_b^2e_c + 2d_{ac}e_a^2e_be_c^2 + 2d_{ac}e_ae_b^3e_c + 2d_{ac}e_ae_b^2e_c^2 + d_{ac}e_ae_be_c^3 + d_{ac}e_b^3e_c^2 +$
$d_{bc}e_a^3e_b^2 + 2d_{bc}e_a^3e_be_c + d_{bc}e_a^3e_c^2 + 2d_{bc}e_a^2e_b^2e_c + 2d_{bc}e_a^2e_be_c^2 + d_{bc}e_ae_b^3e_c + 2d_{bc}e_ae_b^2e_c^2 + d_{bc}e_ae_be_c^3 + e_a^5e_b +$
$e_a^5e_c + e_a^4e_b^2 + 4e_a^4e_be_c + e_a^4e_c^2 + 4e_a^3e_b^3 + 7e_a^3e_b^2e_c + 7e_a^3e_be_c^2 + 4e_a^3e_c^3 + e_a^2e_b^4 + 7e_a^2e_b^3e_c + 12e_a^2e_b^2e_c^2 +$
$7e_a^2e_be_c^3 + e_a^2e_c^4 + e_ae_b^5 + 4e_ae_b^4e_c + 7e_ae_b^3e_c^2 + 7e_ae_b^2e_c^3 + 4e_ae_be_c^4 + e_ae_c^5 + e_b^5e_c + e_b^4e_c^2 + 4e_b^3e_c^3 + e_b^2e_c^4 + e_be_c^5$

$E_{6,45}$ $= d_{ab}e_a^2e_be_c^2+d_{ab}e_ae_b^2e_c^2+d_{ac}e_a^2e_b^2e_c+d_{ac}e_ae_b^2e_c^2+d_{bc}e_a^2e_b^2e_c+d_{bc}e_a^2e_be_c^2+2e_a^4e_b^2+2e_a^4e_be_c+2e_a^4e_c^2+4e_a^3e_b^2e_c+4e_a^3e_be_c^2+2e_a^2e_b^4+4e_a^2e_b^3e_c+8e_a^2e_b^2e_c^2+4e_a^2e_be_c^3+2e_a^2e_c^4+2e_ae_b^4e_c+4e_ae_b^3e_c^2+4e_ae_b^2e_c^3+2e_ae_be_c^4+2e_b^4e_c^2+2e_b^2e_c^4$ kept

$E_{6,46}$ $= e_a^4e_be_c + e_a^3e_b^3 + 2e_a^3e_b^2e_c + 2e_a^3e_be_c^2 + e_a^3e_c^3 + 2e_a^2e_b^3e_c + 3e_a^2e_b^2e_c^2 + 2e_a^2e_be_c^3 + e_ae_b^4e_c + 2e_ae_b^3e_c^2 + 2e_ae_b^2e_c^3 + e_ae_be_c^4 + e_b^3e_c^3$ dropped: $= E_{6,2} - E_{6,4} + E_{6,8} - E_{6,10} + E_{6,14} - E_{6,16} + E_{6,20} - E_{6,22} + E_{6,26} - E_{6,28} + E_{6,32} - E_{6,34} + E_{6,38} - E_{6,40} + E_{6,44}$

$E_{6,47}$ $= e_a^3e_b^2e_c + e_a^3e_be_c^2 + e_a^2e_b^3e_c + e_a^2e_be_c^3 + e_ae_b^3e_c^2 + e_ae_b^2e_c^3$ dropped: $= E_{6,1} - E_{6,5} + E_{6,7} - E_{6,11} + E_{6,13} - E_{6,17} + E_{6,19} - E_{6,23} + E_{6,25} - E_{6,29} + E_{6,31} - E_{6,35} + E_{6,37} - E_{6,41} + E_{6,43}$

$E_{6,48}$ $= e_a^2e_b^2e_c^2$ kept

6 $S(2) \times S(1)$ acting on (a, b, p), $k = 1$

$n = 3$, $k = 1$, $|G| = 2$, $m = 9$ slots. Bursarial outputs: 26; after safe reductions: 23 (exact linear dependencies checked for $c \leq 2$). Quotient dimension (the minimum number of outputs, for generic inputs): 3.

slot j	w_j	degree in L
0	d_{aa}	2
1	d_{bb}	2
2	d_{pp}	2
3	d_{ab}	2
4	d_{ap}	2
5	d_{bp}	2
6	e_a	1
7	e_b	1
8	e_p	1

g	$g \cdot w$
e	$(d_{aa},\ d_{bb},\ d_{pp},\ d_{ab},\ d_{ap},\ d_{bp},\ e_a,\ e_b,\ e_p)$
(ab)	$(d_{bb},\ d_{aa},\ d_{pp},\ d_{ab},\ d_{bp},\ d_{ap},\ e_b,\ e_a,\ e_p)$

$c = 1$

output	expanded	safe reductions
$E_{1,0}$	$= d_{aa} + d_{bb}$	kept
$E_{1,1}$	$= d_{aa} + d_{bb}$	dropped: $= E_{1,0}$
$E_{1,2}$	$= 2d_{pp}$	kept
$E_{1,3}$	$= 2d_{ab}$	kept
$E_{1,4}$	$= d_{ap} + d_{bp}$	kept
$E_{1,5}$	$= d_{ap} + d_{bp}$	dropped: $= E_{1,4}$
$E_{1,6}$	$= e_a + e_b$	kept
$E_{1,7}$	$= e_a + e_b$	dropped: $= E_{1,6}$
$E_{1,8}$	$= 2e_p$	kept

$c = 2$

output	expanded	safe reductions

$E_{2,0}$	$= d_{aa}d_{bb}$	kept
$E_{2,1}$	$= d_{aa}^2 + d_{bb}^2$	kept
$E_{2,2}$	$= d_{aa}d_{bb} + d_{aa}d_{pp} + d_{bb}d_{pp}$	kept
$E_{2,3}$	$= d_{aa}d_{ab} + d_{aa}d_{pp} + d_{ab}d_{bb} + d_{bb}d_{pp}$	kept
$E_{2,4}$	$= d_{aa}d_{ab} + d_{aa}d_{bp} + d_{ab}d_{bb} + d_{ap}d_{bb} + d_{pp}^2$	kept
$E_{2,5}$	$= 2d_{aa}d_{ap} + 2d_{ab}d_{pp} + 2d_{bb}d_{bp}$	kept
$E_{2,6}$	$= d_{aa}d_{bp} + d_{aa}e_b + d_{ab}^2 + d_{ap}d_{bb} + d_{ap}d_{pp} + d_{bb}e_a + d_{bp}d_{pp}$	kept
$E_{2,7}$	$= 2d_{aa}e_a + d_{ab}d_{ap} + d_{ab}d_{bp} + d_{ap}d_{pp} + 2d_{bb}e_b + d_{bp}d_{pp}$	kept
$E_{2,8}$	$= d_{aa}e_b + d_{aa}e_p + d_{ab}d_{ap} + d_{ab}d_{bp} + d_{ap}d_{bp} + d_{bb}e_a + d_{bb}e_p + d_{pp}e_a + d_{pp}e_b$	kept
$E_{2,9}$	$= d_{aa}e_p + d_{ab}e_a + d_{ab}e_b + d_{ap}^2 + d_{bb}e_p + d_{bp}^2 + d_{pp}e_a + d_{pp}e_b$	kept
$E_{2,10}$	$= d_{ab}e_a + d_{ab}e_b + d_{ap}d_{bp} + d_{ap}e_b + d_{bp}e_a + 2d_{pp}e_p$	kept
$E_{2,11}$	$= 2d_{ab}e_p + 2d_{ap}e_a + 2d_{bp}e_b$	kept
$E_{2,12}$	$= d_{ap}e_b + d_{ap}e_p + d_{bp}e_a + d_{bp}e_p + e_ae_b$	kept
$E_{2,13}$	$= d_{ap}e_p + d_{bp}e_p + e_a^2 + e_b^2$	kept
$E_{2,14}$	$= e_ae_b + e_ae_p + e_be_p$	kept
$E_{2,15}$	$= e_ae_p + e_be_p$	kept
$E_{2,16}$	$= e_p^2$	kept